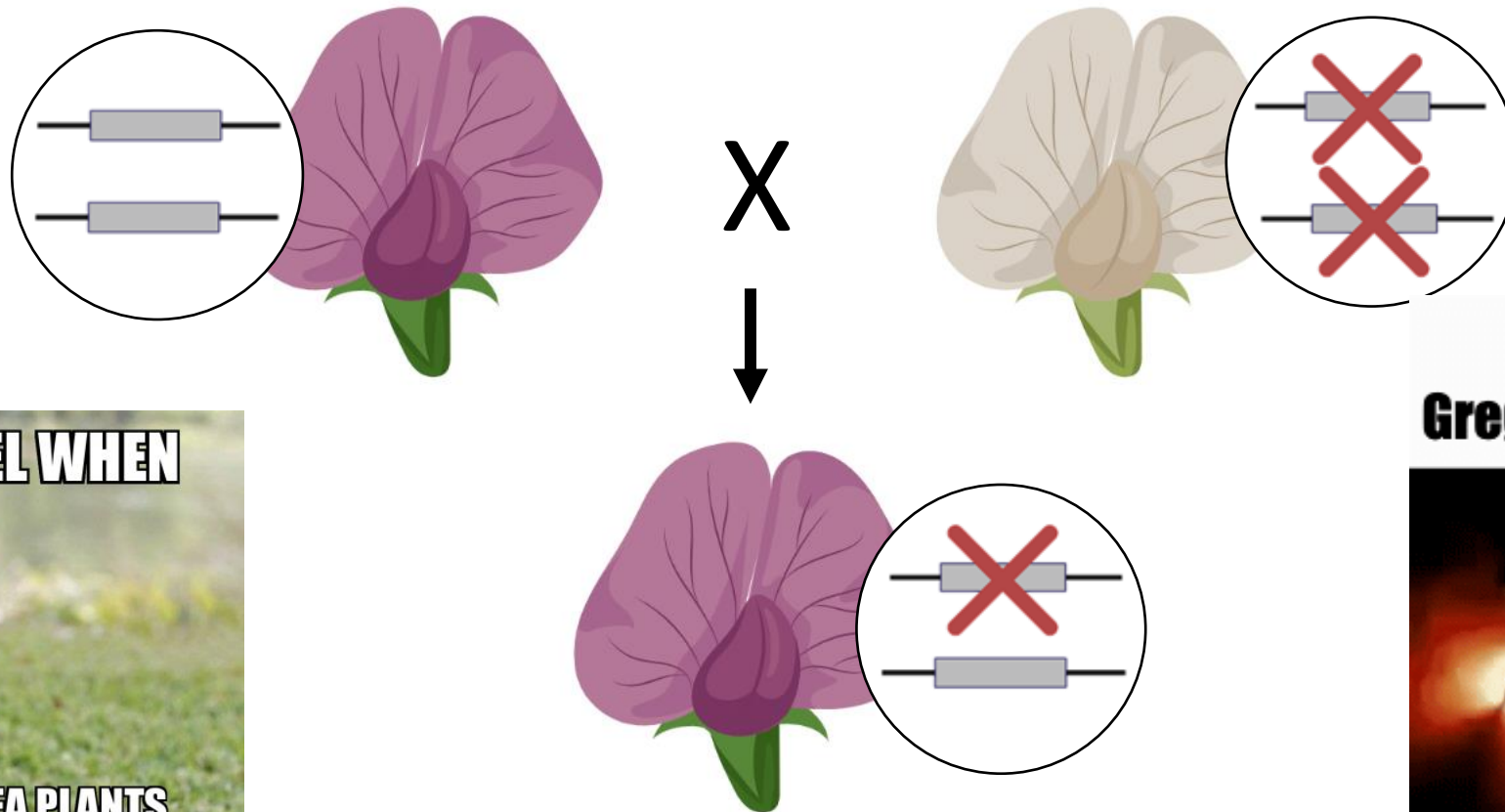


“Transmission genetics”

The transmission of genetic information from one generation to the next



GREGOR MENDEL WHEN

HE CROSSED TWO PEA PLANTS

makeameme.org

Gregor Mendel in 1865



imgflip.com



ChatGPT

what is the best university for plant sciences and plant genetics,
ranked

what is the best university for plant sciences and plant genetics, ranked

Top Universities & Rankings

Below are several sources and examples of universities that frequently rank at or near the top in plant science / agriculture / genetics:

University	Strengths / Notes
University of California, Davis (USA)	Very strong in plant sciences, agriculture, breeding, genetics, and related areas
University of Florida (USA)	Strong in agriculture, plant physiology, breeding
Wageningen University & Research (Netherlands)	Globally known for agriculture, plant breeding, food sciences
China Agricultural University / Nanjing Agricultural University / Huazhong Agricultural University (China)	Strong in agricultural and plant biology research in China / Asia
Cornell University (USA)	Excellent plant breeding & genetics programs
University of California, Berkeley (USA)	Strong in plant biology, botany, molecular biology
University of Wisconsin–Madison (USA)	Good reputation in plant genetics / breeding



what is the best university for horses, ranked

Top programs / universities

Here are examples of institutions often regarded among the strongest:

Institution	Strengths / Notes
Colorado State University (USA)	Called the "most comprehensive equine program in the United States" in research, teaching, and public service. College of Agricu...
University of Kentucky (USA)	Has strong equine science & management programs and is located in a major horse region. horseillustrated.c... +1
Hartpury University (UK)	Strong in equine and sport / animal science programs. https://www.educ... +1
University of Edinburgh / Royal (Dick) School of Veterinary Studies (UK)	Excellent for veterinary and equine research. The Dick Vet is highly rated in veterinary & agriculture science research in the UK. Wikipedia
Texas A&M University (USA)	Prominent in equine / animal science circles (often listed among top U.S. equestrian colleges). College Flight Path +1
University of New Hampshire (USA)	Ranked high in "horse husbandry / equine science & management" discipline by College Factual. College Factual
University of Findlay (USA)	Frequent listing in top U.S. schools for equine studies. College Factual +1
Emory & Henry College (USA)	Noted for its equine center and hands-on program. Emory & Henry U... +1



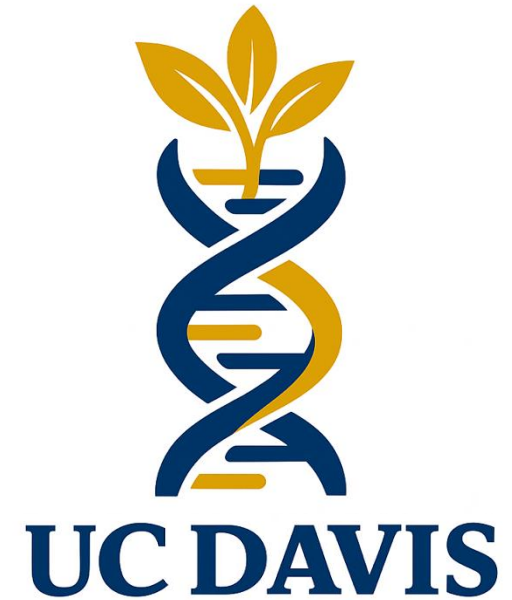
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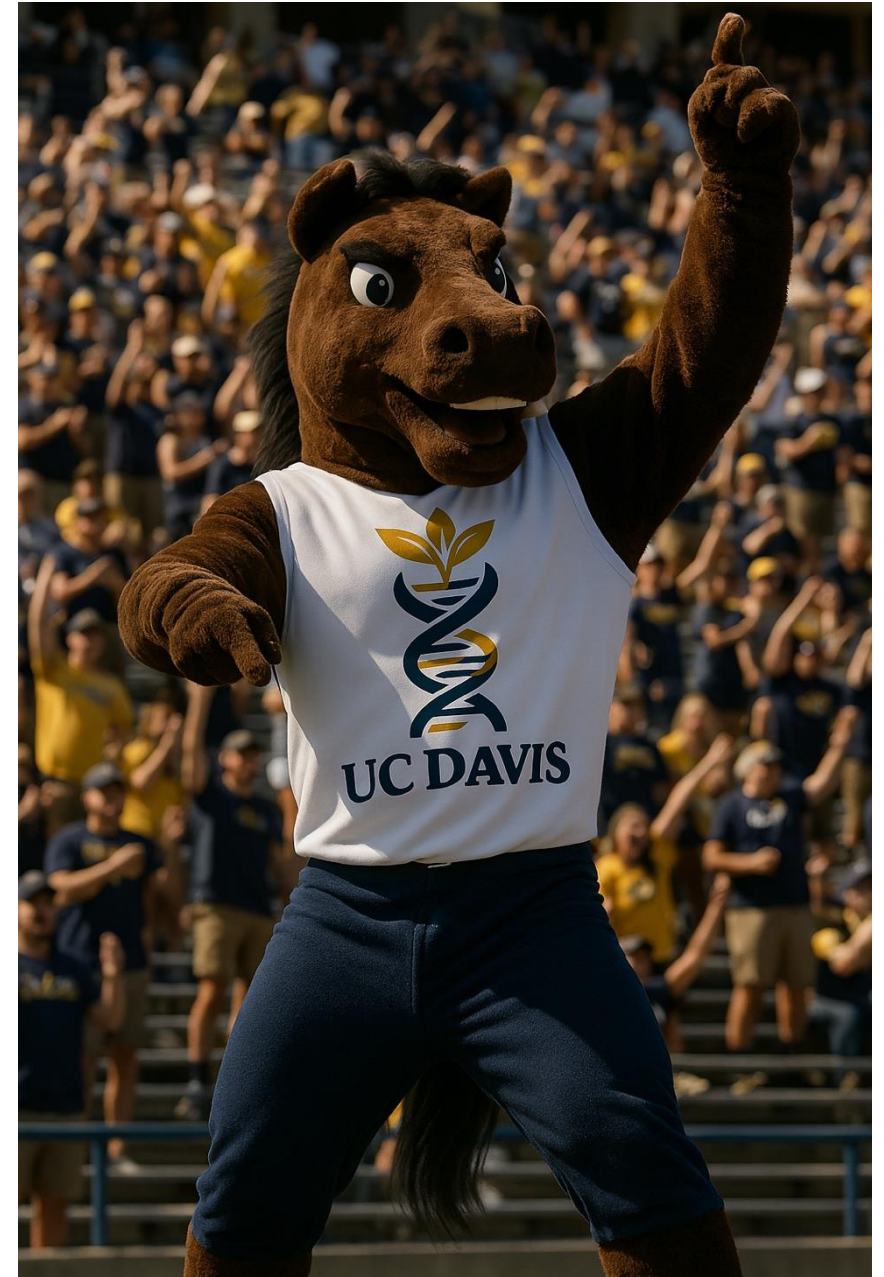


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what is the best university for plant sciences and plant genetics, ranked

Top Universities & Rankings

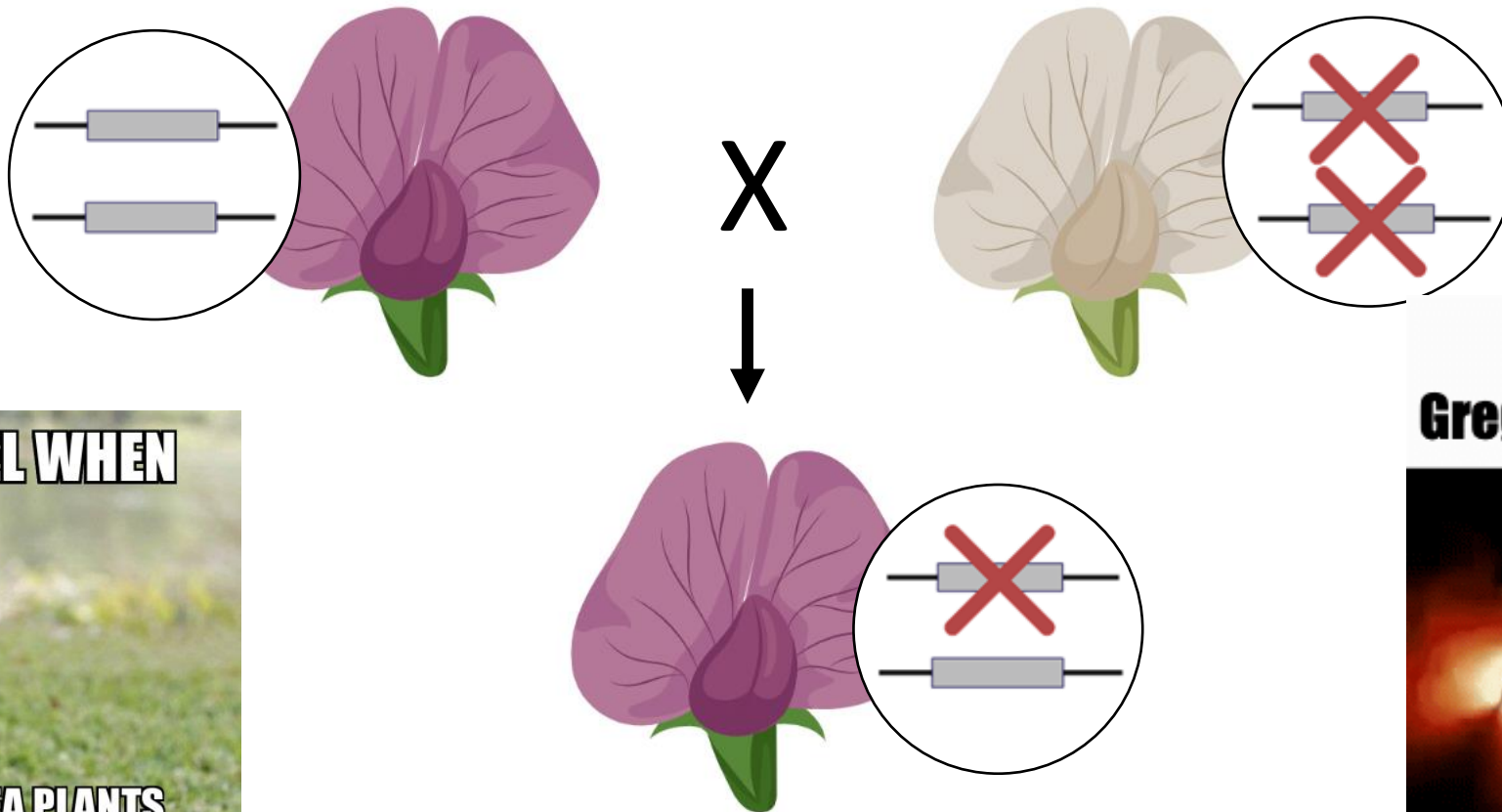
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makeameme.org

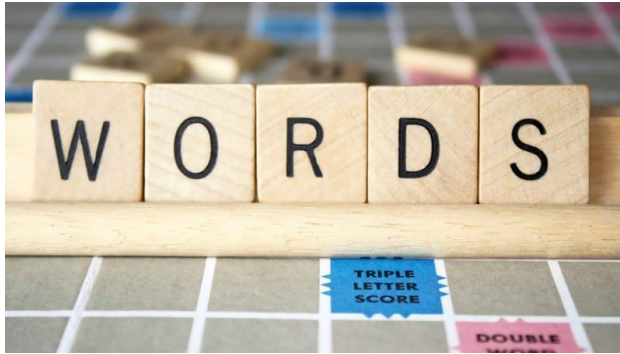
Gregor Mendel in 1865



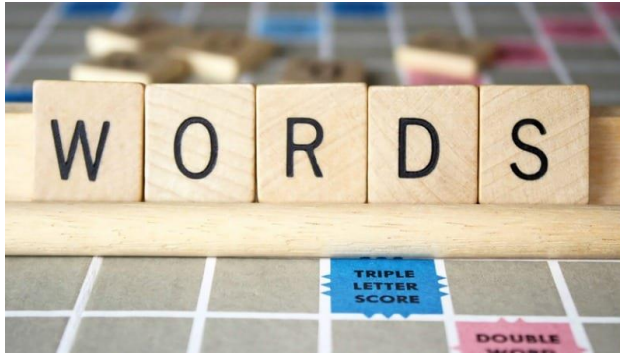
imgflip.com

First some logistical details

- Reminder to do daily reflection on Canvas
- Reminder that weekly Quiz is online and due before class next Tuesday



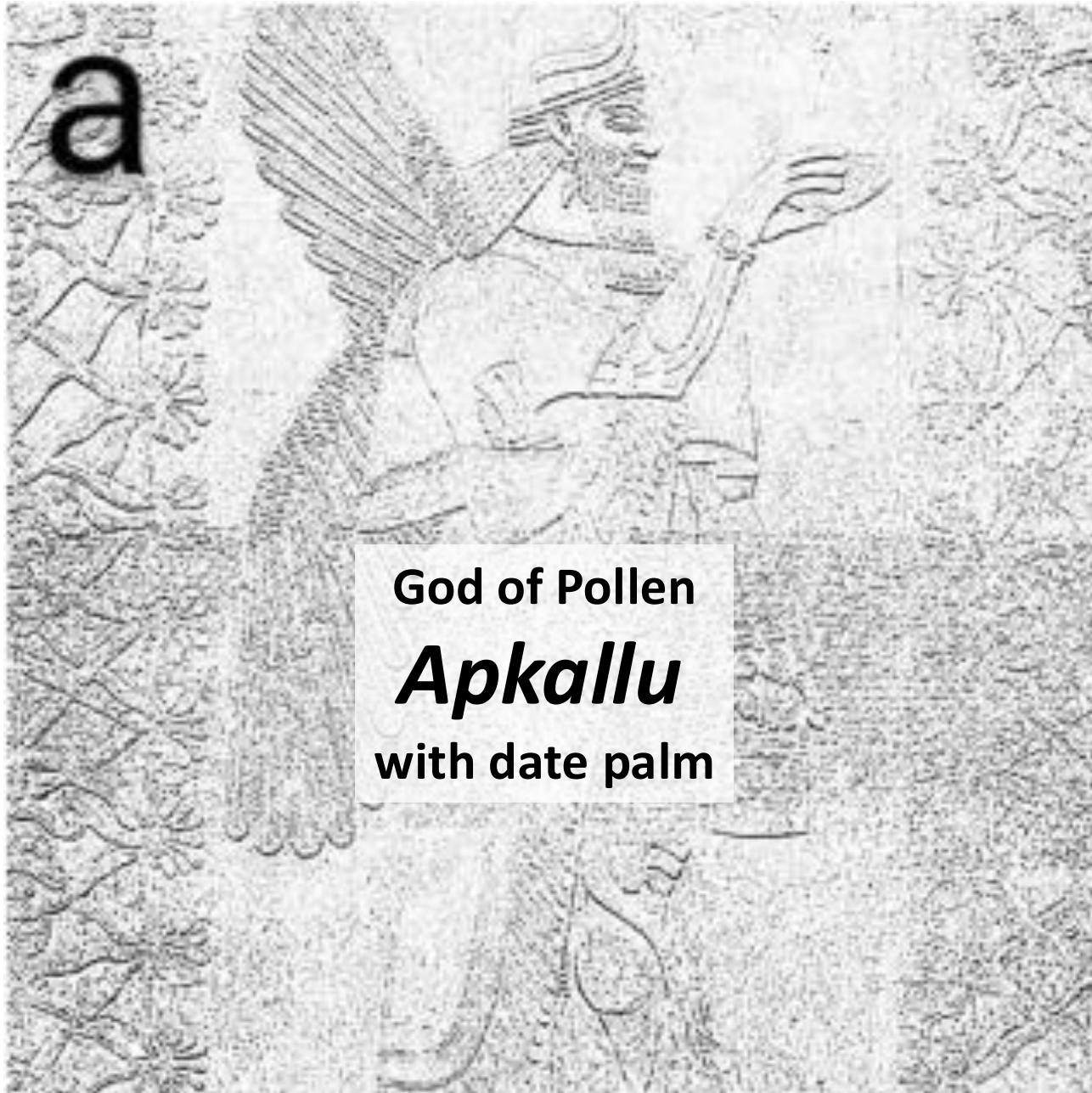
- 1. Gene:** A segment of DNA that codes for a specific protein or RNA molecule, and thus determines a specific characteristic.
- 2. Allele:** One of two or more alternative forms of a gene.
- 3. Trait:** A measurable characteristic or feature of an organism (e.g. flower color)
- 4. Phenotype:** The observed trait for a particular organism (e.g. purple flower, white flower).
- 5. Dominant:** Allele whose phenotype is seen in an individual even if only one copy of is present.
- 6. Recessive:** An allele whose phenotype is seen only when two are present.
- 7. Segregation:** The separation of paired alleles during the formation of gametes, ensuring that each gamete receives only one allele for each gene.
- 8. Assortment:** The independent distribution of genes into gametes during meiosis; based on Mendel's law of independent assortment.
- 9. F1:** The first generation of offspring produced from a cross of two parent organisms.
- 10. F2:** The second generation of offspring, produced from crossing two F1 individuals.
- 11. Blending inheritance:** An outdated theory suggesting that offspring have a blend or intermediate of the parents' traits.
- 12. Particulate inheritance:** The concept that inheritance is based on discrete units (alleles) that are passed from one generation to the next.
- 13. True breeding:** Refers to organisms that produce offspring identical to themselves if allowed to self-pollinate.
- 14. Inbred:** Resulting from or involved in breeding closely related individuals.
- 15. Hybrid:** An offspring resulting from the cross between parents of different genotypes.
- 16. Homozygous:** Having two identical alleles for a particular gene.
- 17. Heterozygous:** Having two different alleles for a particular gene.
- 18. Cross:** The mating of two organisms.
- 19. Dihybrid cross:** A cross between two individuals that differ in two genes.
- 20. Trihybrid cross:** A cross between two individuals that differ in three genes.



Learning objective: **Predict genotype and phenotype of offspring from crosses**

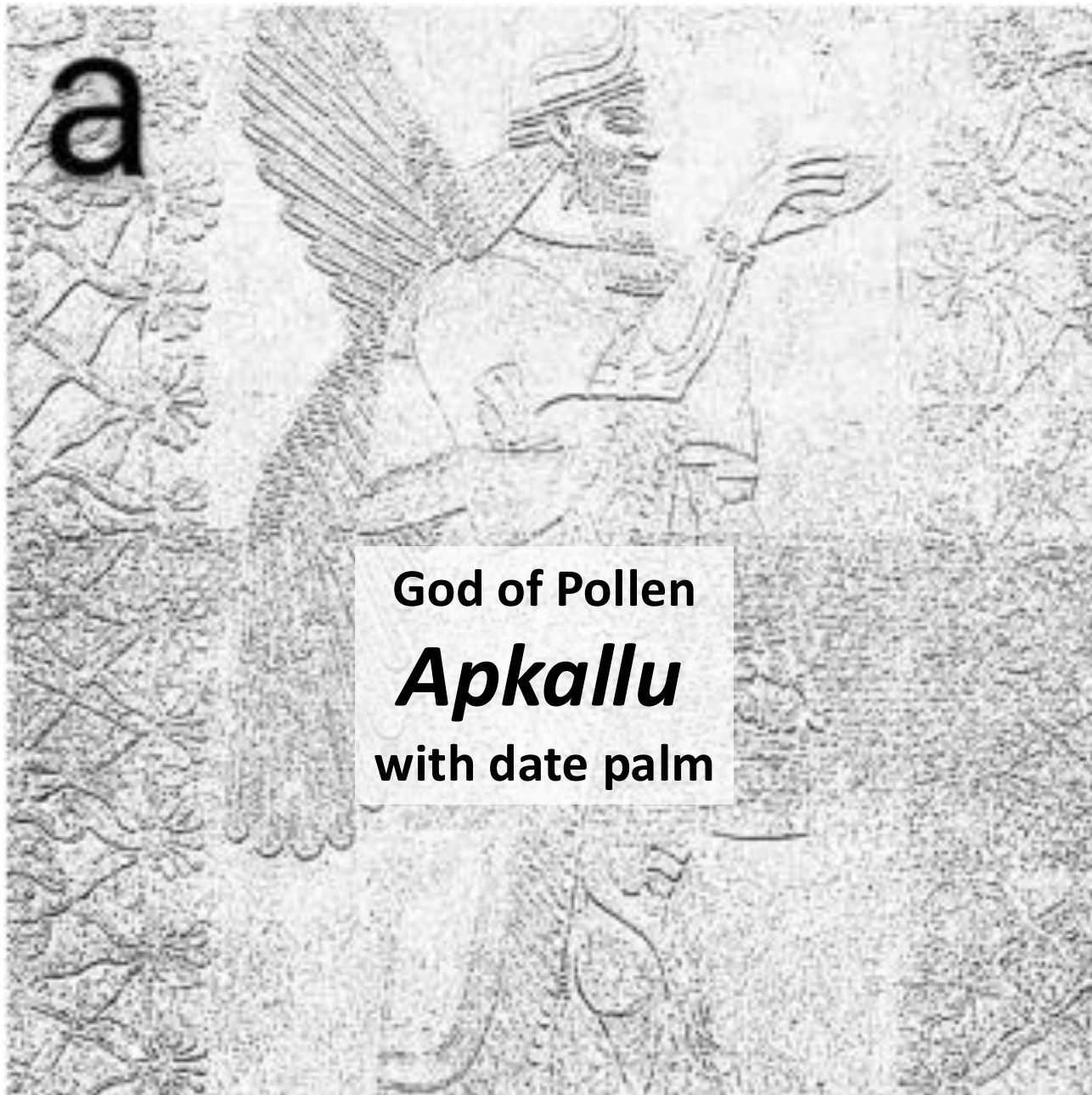
- Particulate (allelic) inheritance
- Allele Dominance
- Allele Segregation

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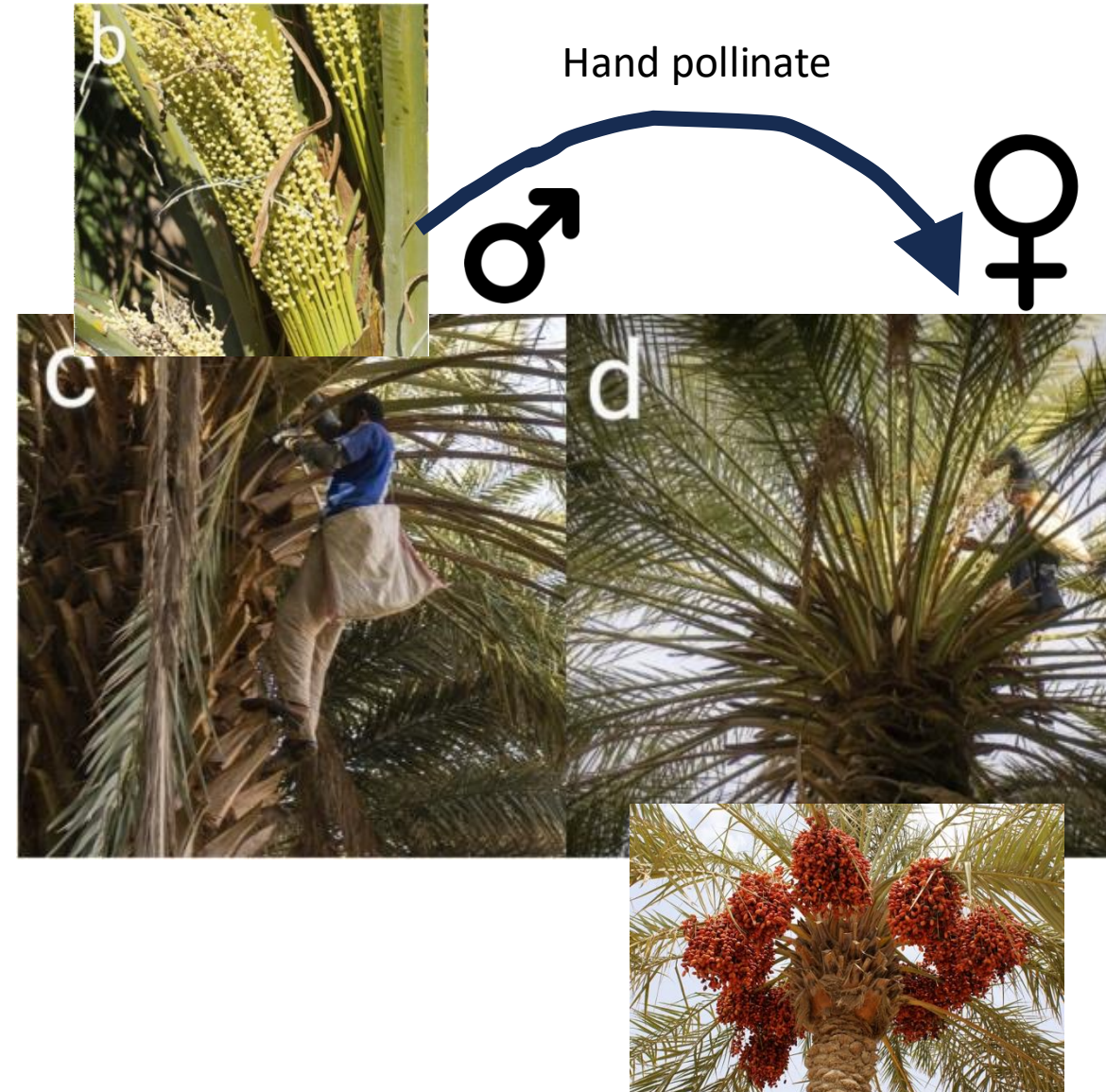


God of Pollen
Apkallu
with date palm

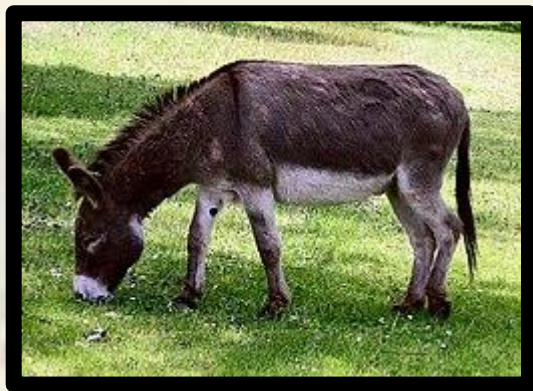
**Humans have been
interested in inheritance
for thousands of years**



Dioecious date palm

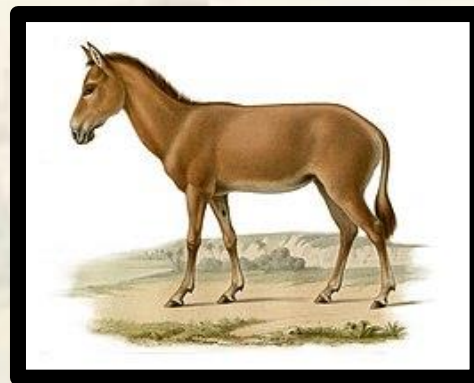


Making hybrids 5000 years ago



Donkey

X
↓



Syrian onager

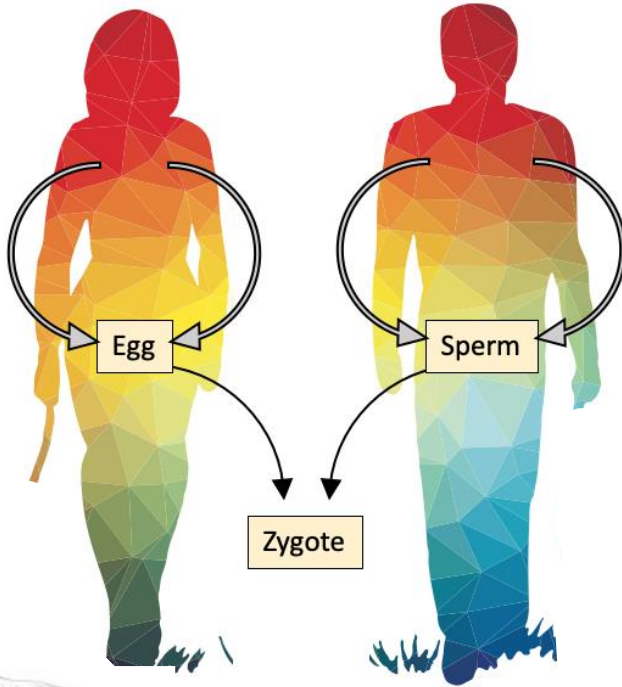


Kunga

Why do offspring look like their parents?

Pangenesis theory

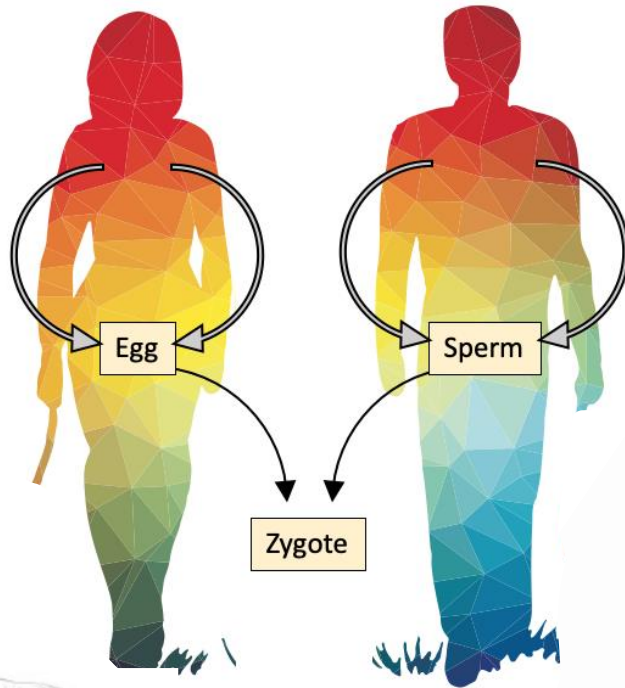
All cells (both germ-line and soma) contribute to heritable information.



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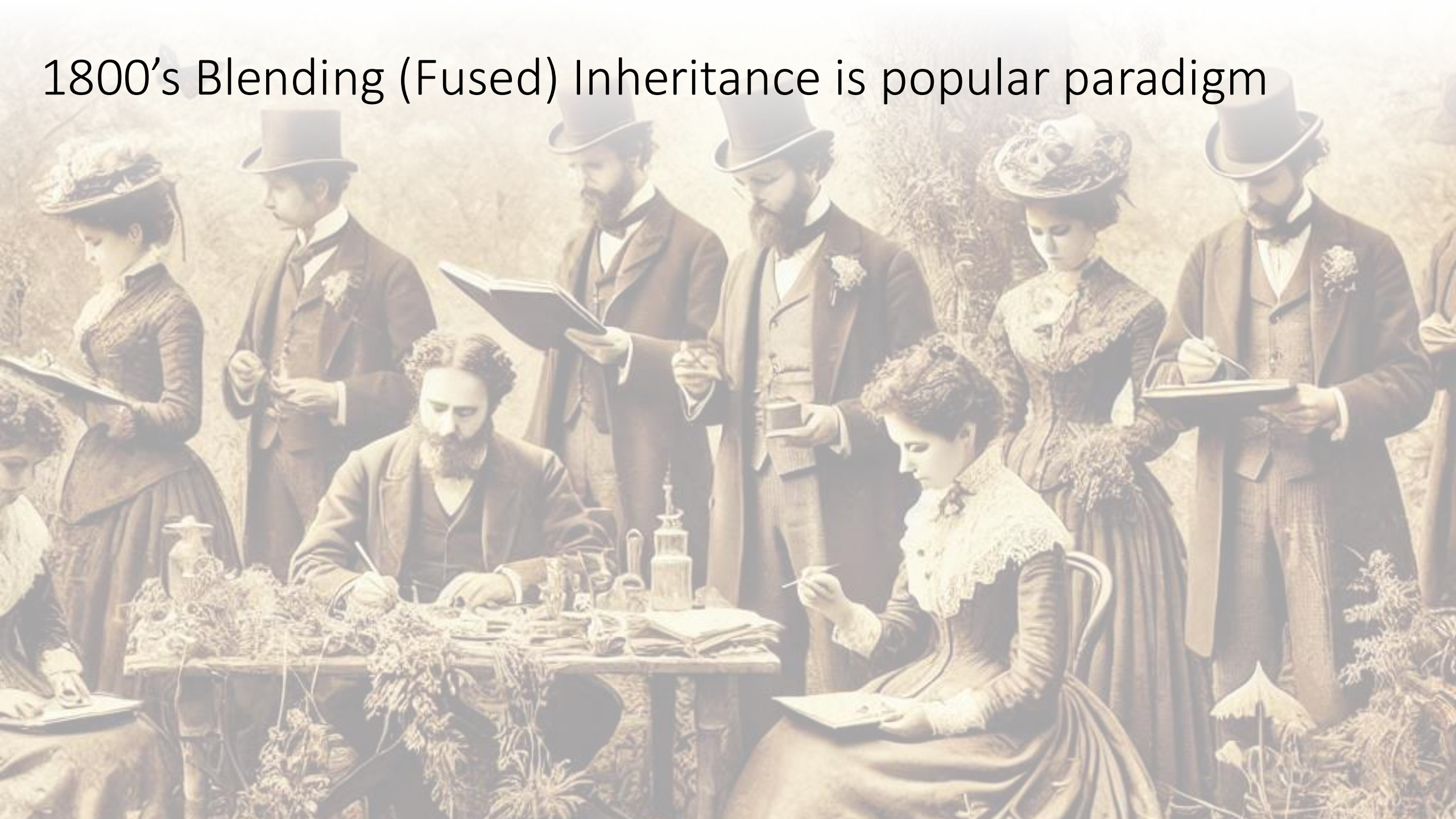
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Preformation?



1800's Blending (Fused) Inheritance is popular paradigm



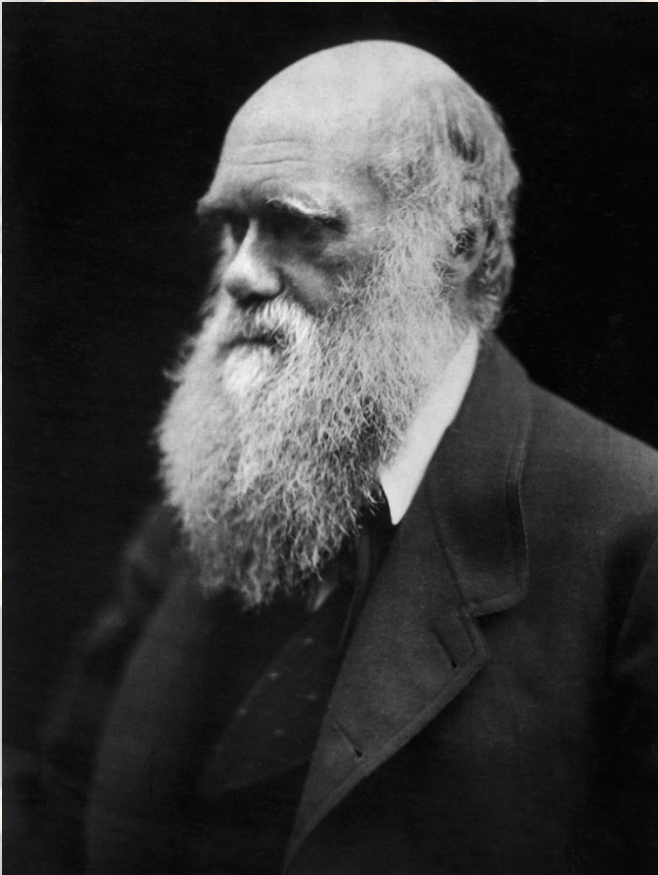
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Parent 1

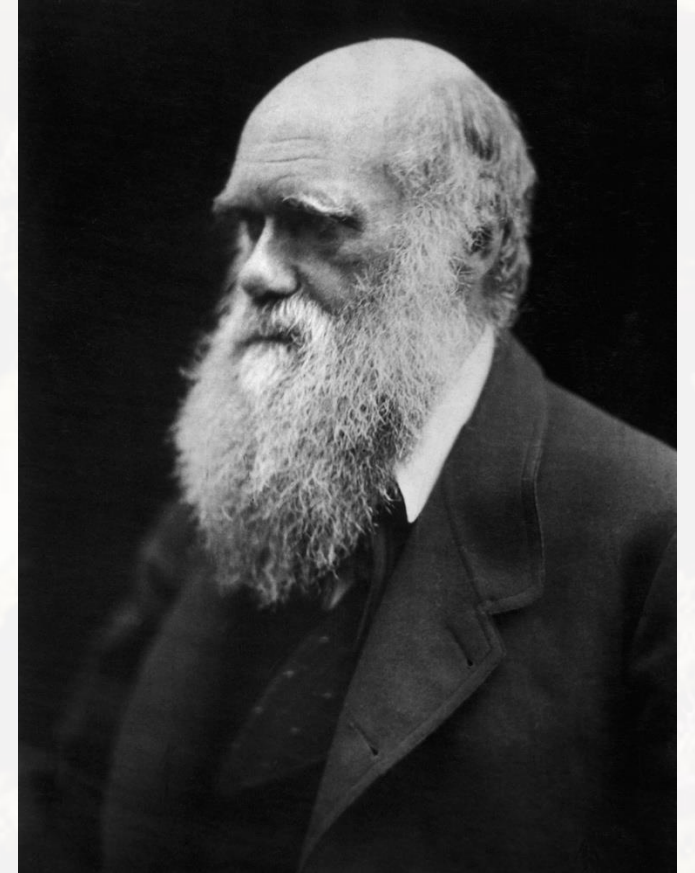
Parent 2

Offspring





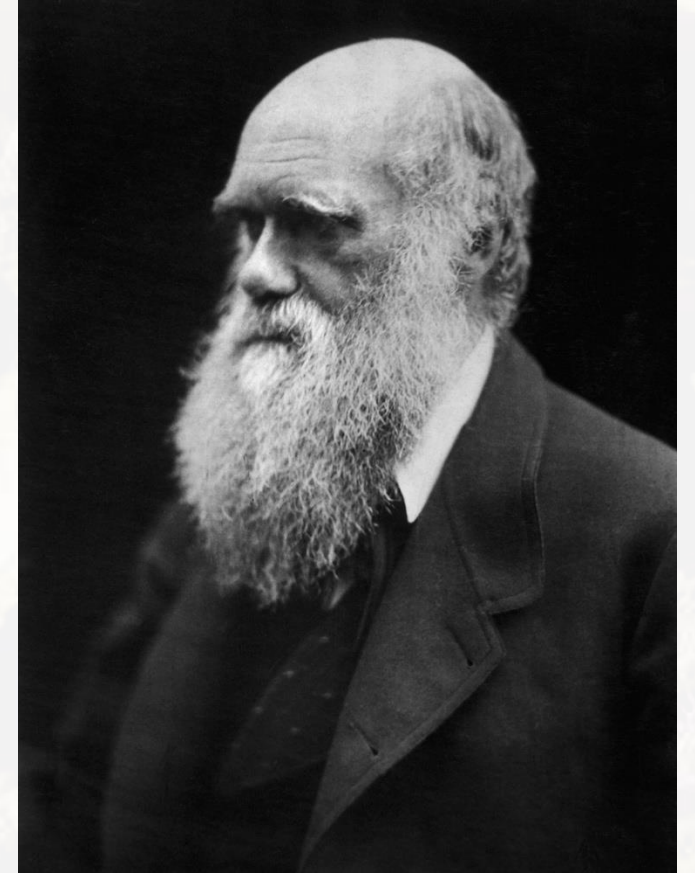
Blending Inheritance was a major problem for Darwin



Blending Inheritance was a major problem for Darwin

Evolution by Natural Selection

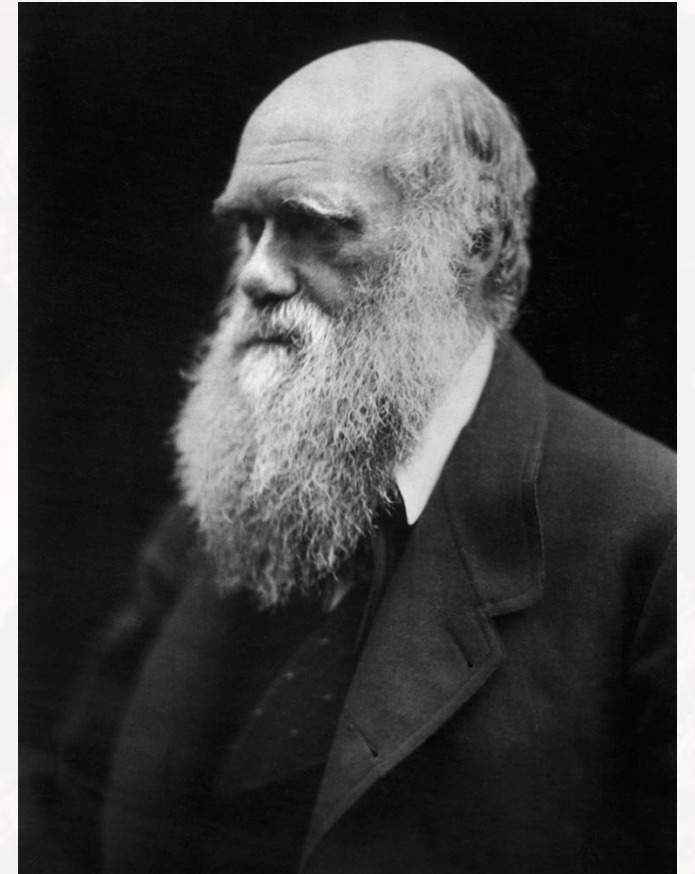
1. Natural Variation: Individuals differ in traits.



Blending Inheritance was a major problem for Darwin

Evolution by Natural Selection

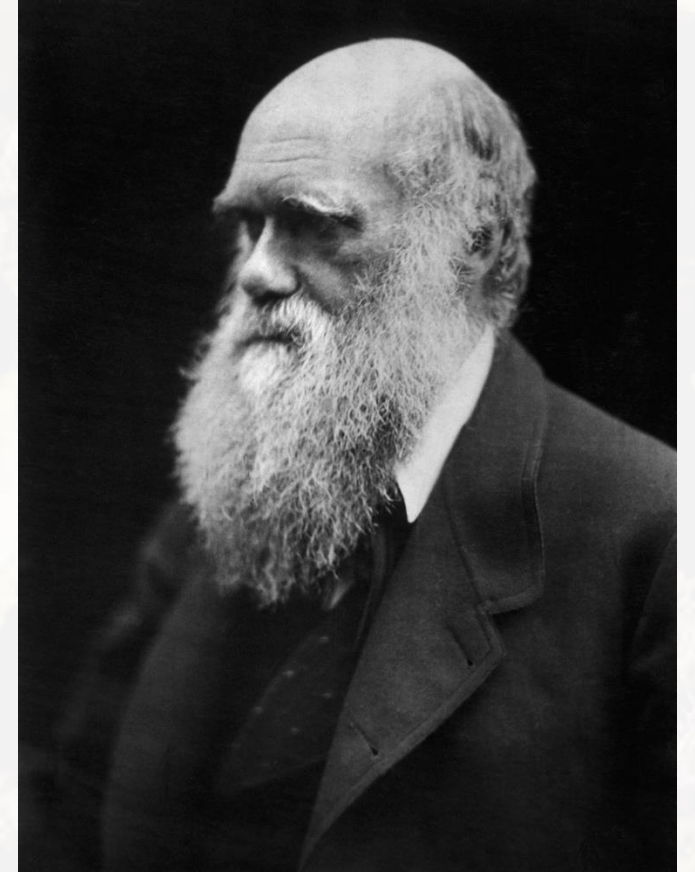
1. **Natural Variation:** Individuals differ in traits.
2. **Competition:** not enough resources for all to survive and reproduce.



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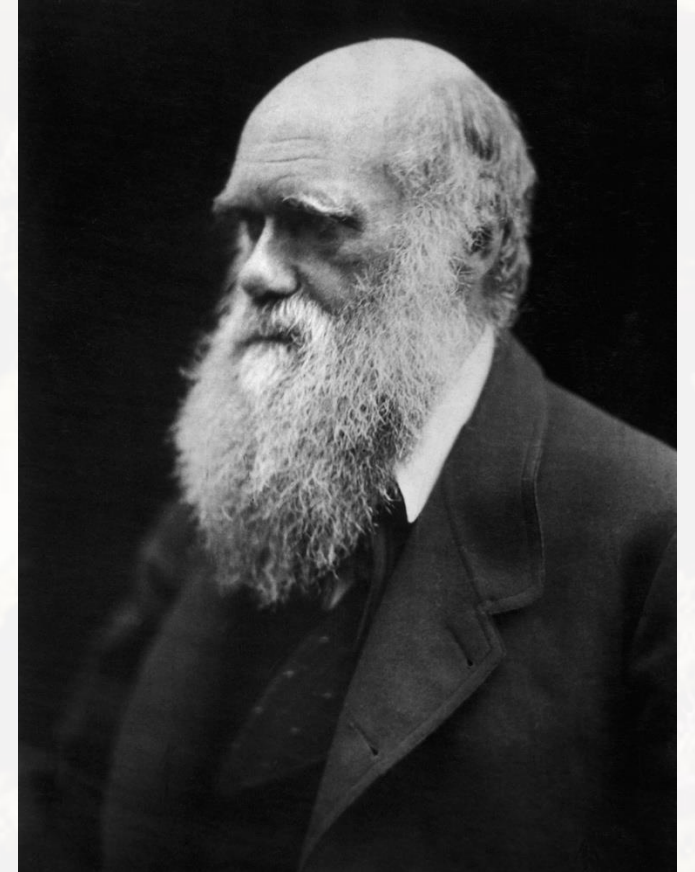
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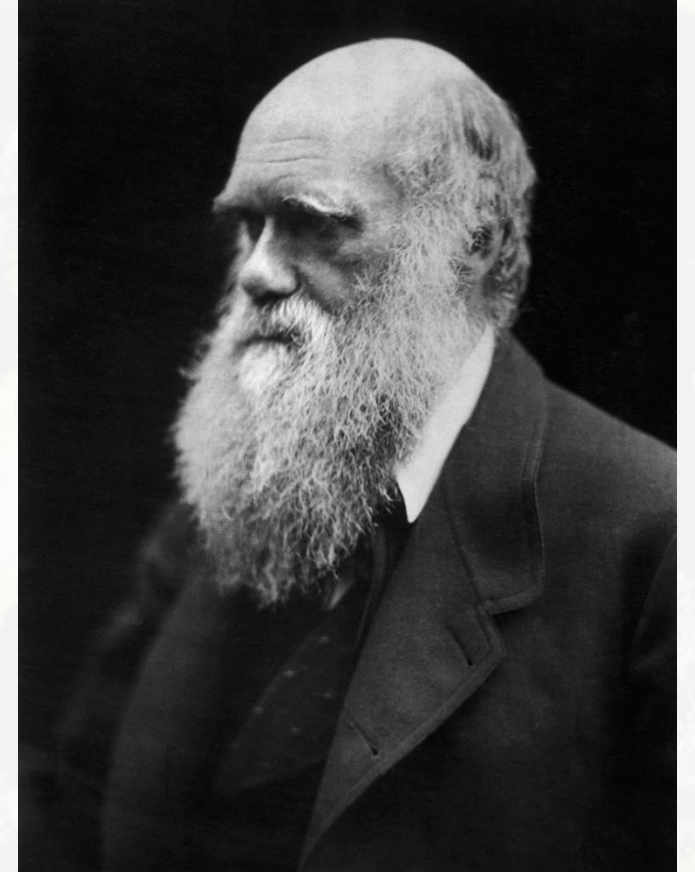


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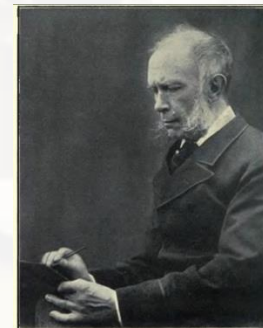
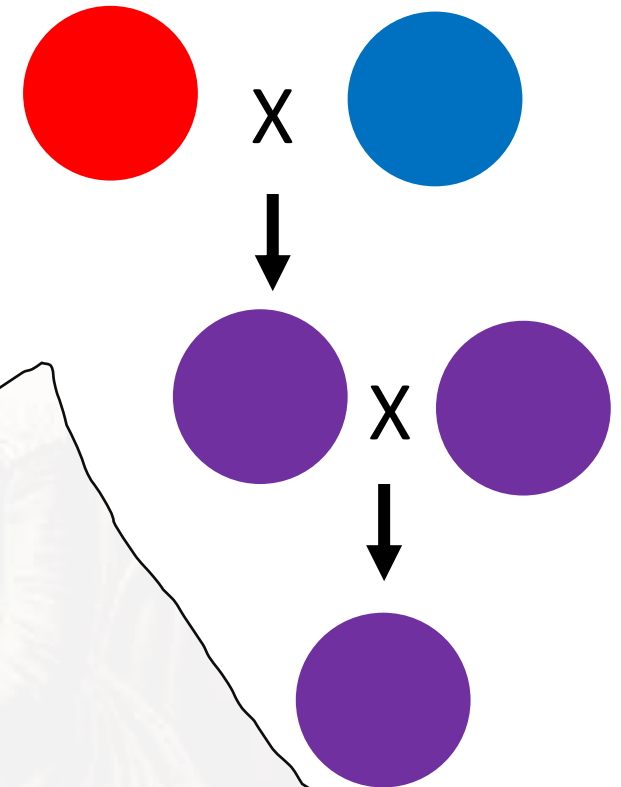


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Evolution by Natural Selection

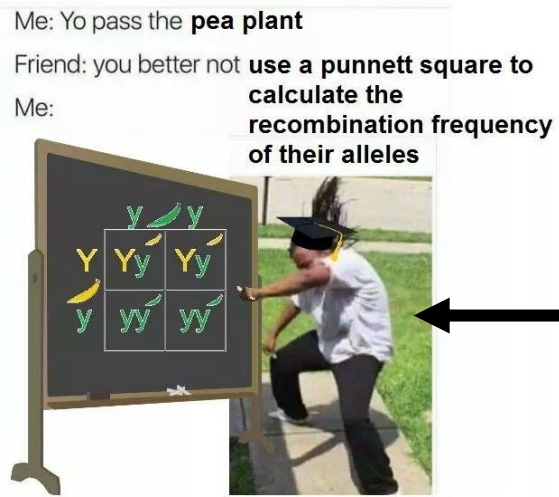
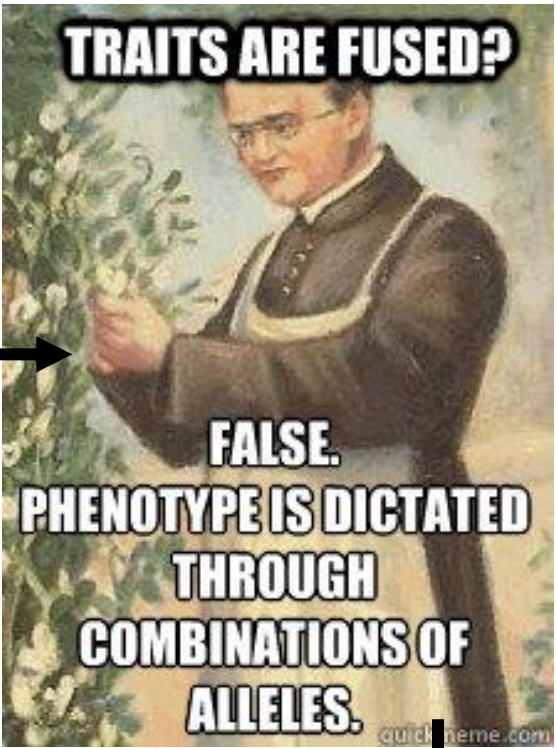
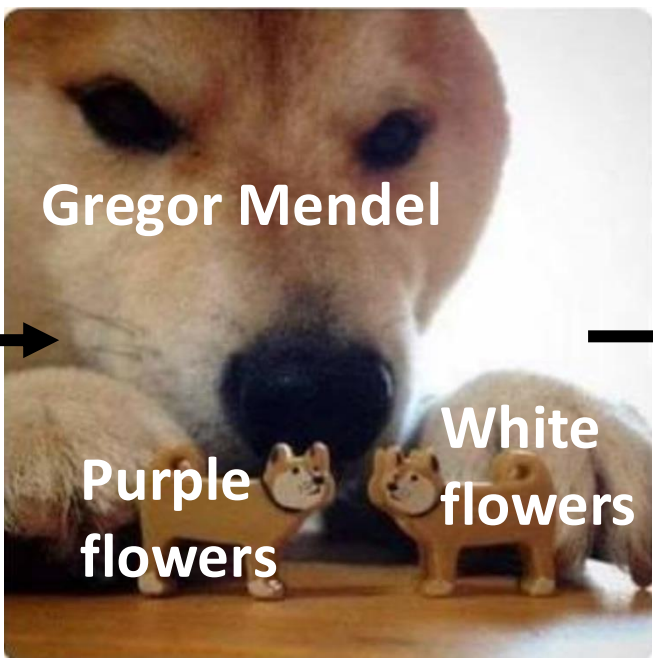
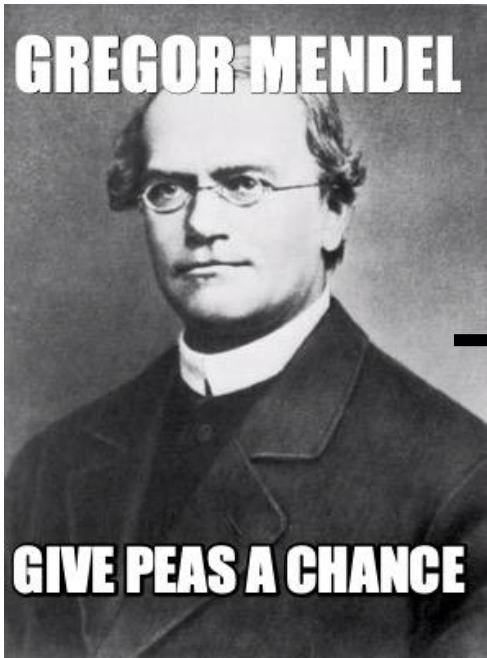
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Blending inheritance rapidly removes natural variation



?



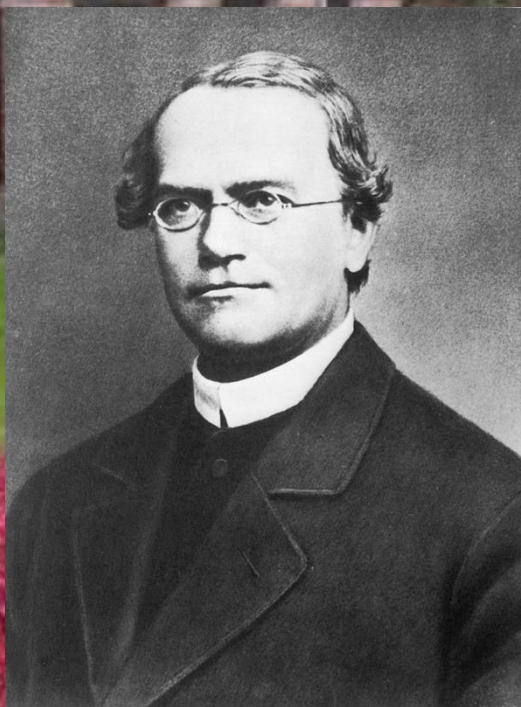


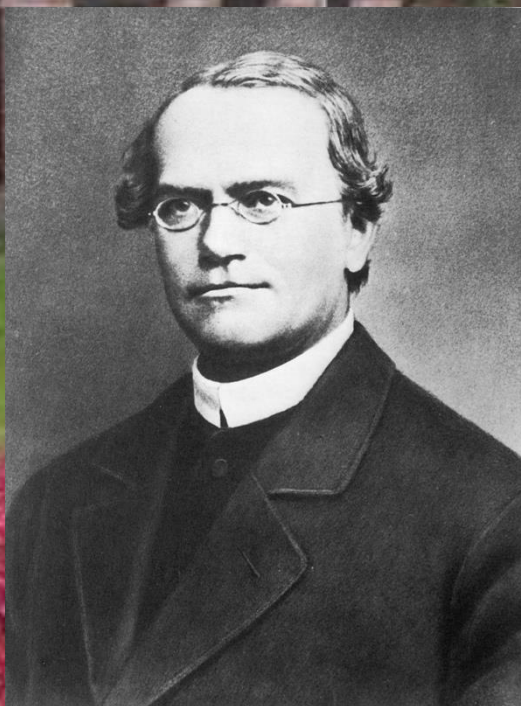
When Mendel's work on inheritance was discarded earlier but later recognised & applauded by Scientists



GREGOR MENDEL EXPLAINING HIS PEA STUFF TO MORAVIAN FARMERS







X



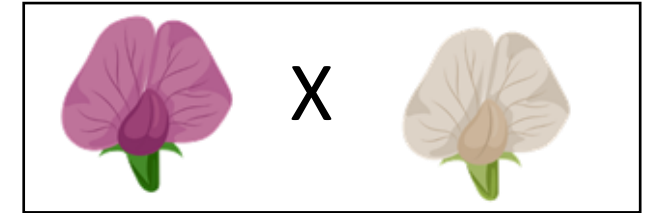
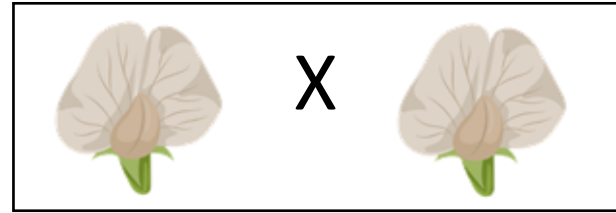
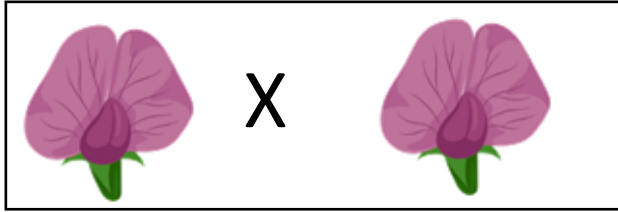
X



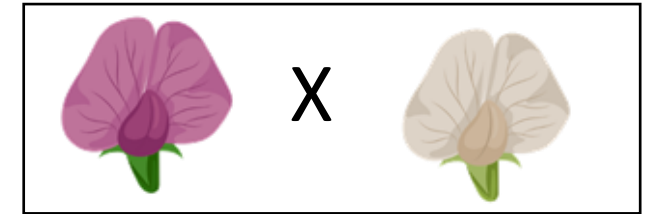
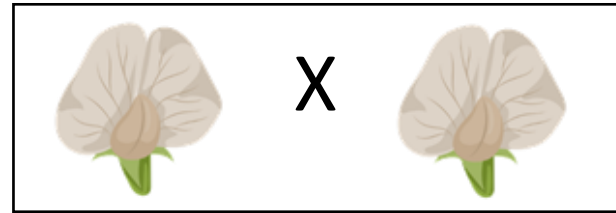
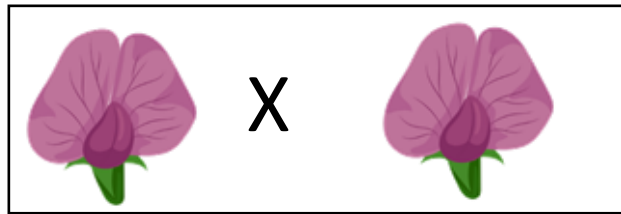
X



What are possible phenotypes of offspring?



What are possible phenotypes of offspring?



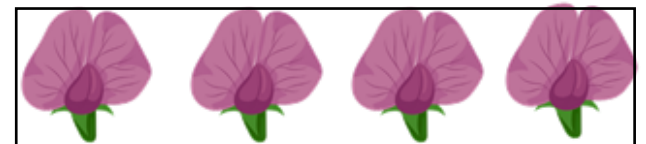
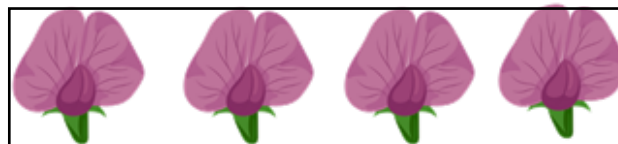
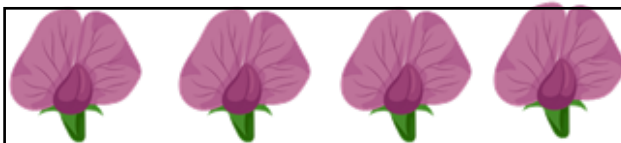
Blend (pink)



Mix



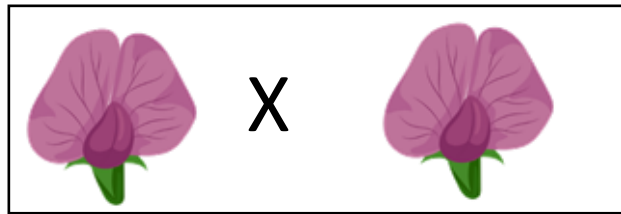
All purple



All white



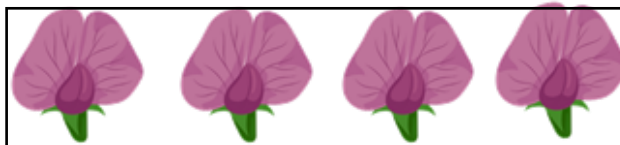
Mendel's Observations



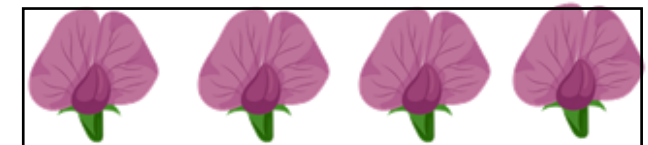
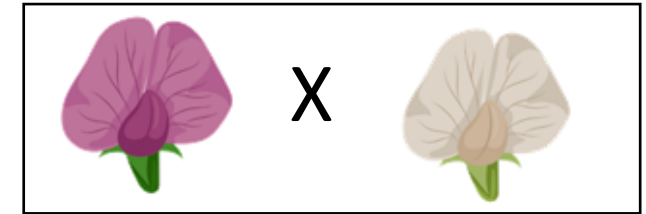
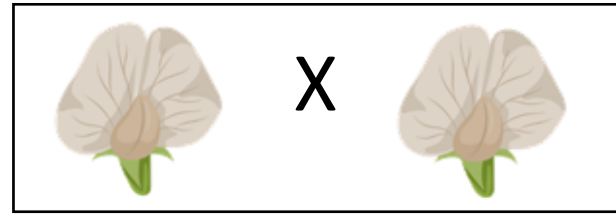
Mix

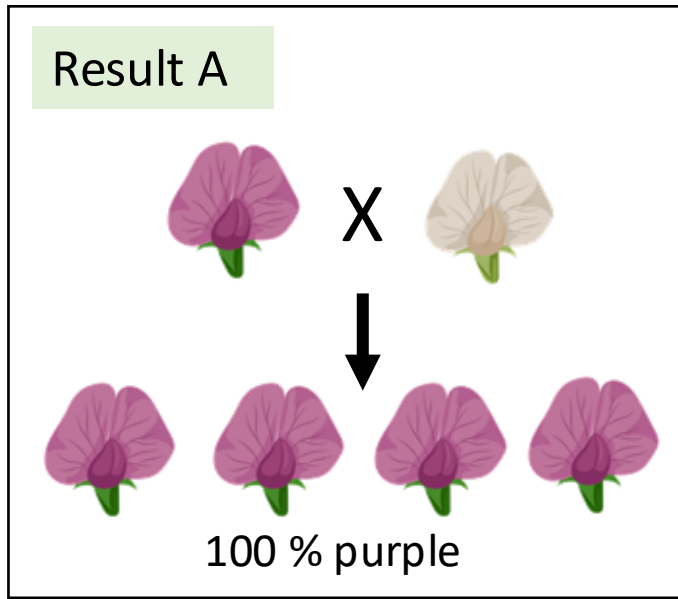


All purple



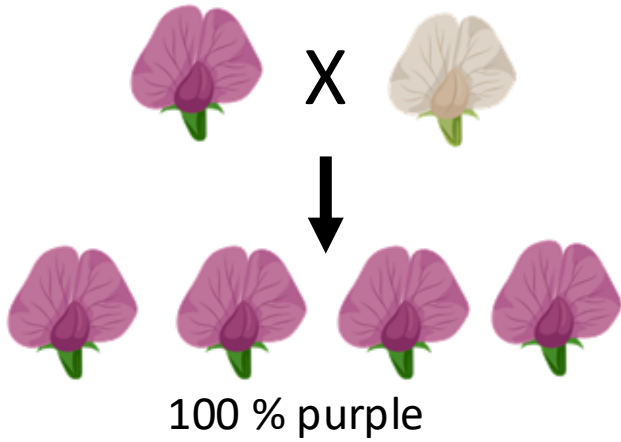
All white



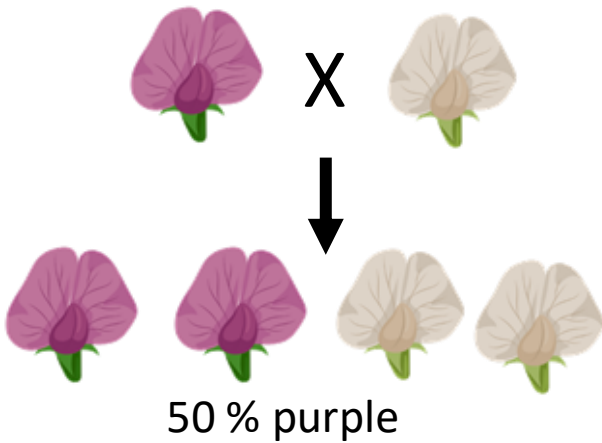


Note: hundreds of offspring were counted per cross (not 4 as illustrated)

Result A

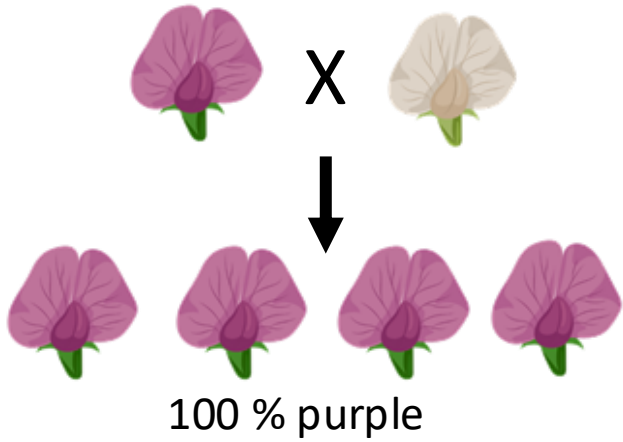


Result B

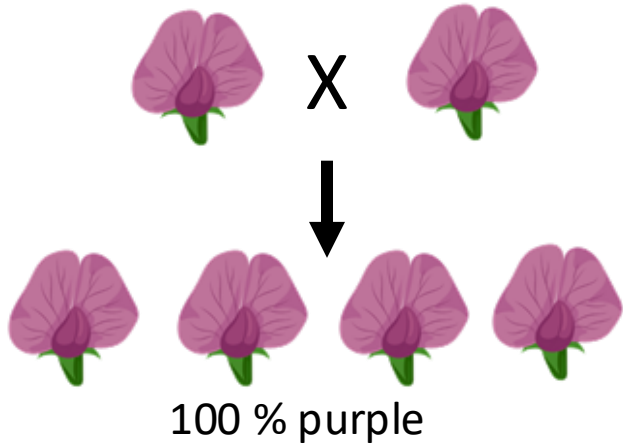


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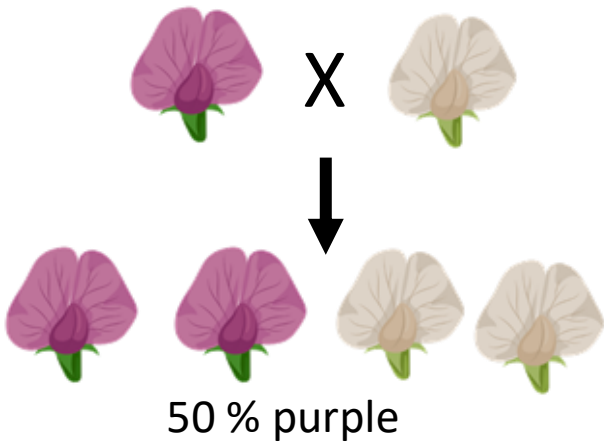
Result A



Result C

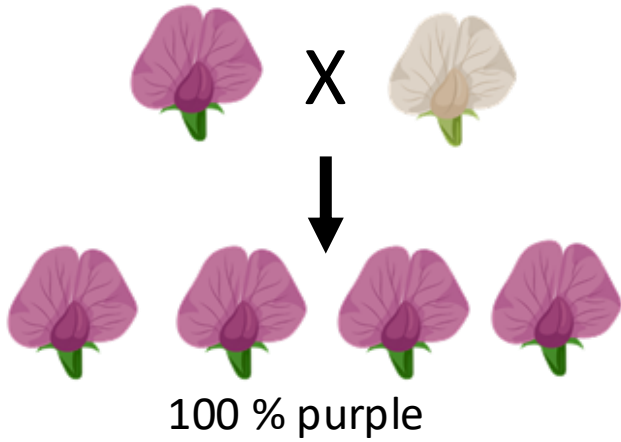


Result B

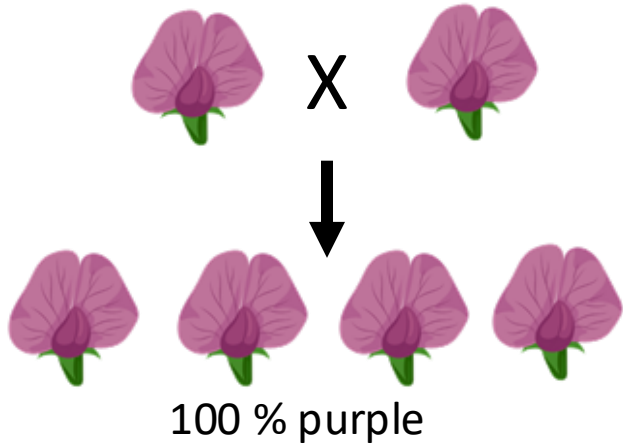


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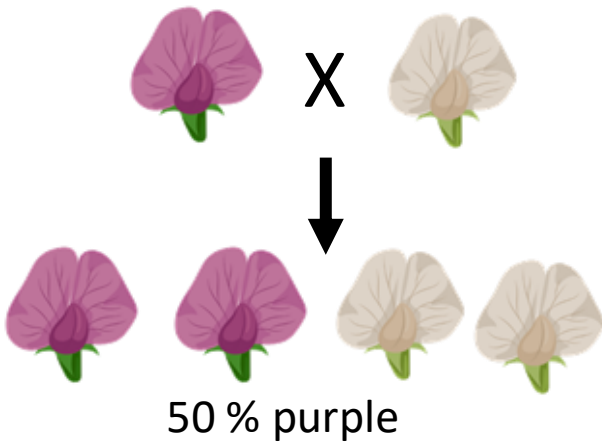
Result A



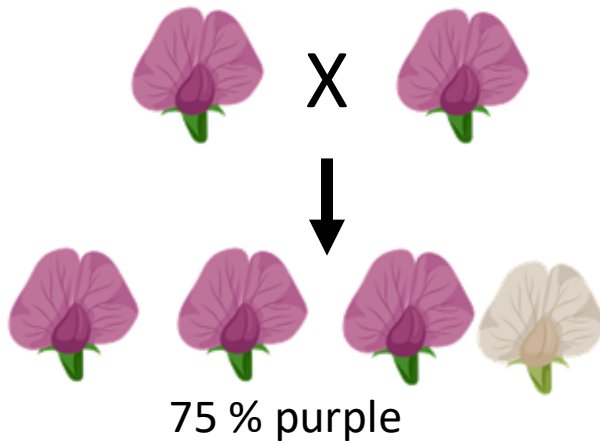
Result C



Result B

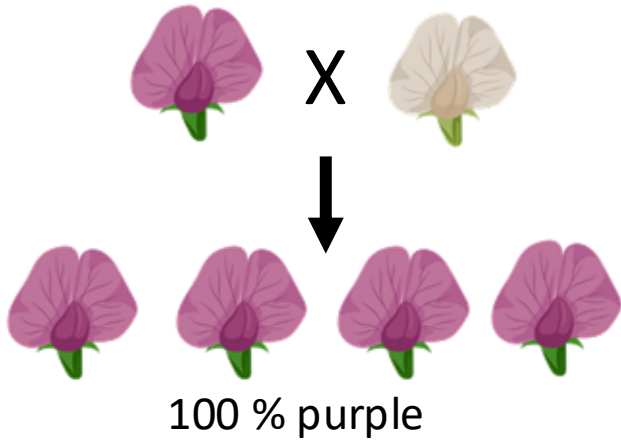


Result D

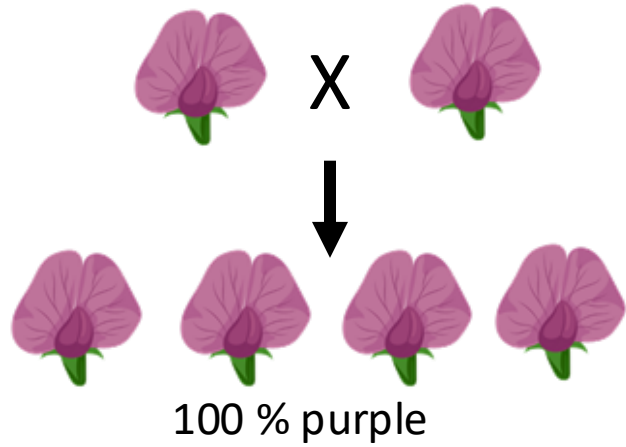


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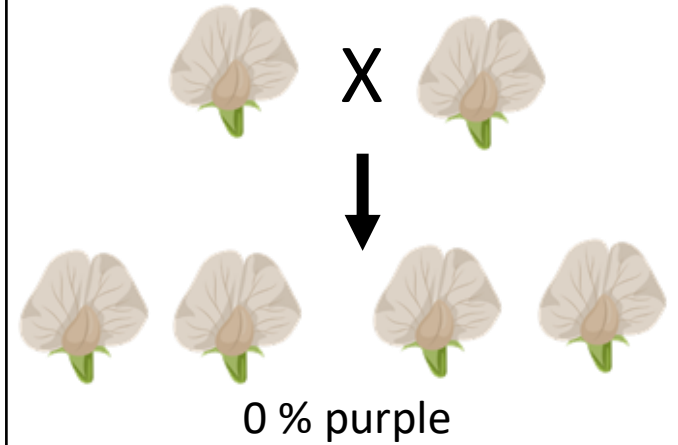
Result A



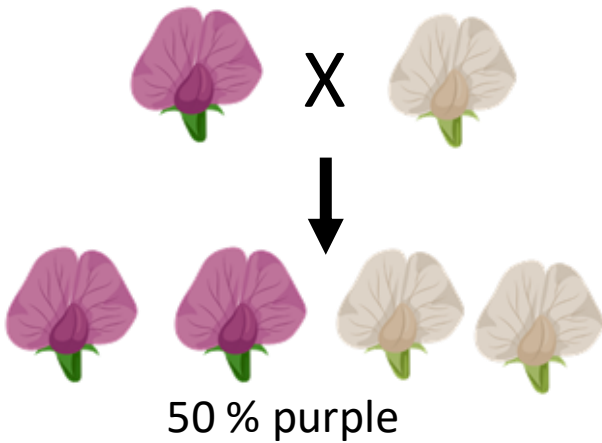
Result C



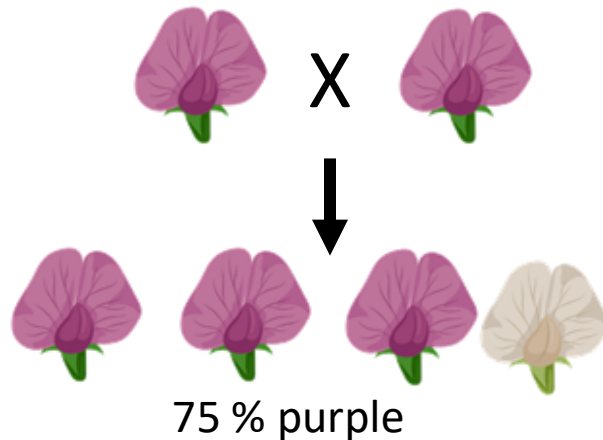
Result E



Result B

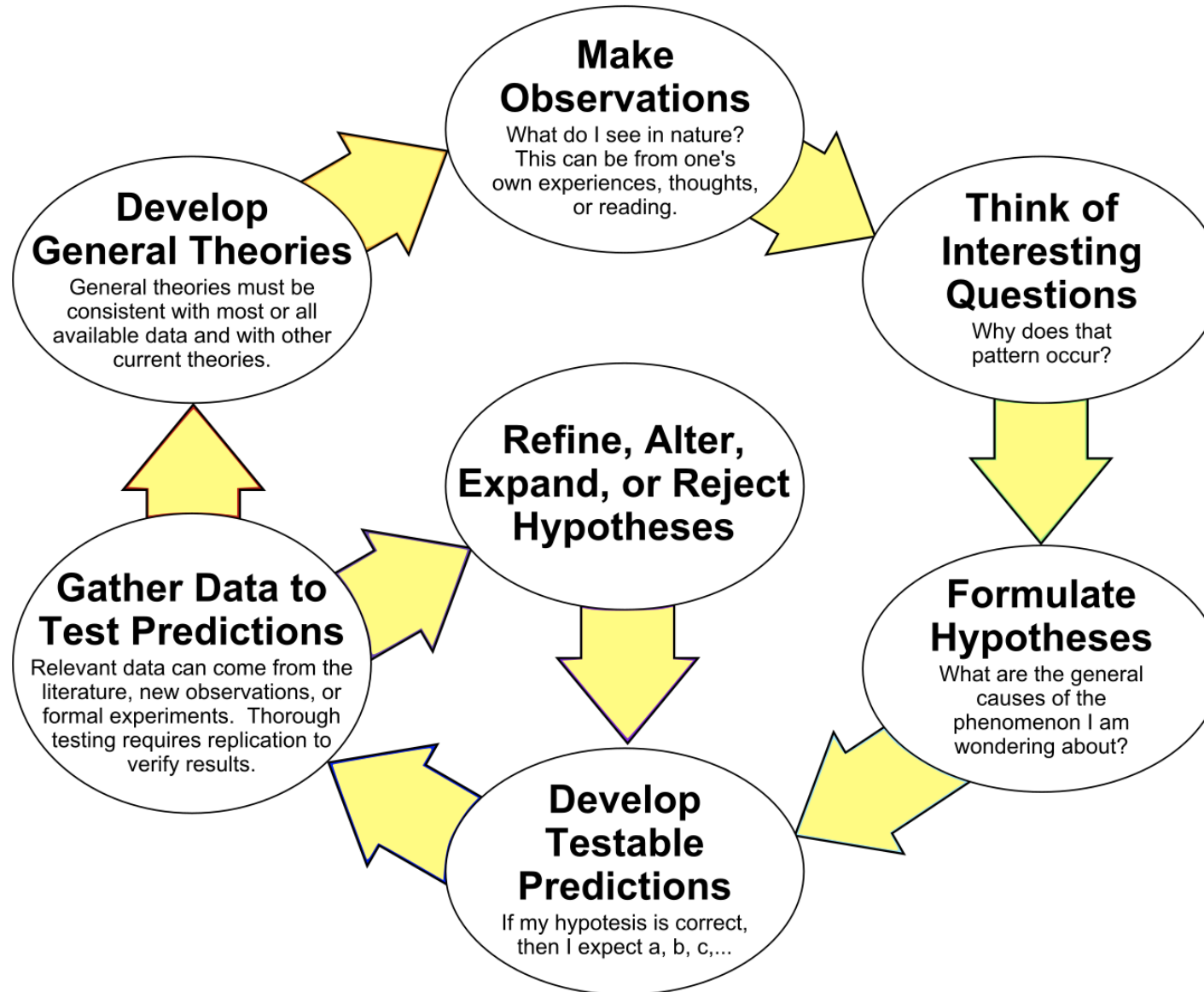


Result D



Note: hundreds of offspring were counted per cross (not 4 as illustrated)

The Scientific Method as an Ongoing Process



Hypothesis

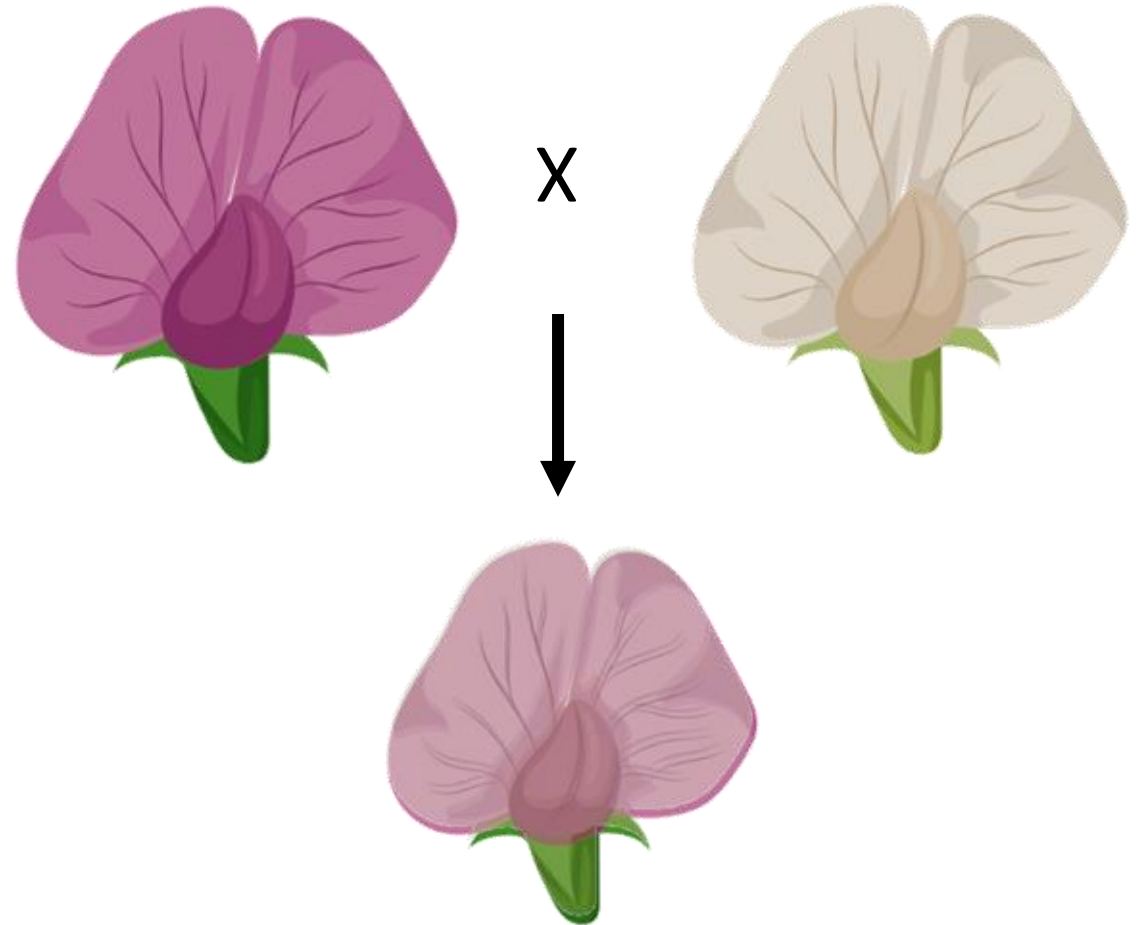
A claim that is logical and can be rejected by experiments.



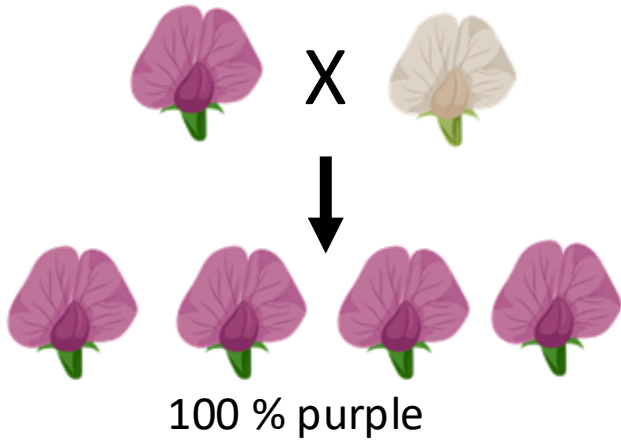


Main paradigm: Blending inheritance.

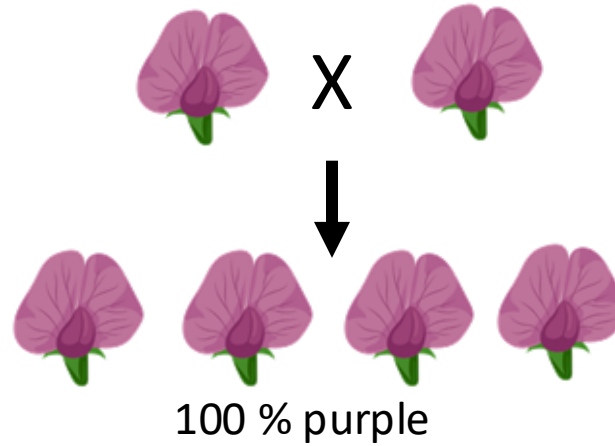
Hypothesis: Offspring phenotype is a blend of parents.



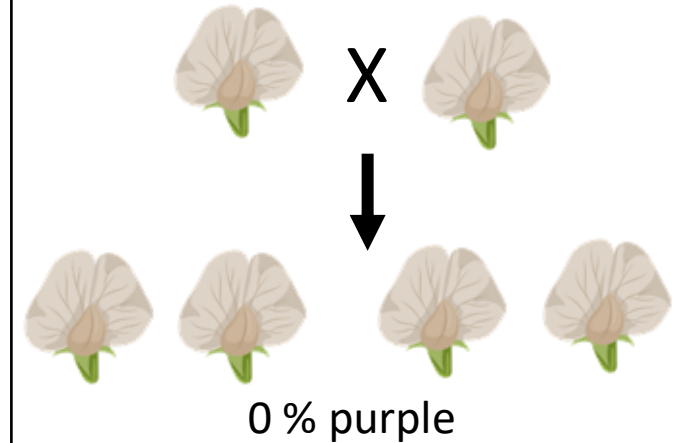
Result A



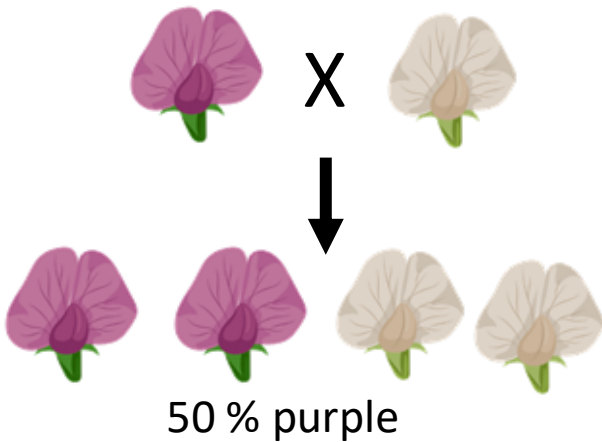
Result C



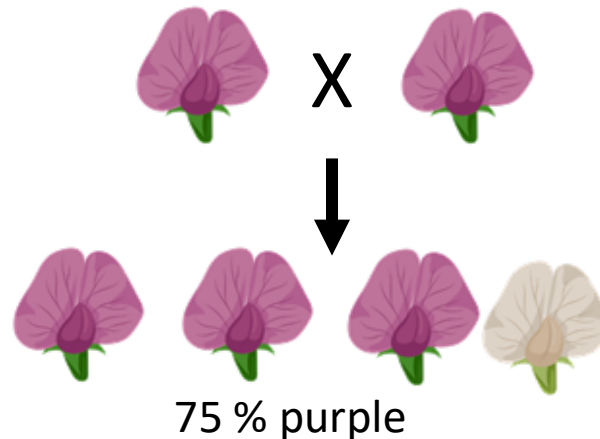
Result E



Result B



Result D

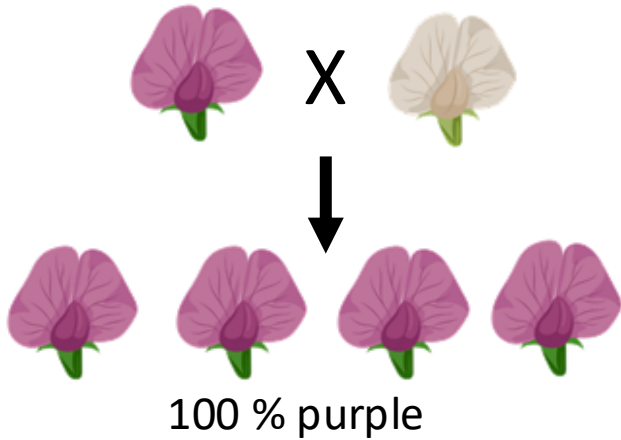


Hypothesis: Offspring phenotype is a blend of parents.

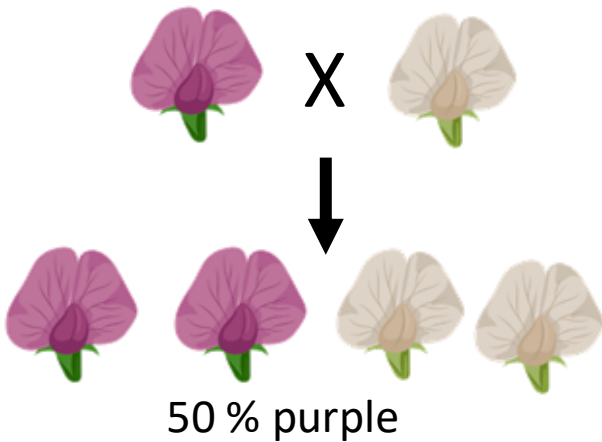
Which results are inconsistent with this hypothesis?

Note: hundreds of offspring were counted per cross (not 4 as illustrated)

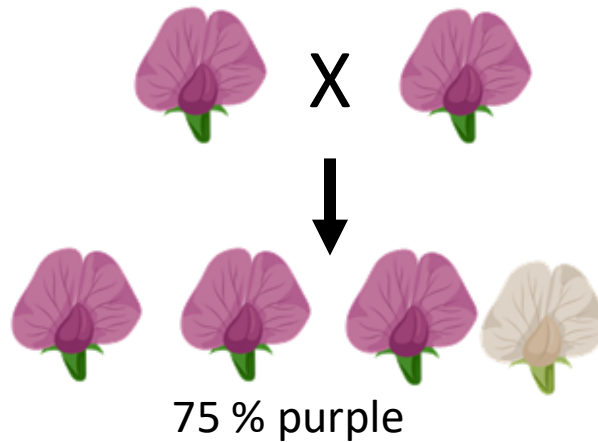
Result A



Result B



Result D



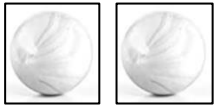
Hypothesis: Offspring phenotype is a blend of parents.

Which results are inconsistent with this hypothesis?

Note: hundreds of offspring were counted per cross (not 4 as illustrated)

Alternative paradigm: particulate inheritance. But how does it work?

Hypothesis: Each individual has two “particles” (one from each parent) that determines their phenotype.



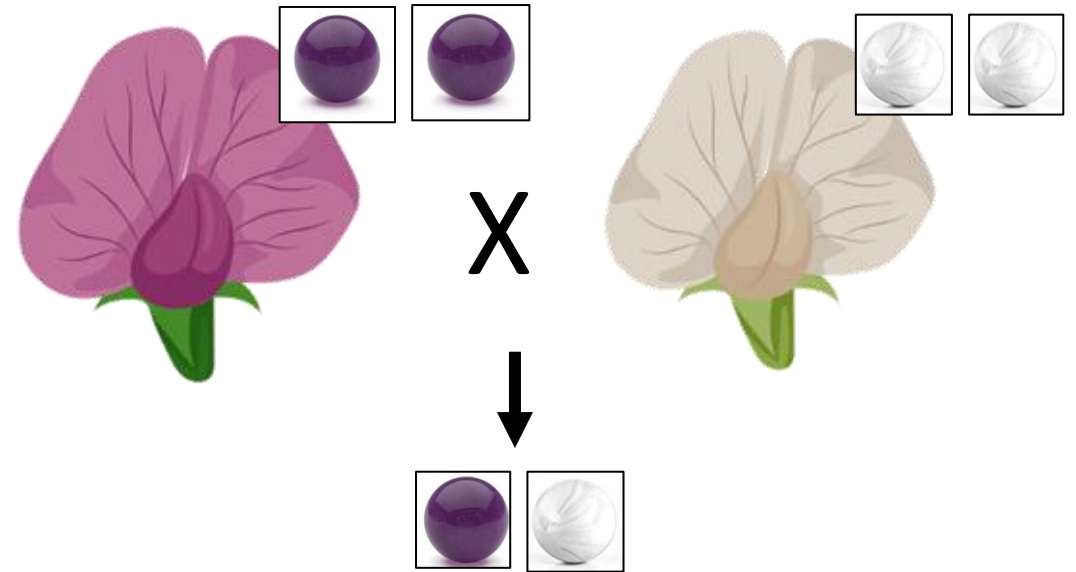
“Homozygous”



“Homozygous”

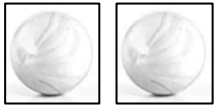


“Heterozygous”



Alternative paradigm: particulate inheritance. But how does it work?

Hypothesis: Each individual has two “particles” (one from each parent) that determines their phenotype.



“Homozygous”

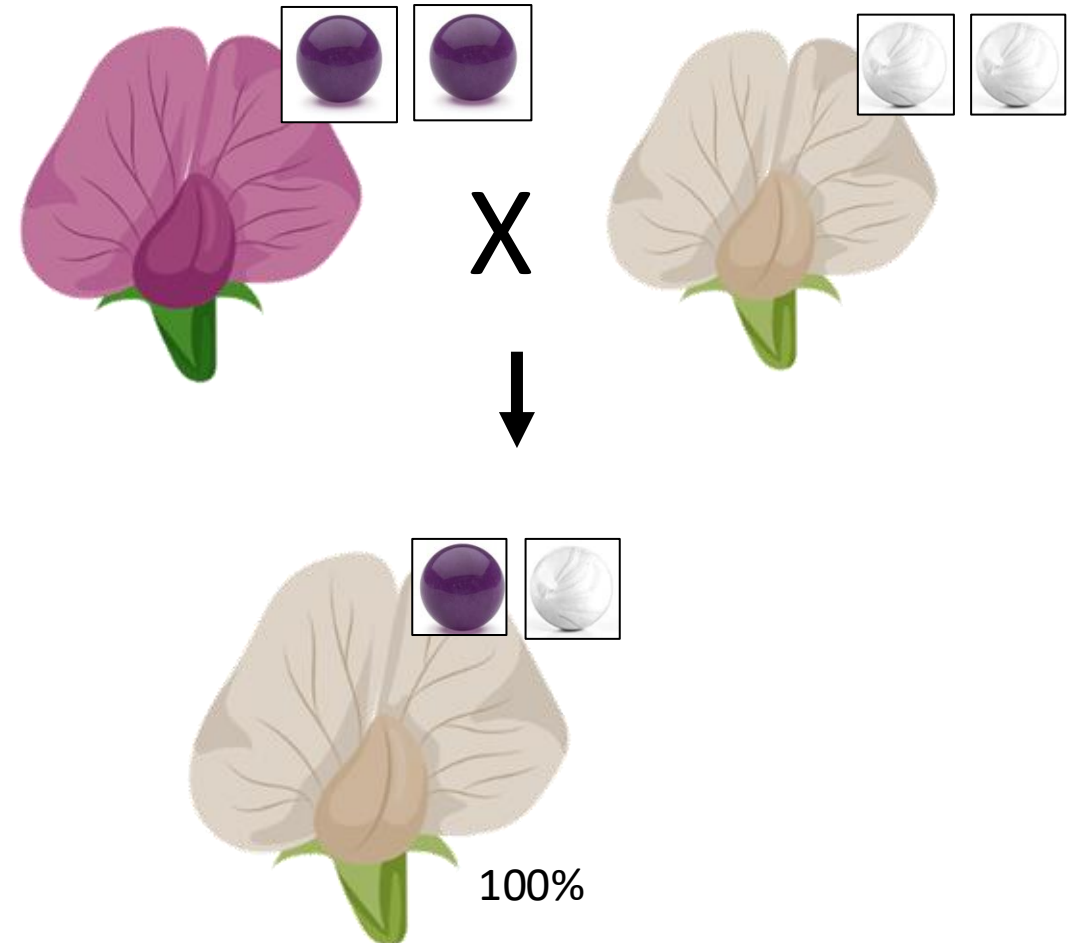


“Homozygous”

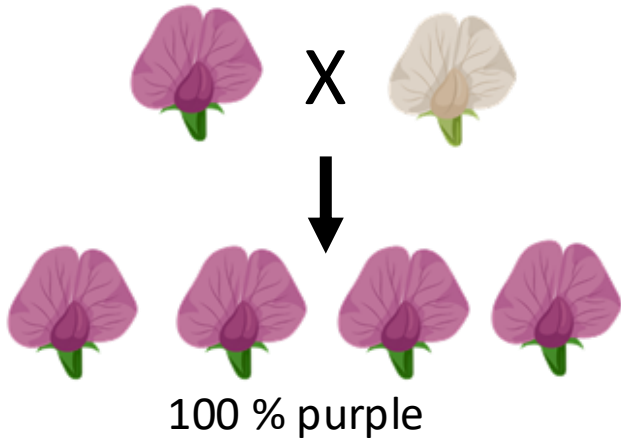


“Heterozygous”

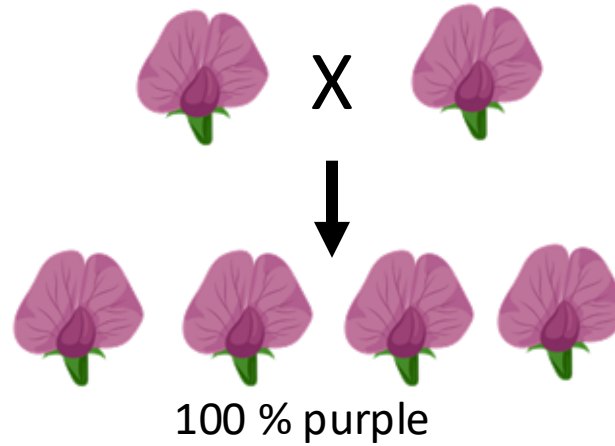
Hypothesis: In heterozygous individuals, the phenotype is always white. (white is dominant allele)



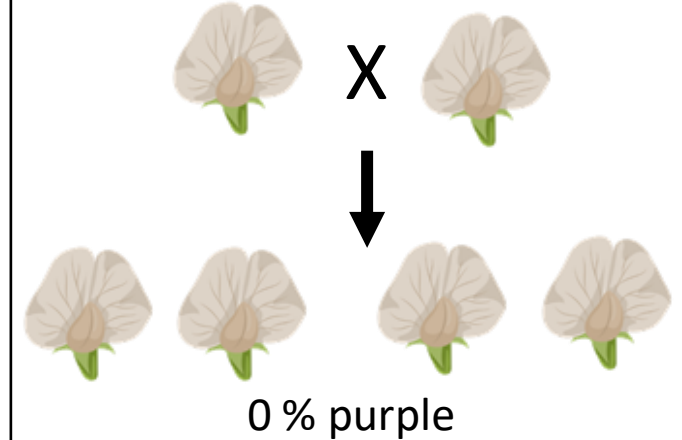
Result A



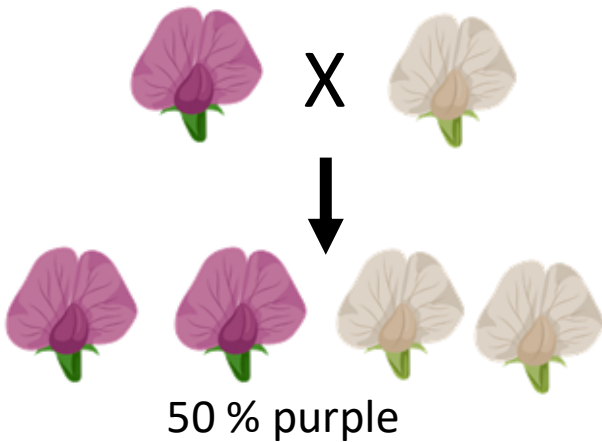
Result C



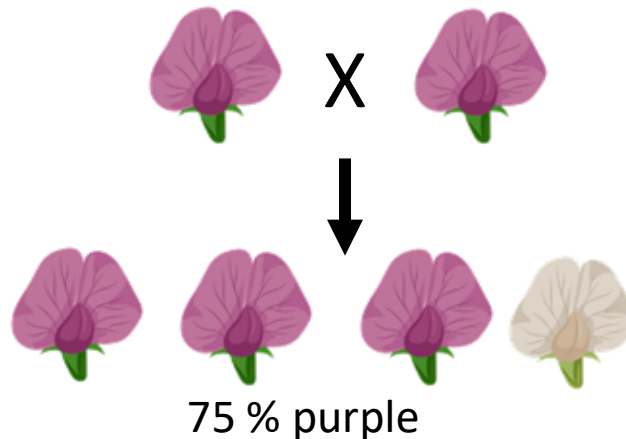
Result E



Result B



Result D

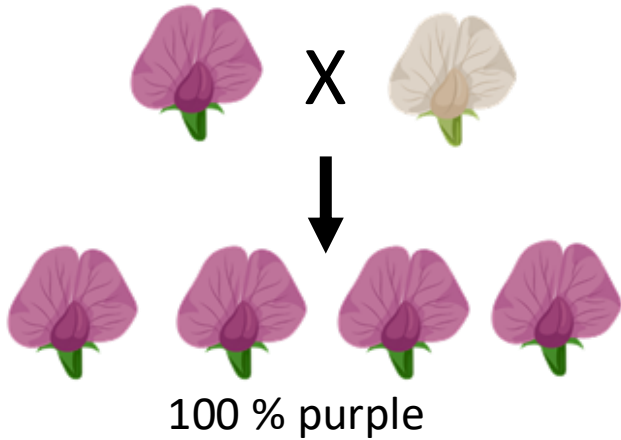


Hypothesis: White is dominant allele.

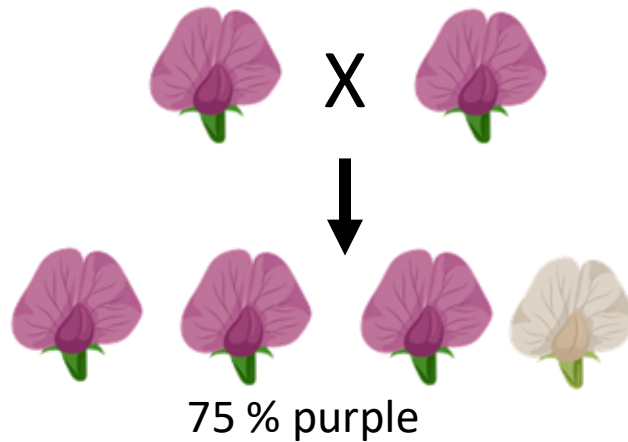
Which results are inconsistent with this hypothesis?

Note: hundreds of offspring were counted per cross (not 4 as illustrated)

Result A



Result D



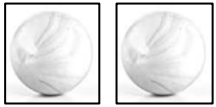
Hypothesis: White is dominant allele.

Which results are inconsistent with this hypothesis?

Note: hundreds of offspring were counted per cross (not 4 as illustrated)

Alternative paradigm: particulate inheritance. But how does it work?

Hypothesis: Each individual has two “particles” (one from each parent) that determines their phenotype.



“Homozygous”

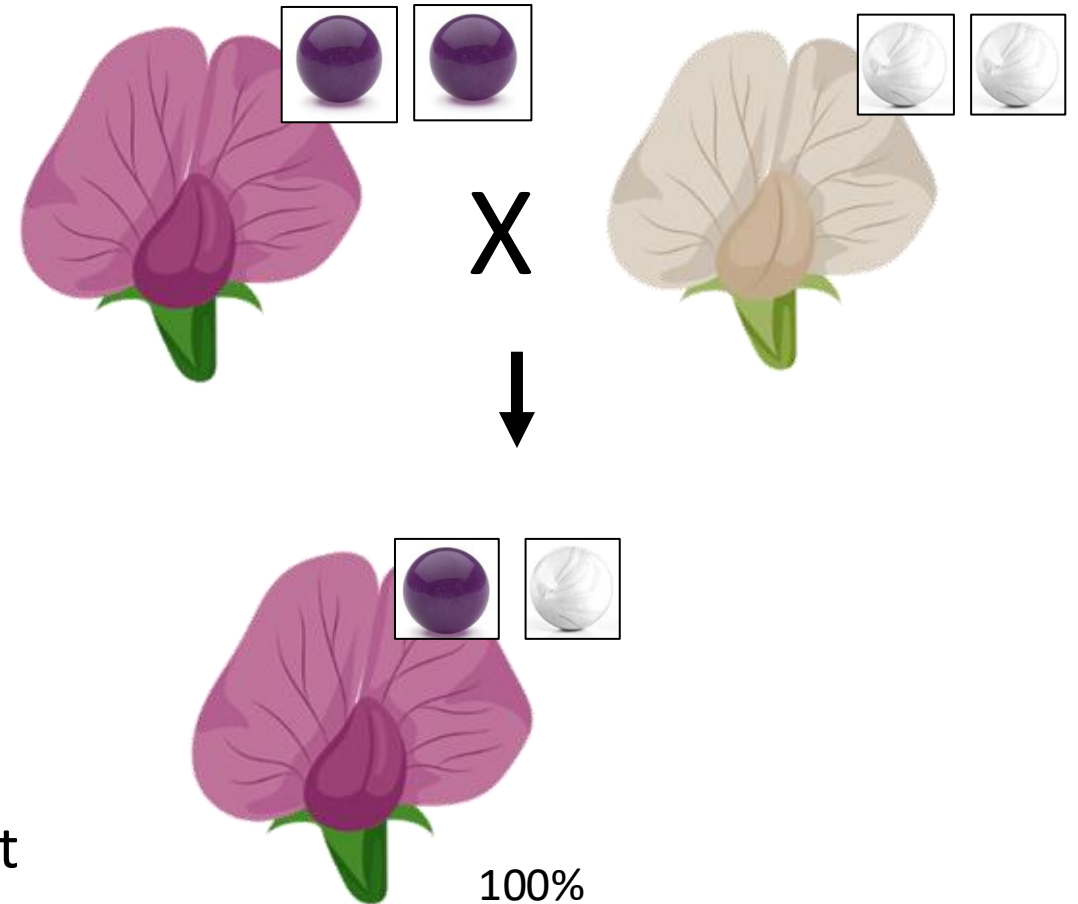


“Homozygous”

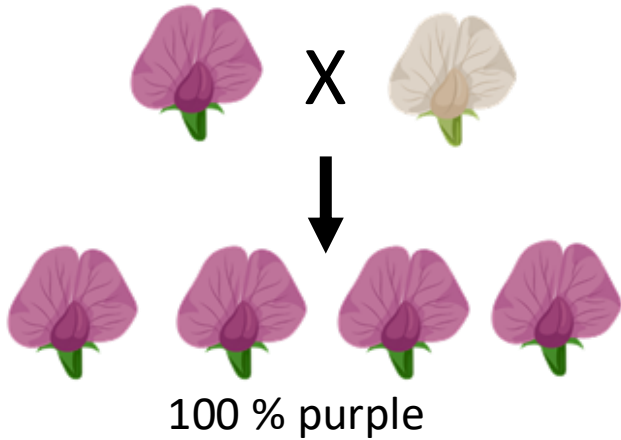


“Heterozygous”

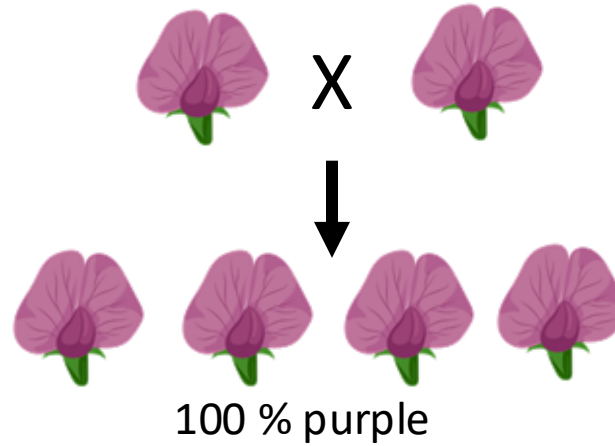
Hypothesis: In heterozygous individuals, the phenotype is always purple. (purple is dominant allele)



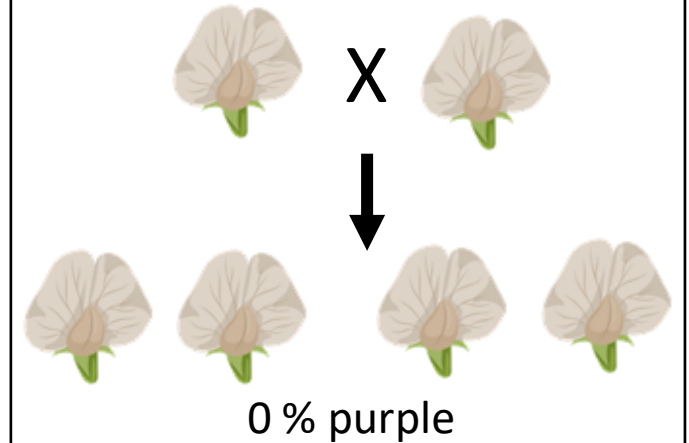
Result A



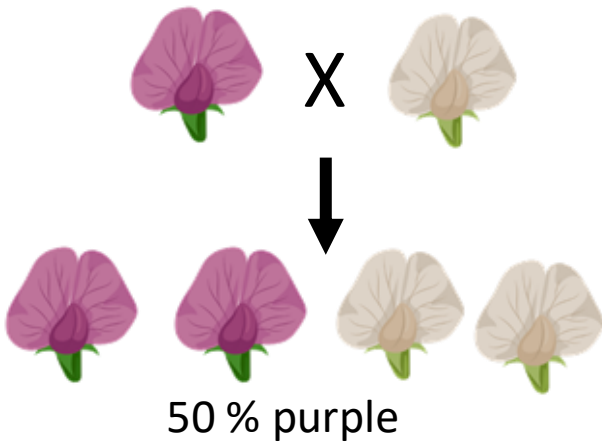
Result C



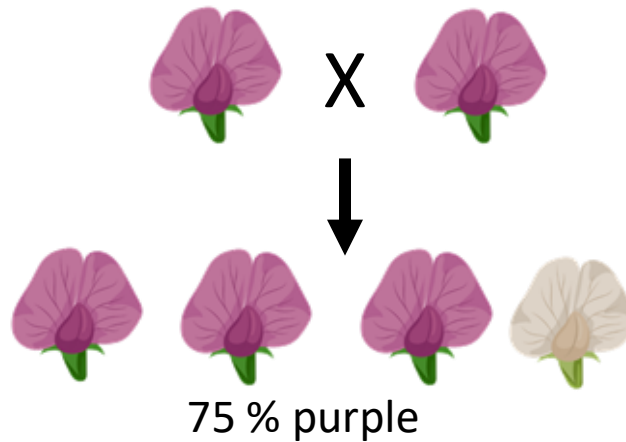
Result E



Result B



Result D

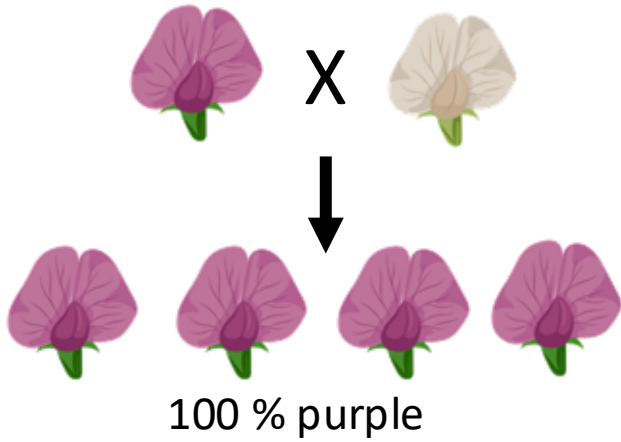


Hypothesis: Purple is dominant allele.

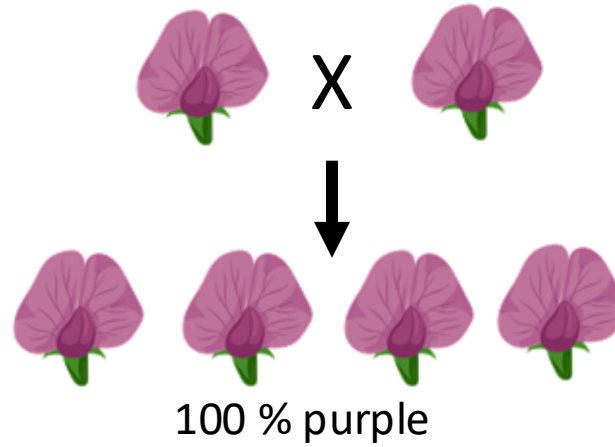
Which results are inconsistent with this hypothesis?

Note: hundreds of offspring were counted per cross (not 4 as illustrated)

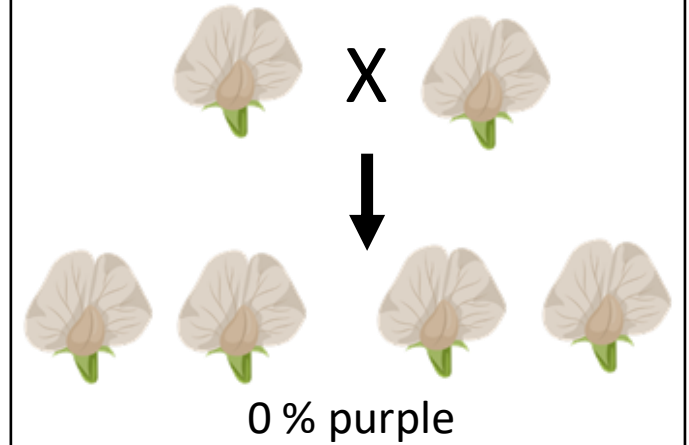
Result A



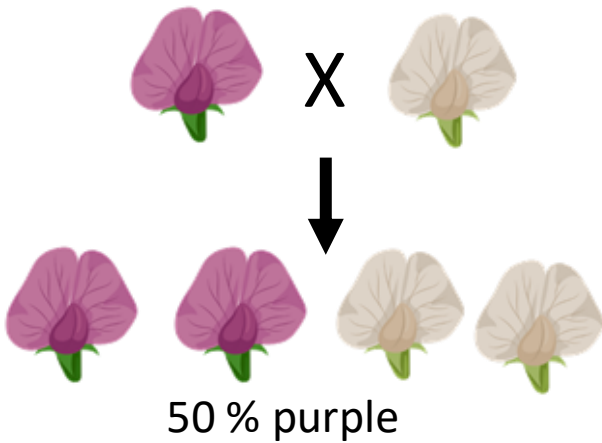
Result C



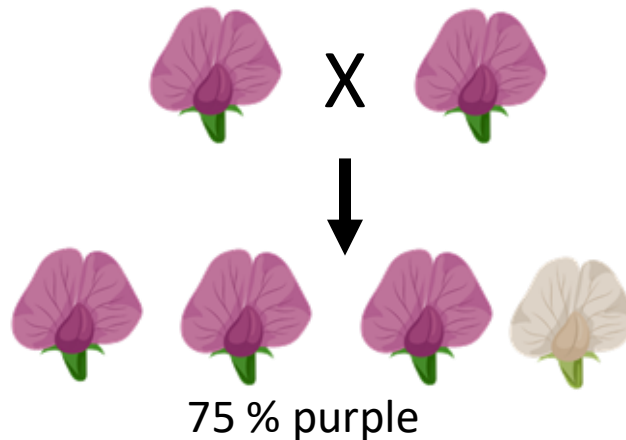
Result E



Result B



Result D

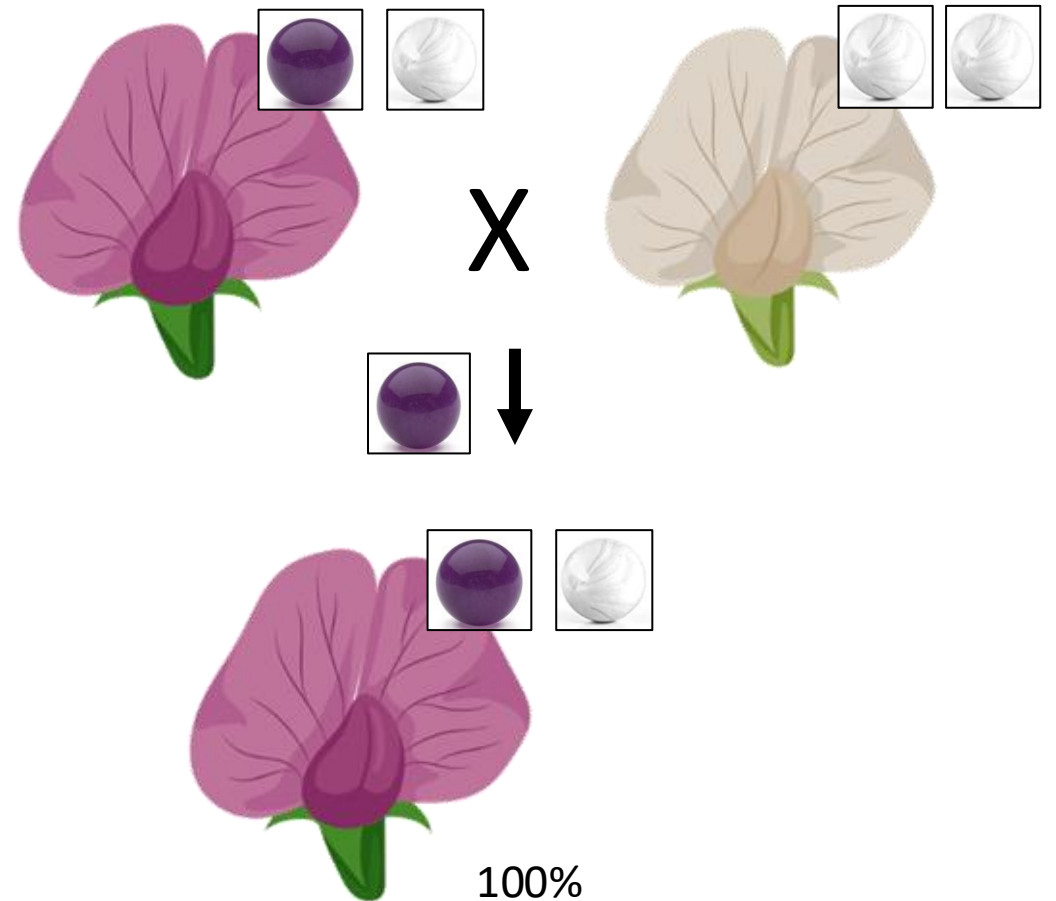


- **Purple is dominant**

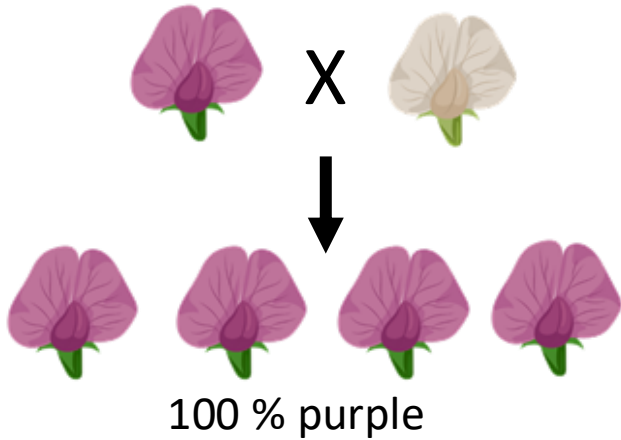
Note: hundreds of offspring were counted per cross (not 4 as illustrated)

Alternative paradigm: particulate inheritance. But how does it work?

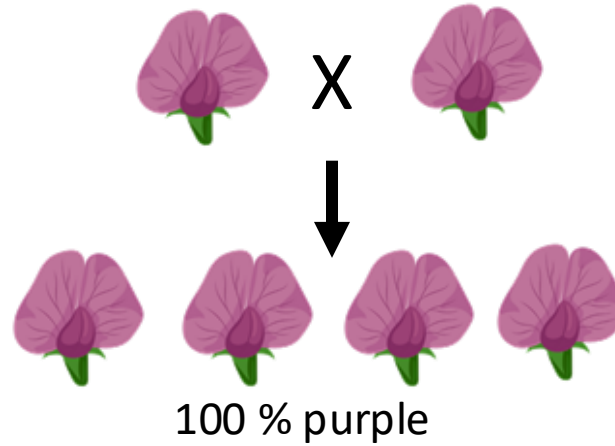
Hypothesis: Parents with purple only give purple allele to offspring.



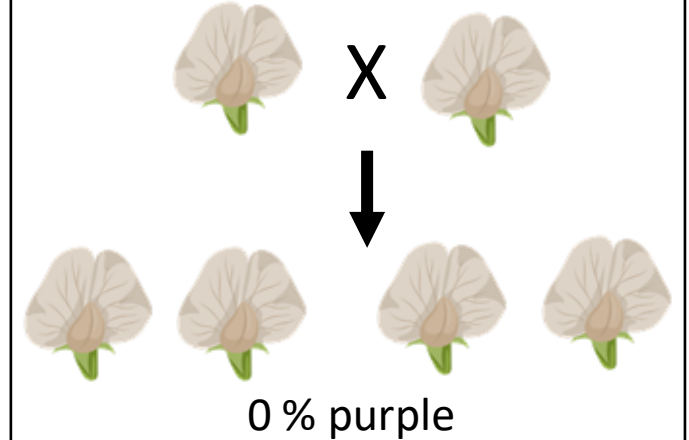
Result A



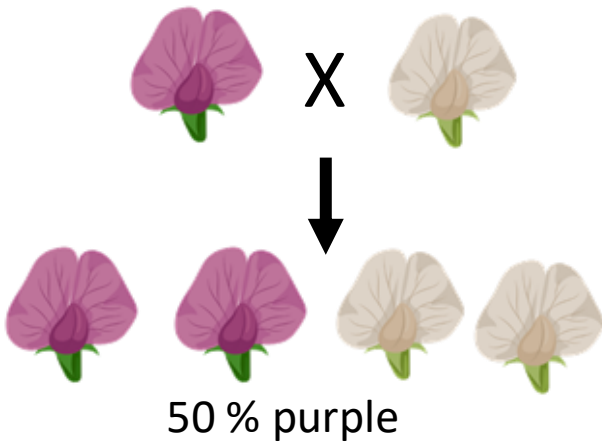
Result C



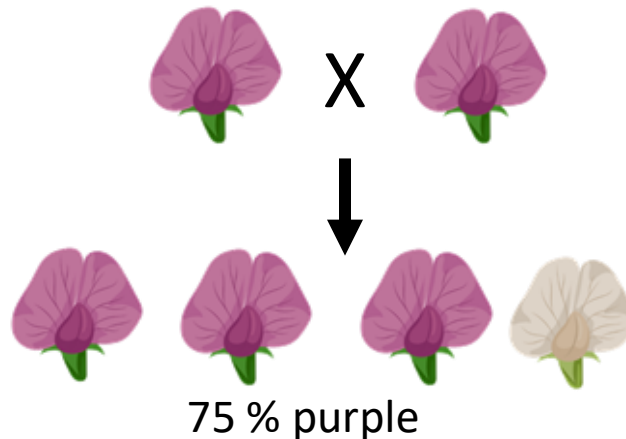
Result E



Result B



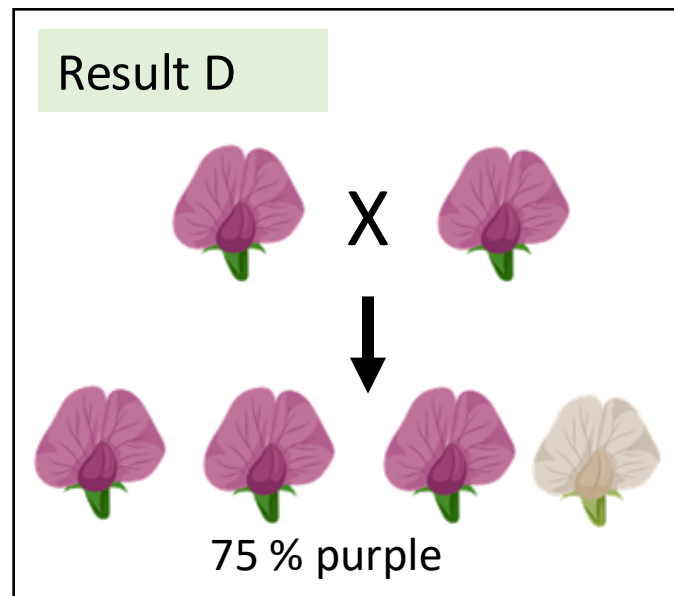
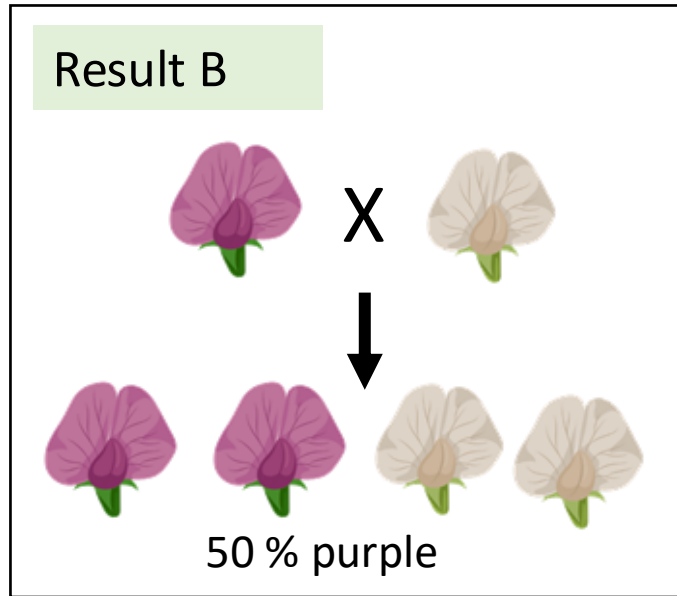
Result D



Hypothesis: Parents with purple only give purple allele to offspring.

Which results are inconsistent with this hypothesis?

Note: hundreds of offspring were counted per cross (not 4 as illustrated)

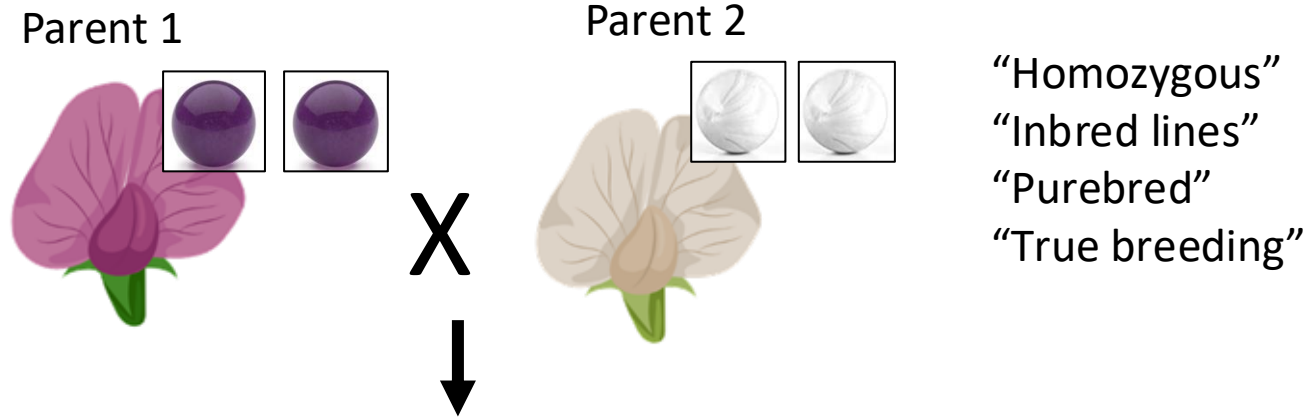


Hypothesis: Parents with purple only give purple allele to offspring.

Which results are inconsistent with this hypothesis?

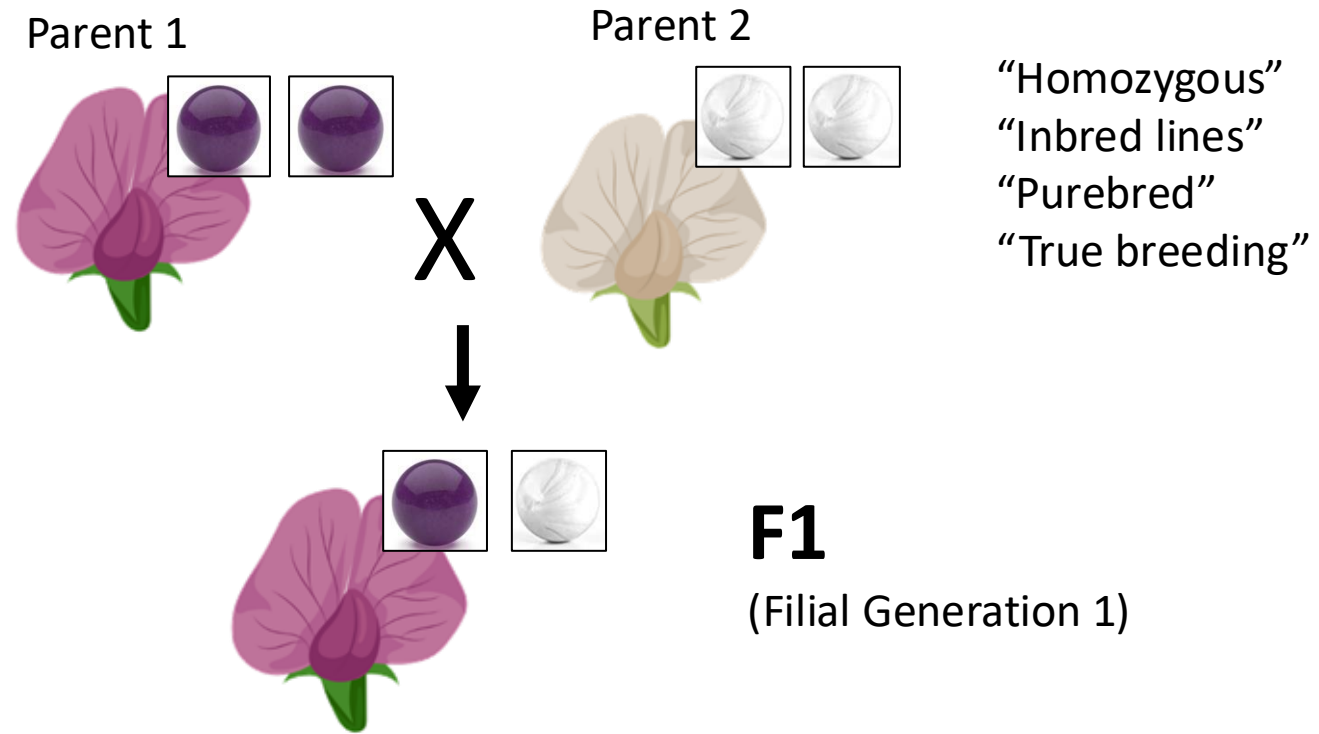
Note: hundreds of offspring were counted per cross (not 4 as illustrated)

Hypothesis: Parents
randomly pass one of their
particles to their offspring.

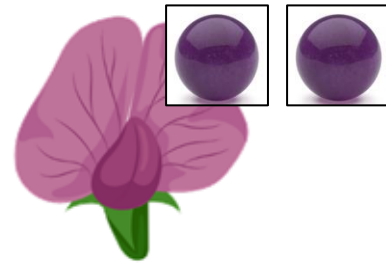


Hypothesis: Parents randomly pass one of their particles to their offspring.

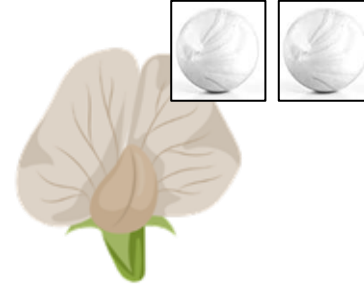
Hypothesis: Parents randomly pass one of their particles to their offspring.



Parent 1



Parent 2



X



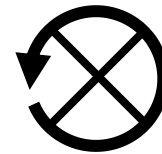
“Homozygous”
“Inbred lines”
“Purebred”
“True breeding”

Hypothesis: Parents randomly pass one of their particles to their offspring.



F1

(Filial Generation 1)

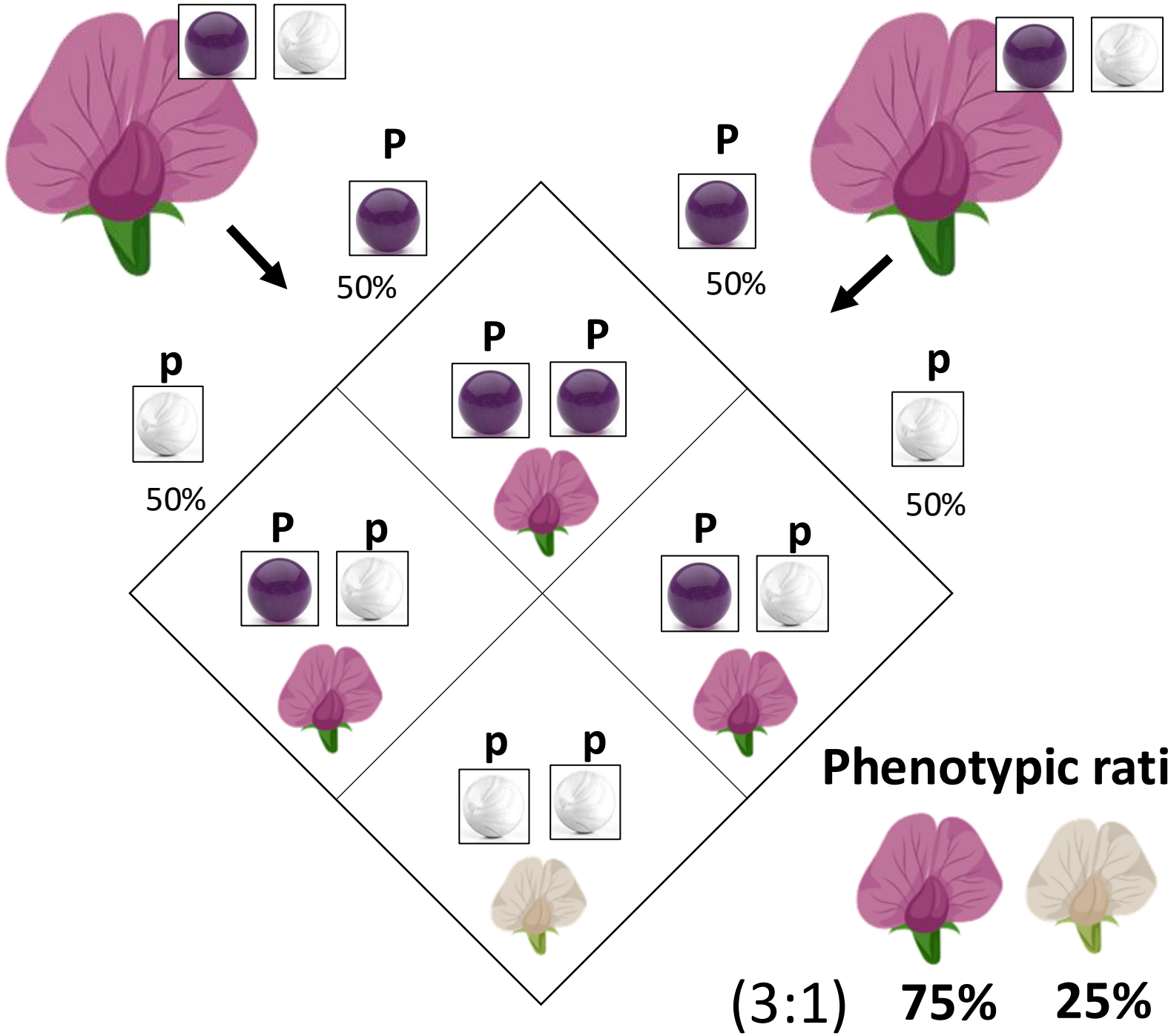


Self pollination

What is the expected ratio of purple to white flowered offspring?

F2

(Filial Generation 2)



Monohybrid Cross

$P/p \times P/p$

	P	p
P	P/P	P/p
p	P/p	p/p

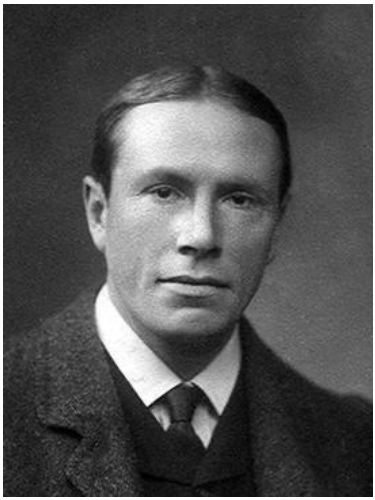
Genotypic ratios

PP: 25%

Pp: 50%

pp=25%

(1:2:1)



Reginald Punnett

Punnett Squares

$P/p \times P/p$
P p

P	P/P	P/p
p	P/p	p/p

$P/P \times p/p$
P p

P	P/p	P/p
P	P/p	P/p

$P/P \times P/p$
P p

P	P/P	P/p
P	P/P	P/p

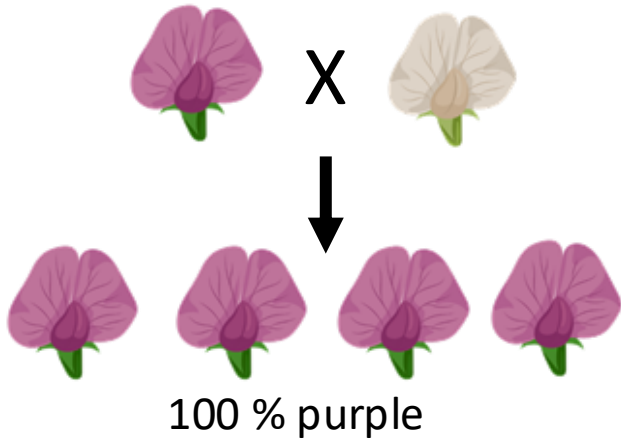
$P/p \times p/p$
P p

P	P/p	P/p
p	p/p	p/p

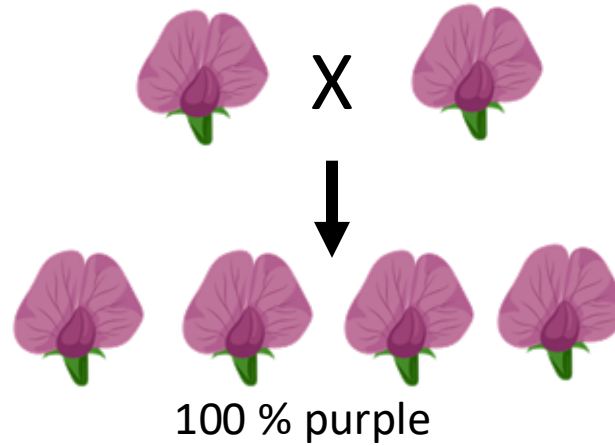
$p/p \times p/p$
p p

p	p/p	p/p
p	p/p	p/p

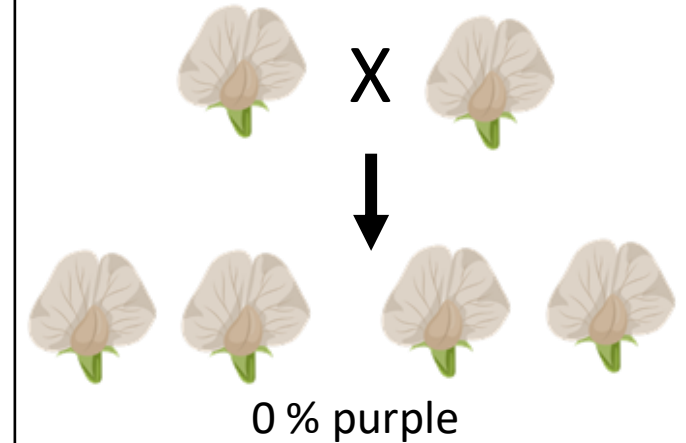
Result A



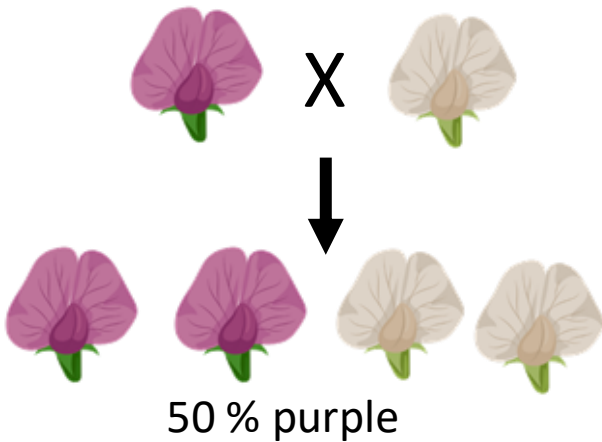
Result C



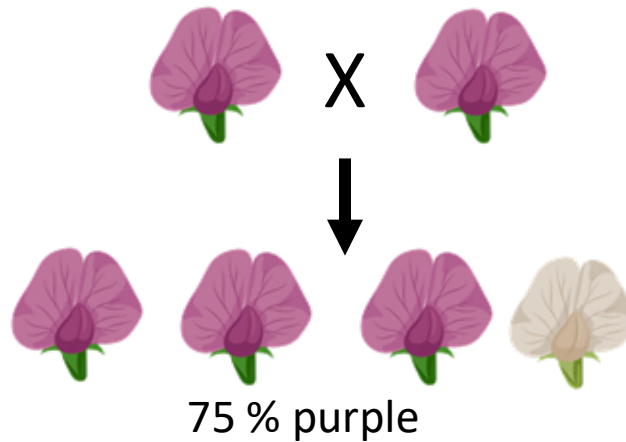
Result E



Result B



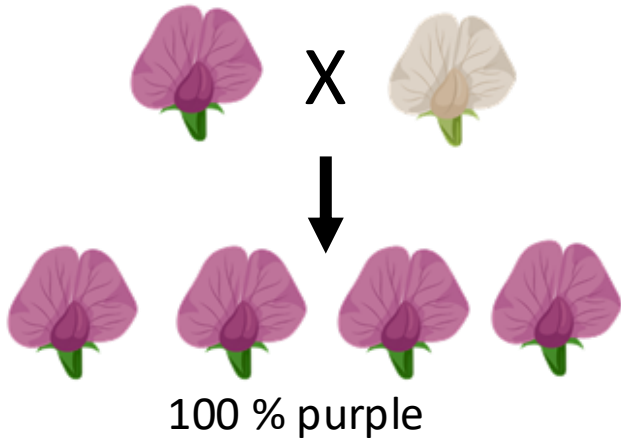
Result D



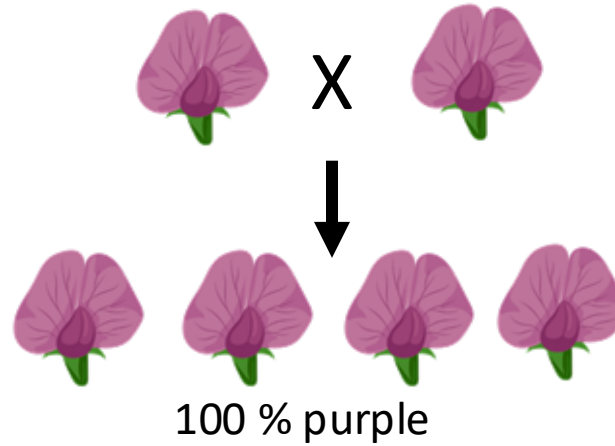
- **Purple is dominant**
- **Parents contribute 1 random allele to their offspring**

Note: hundreds of offspring were counted per cross (not 4 as illustrated)

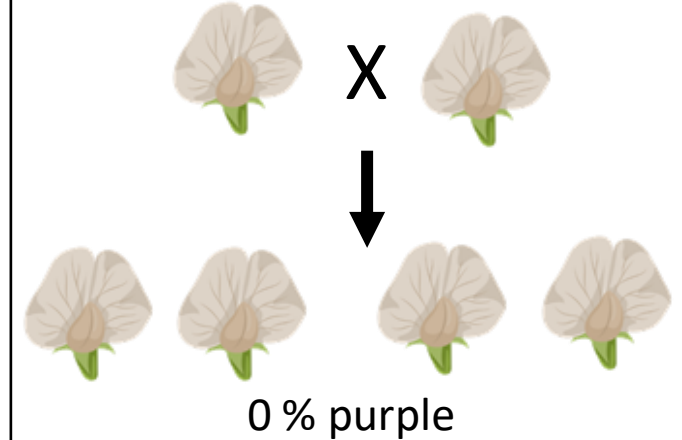
Result A



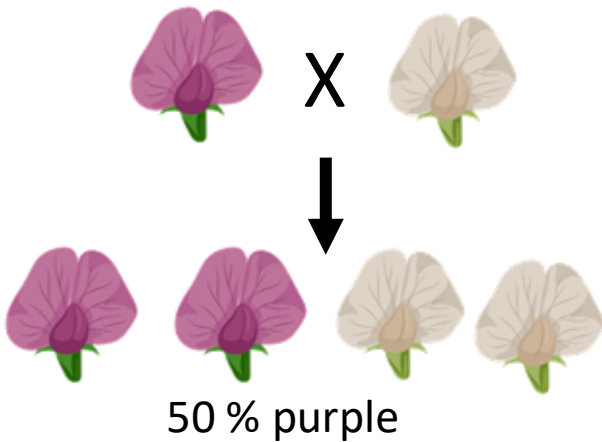
Result C



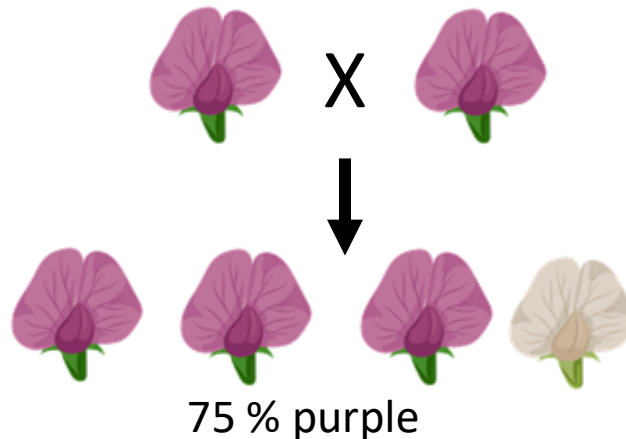
Result E



Result B



Result D

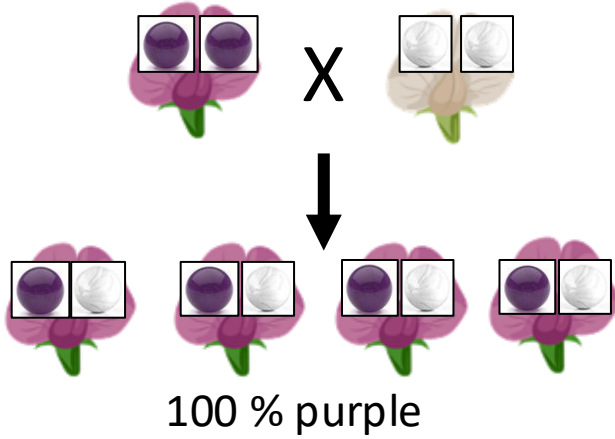


What are the genotypes (alleles) of the individuals in these experiments?

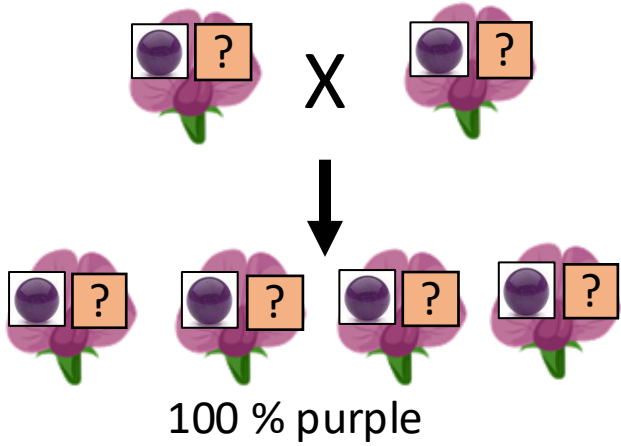
Can you be certain of all of them?

Note: hundreds of offspring were counted per cross (not 4 as illustrated)

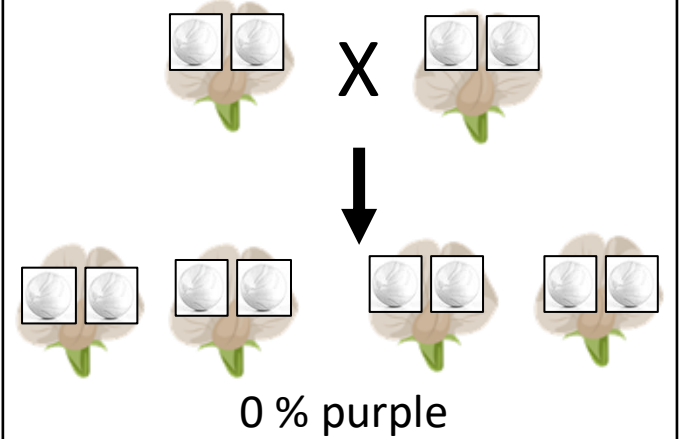
Result A



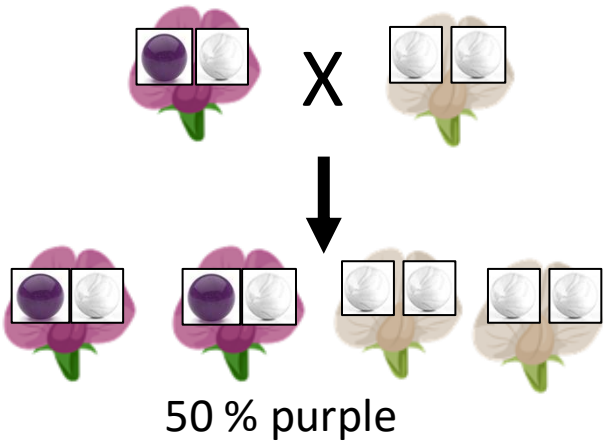
Result C



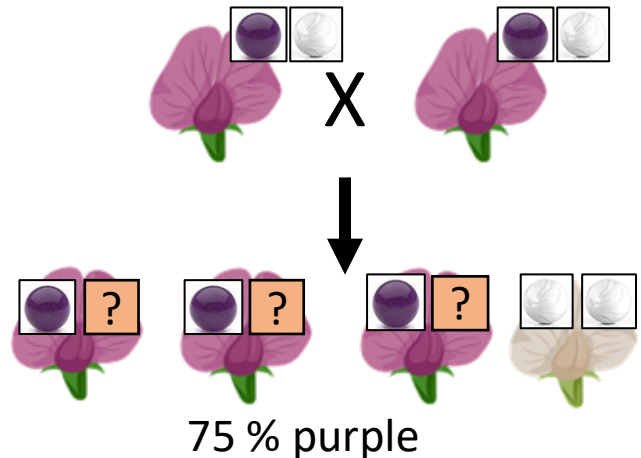
Result E



Result B



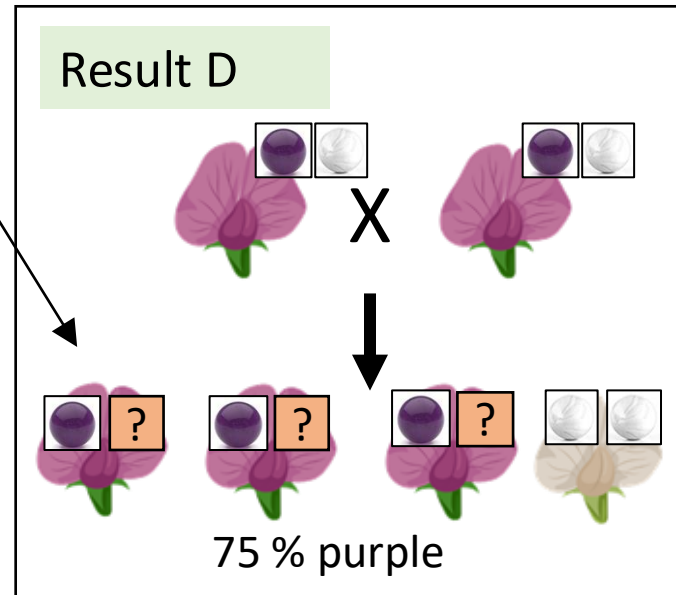
Result D

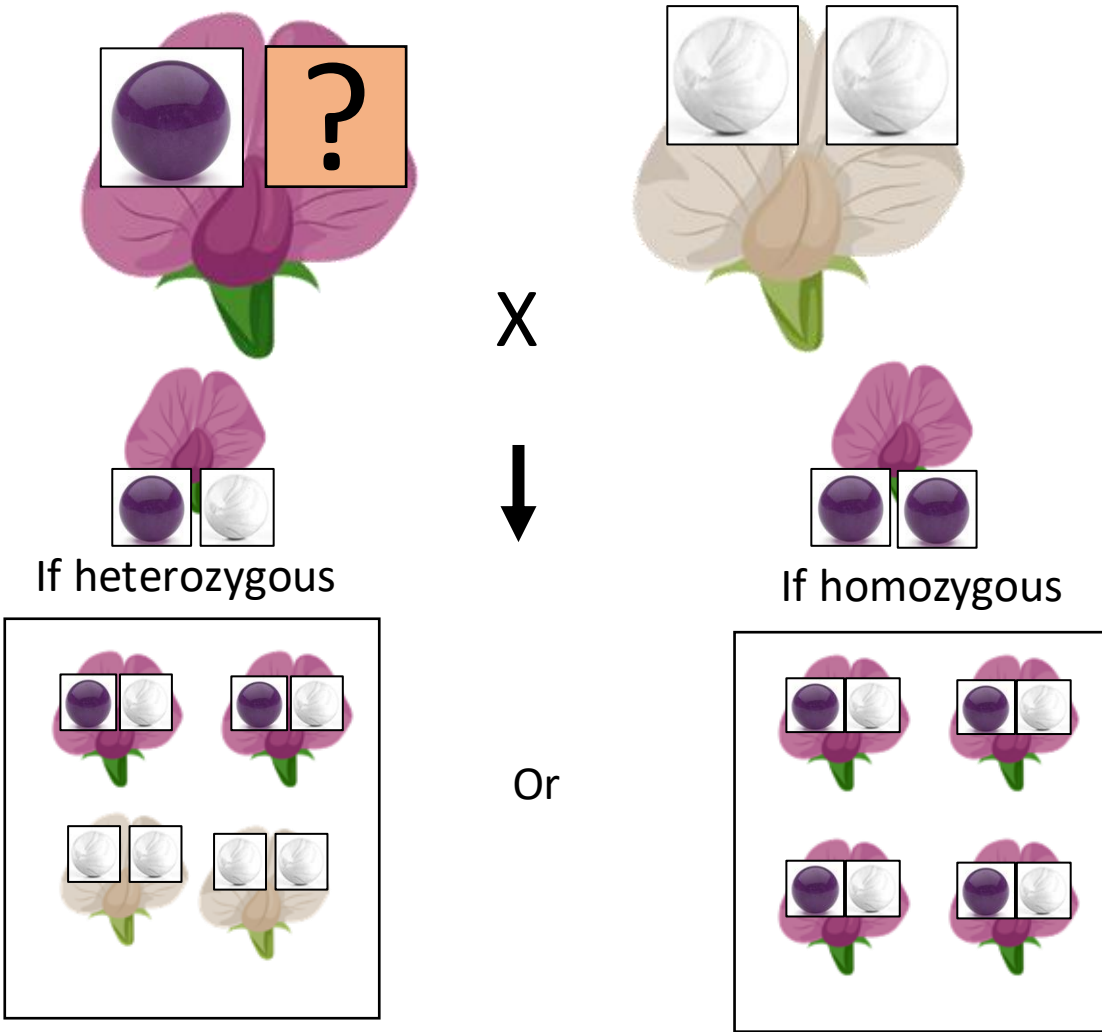


Note: hundreds of offspring were counted per cross (not 4 as illustrated)

What is the most probable
genotype of this
individual?

What is an efficient
experiment to test your
hypothesis?



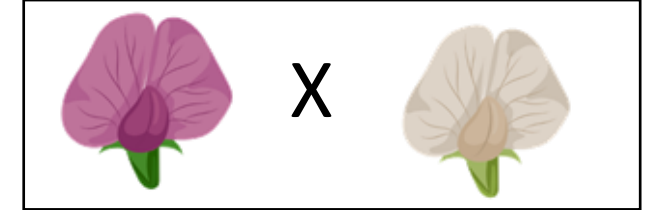
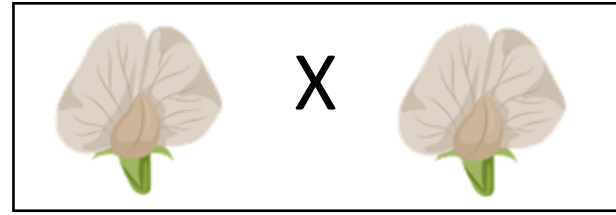
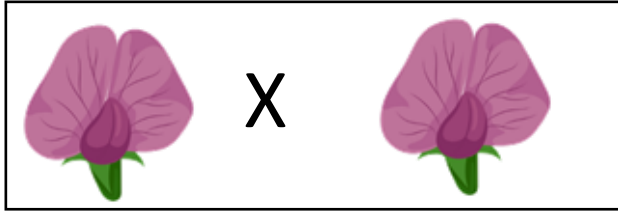


*** If all you know is a plant has a dominant phenotype, you can't be certain of its genotype**

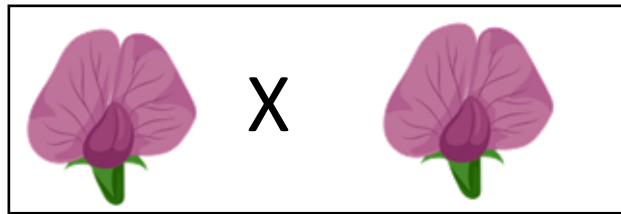
Testcross:

Used to determine unknown genotype.
Cross an unknown genotype with a homozygous recessive

Which of these crosses can we expect to predict the phenotypes of offspring?



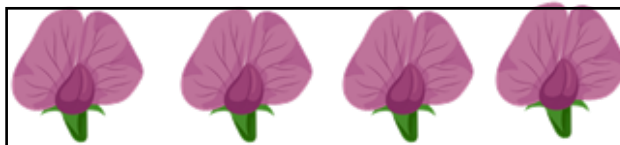
Which of these crosses can we expect to predict the phenotypes of offspring?



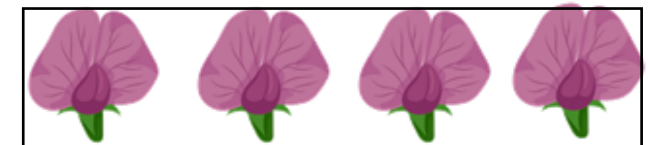
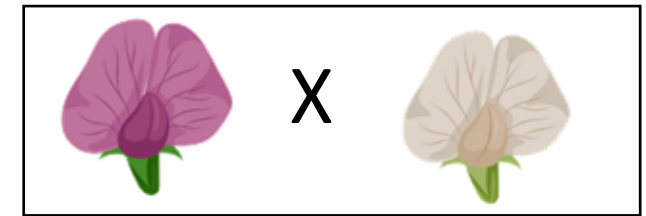
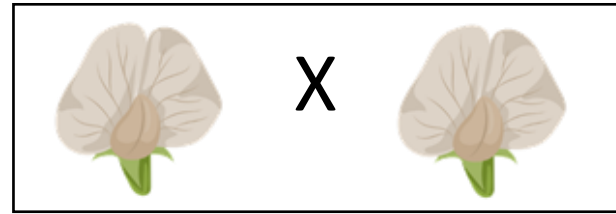
Mix



All purple



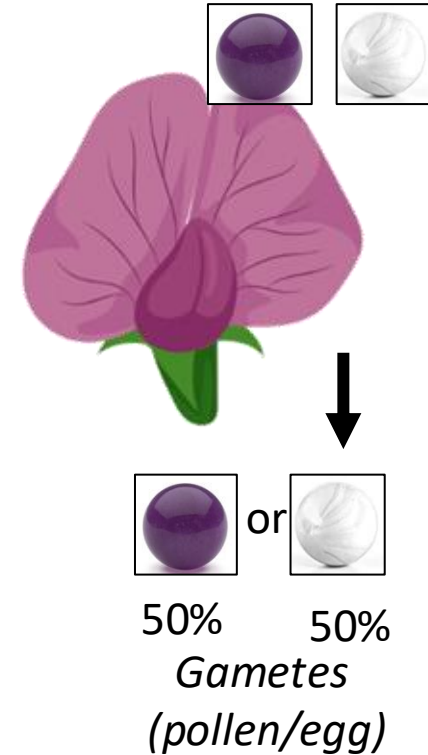
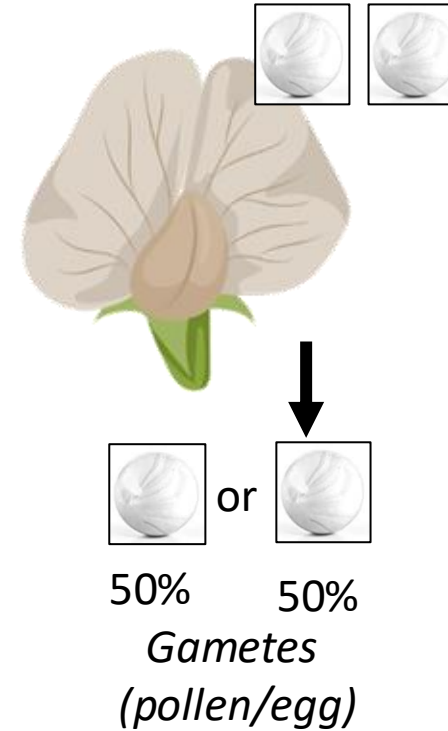
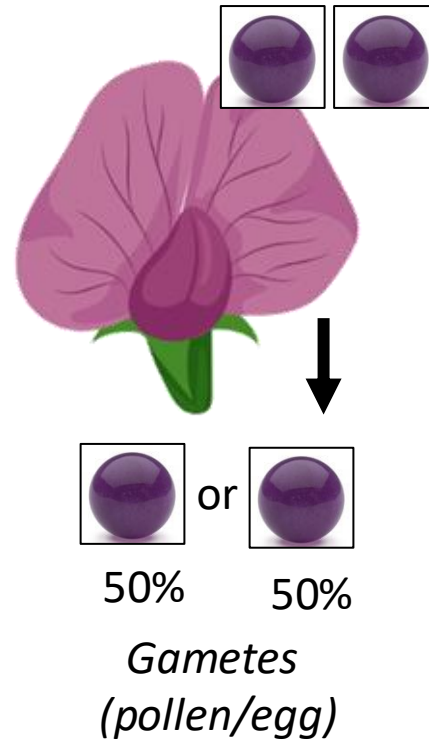
All white



Mendel's conclusions

For a given trait:

- Each individual has two alleles that determines their phenotype.
- Heterozygous individuals show the phenotype of the dominant allele.
- Parents randomly pass one allele to their offspring (segregation of alleles).

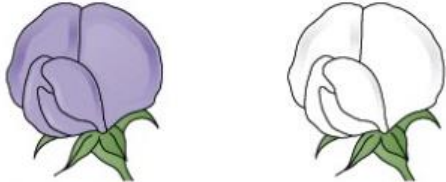




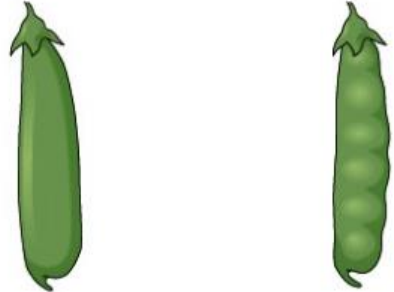
Round or wrinkled ripe seeds



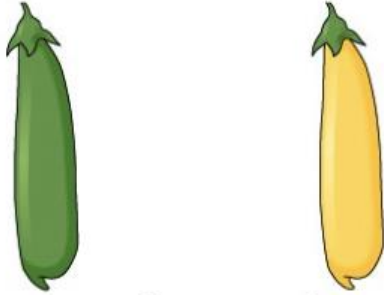
Yellow or green seeds



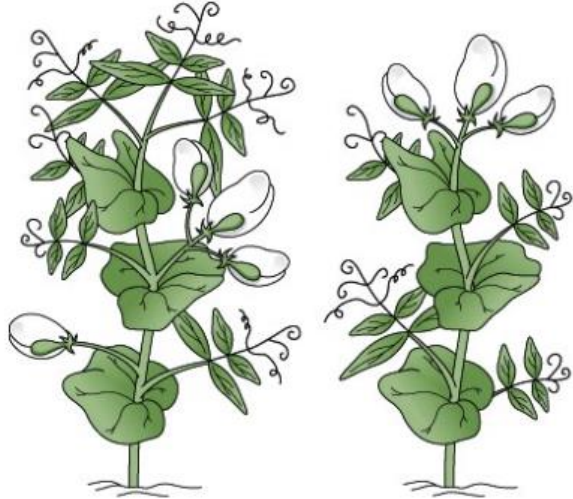
Purple or white petals



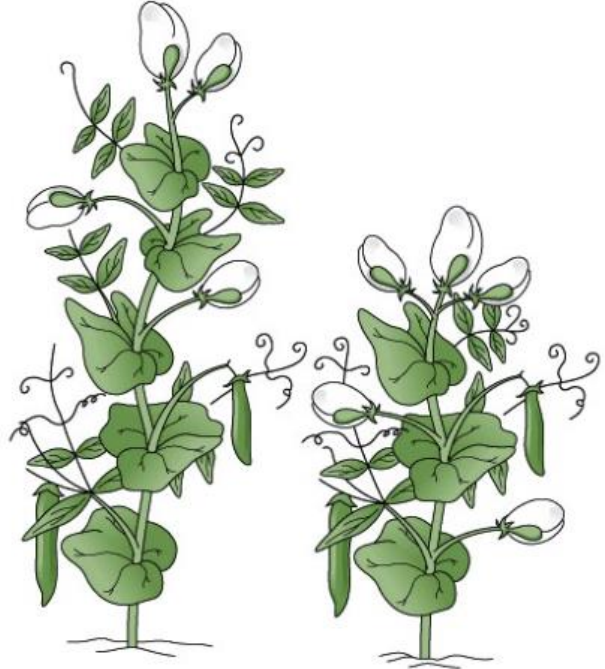
Inflated or pinched ripe pods



Green or yellow unripe pods



Axial or terminal flowers



Long or short stems

Figure 2-9
Introduction to Genetic Analysis, Ninth Edition
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Table 2-1 Results of All Mendel's Crosses in Which Parents Differed in One Character

Parental phenotype	F ₁	F ₂	F ₂ ratio
1. round × wrinkled seeds	All round	5474 round; 1850 wrinkled	2.96:1
2. yellow × green seeds	All yellow	6022 yellow; 2001 green	3.01:1
3. purple × white petals	All purple	705 purple; 224 white	3.15:1
4. inflated × pinched pods	All inflated	882 inflated; 299 pinched	2.95:1
5. green × yellow pods	All green	428 green; 152 yellow	2.82:1
6. axial × terminal flowers	All axial	651 axial; 207 terminal	3.14:1
7. long × short stems	All long	787 long; 277 short	2.84:1

Table 2-1

Introduction to Genetic Analysis, Ninth Edition

© 2008 W. H. Freeman and Company

Blending Inheritance was a major problem for Darwin

Evolution by Natural Selection

1. **Natural Variation:** Individuals differ in traits.

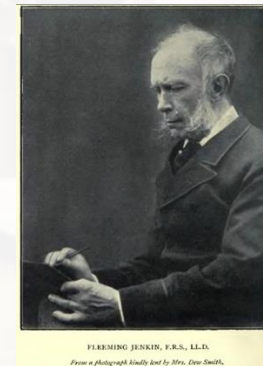
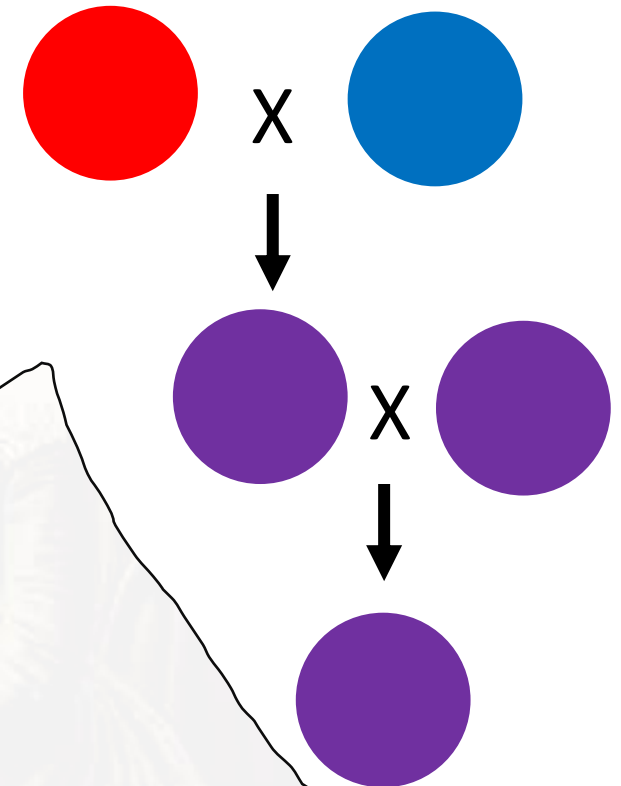
2. **Competition:** not enough resources for all to survive and reproduce.

3. **Differential success:** some variants do better than others.

4. **Inheritance:** traits passed to next generation.

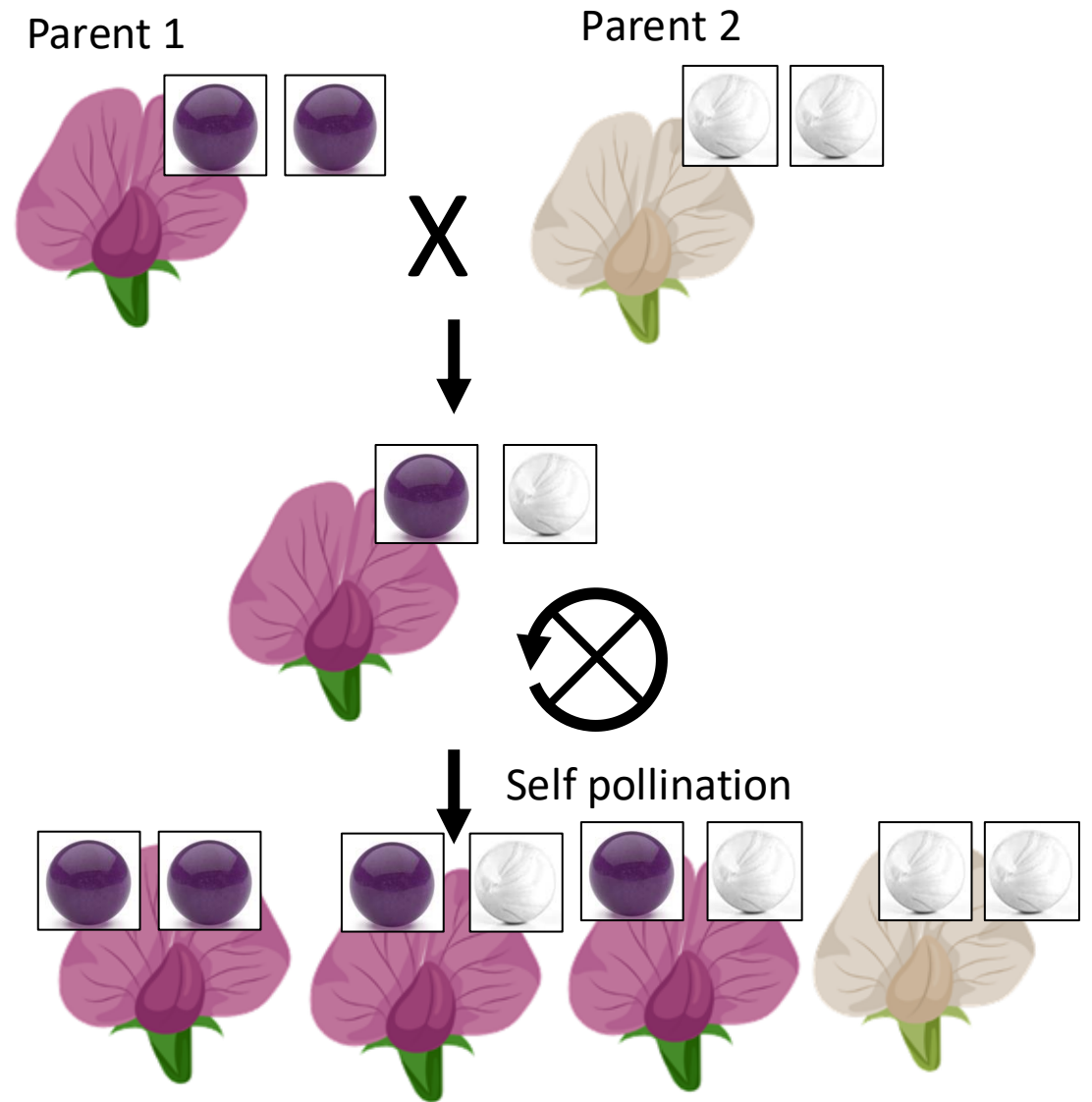
?

Blending inheritance rapidly removes natural variation





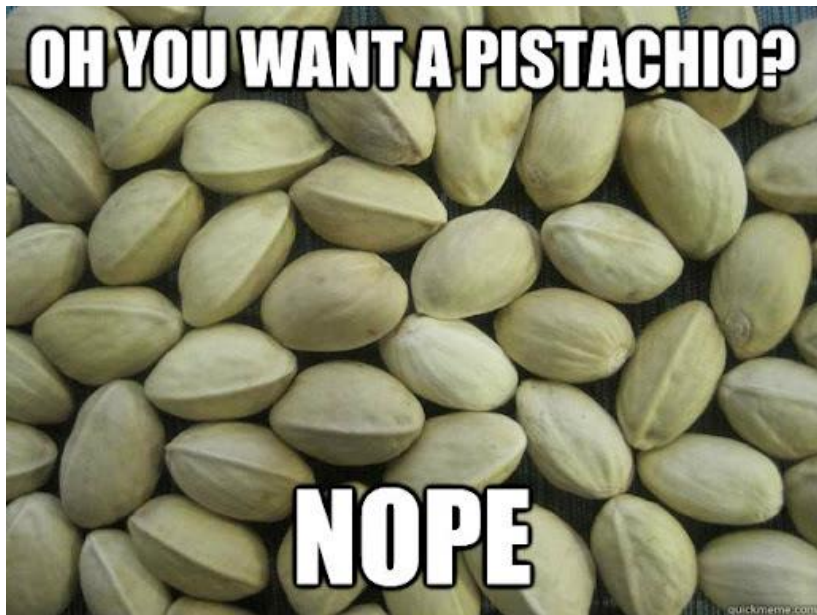
Variation is not lost due to breeding







Applying Mendel's Theory



Which is dominant?



Closed

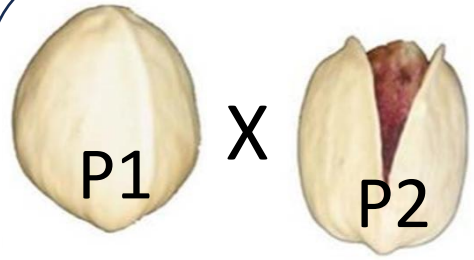


Open

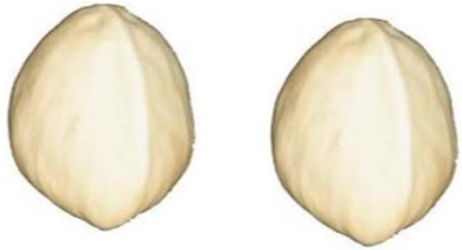
- You start with 2 trees: 1 produces closed nuts and 1 produces open nuts.
- You don't know their genotypes.
- Let's assume pistachio has same mating system as peas (i.e., can self) and the phenotype is definitely determined by a single gene.
- You propose to the California Pistachio Research Board to use crosses to determine which is dominant.
- **It takes 5 years from planting for pistachio trees to produce nuts.**

How would you cross plants to definitely determine which phenotype is dominant within 5 years? And what results are you looking for?

Scenario 1

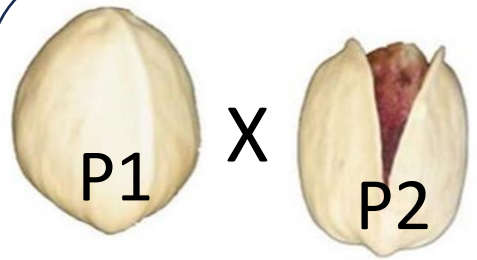


↓ 5 years

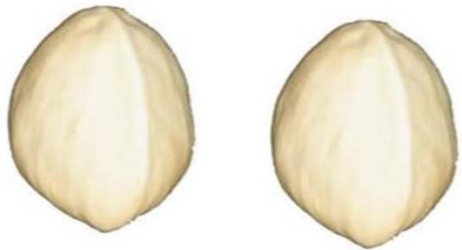


Lucky!

Scenario 1

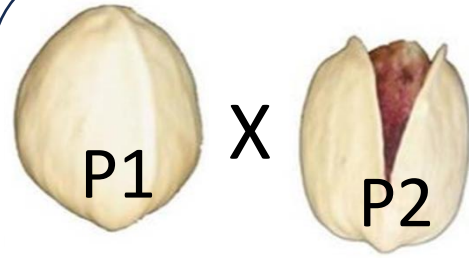


↓ 5 years

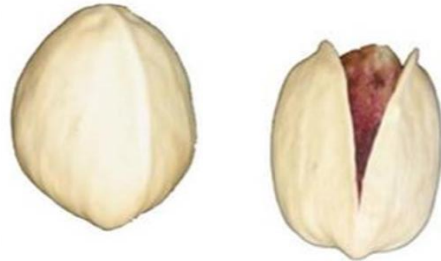


Lucky!

Scenario 2

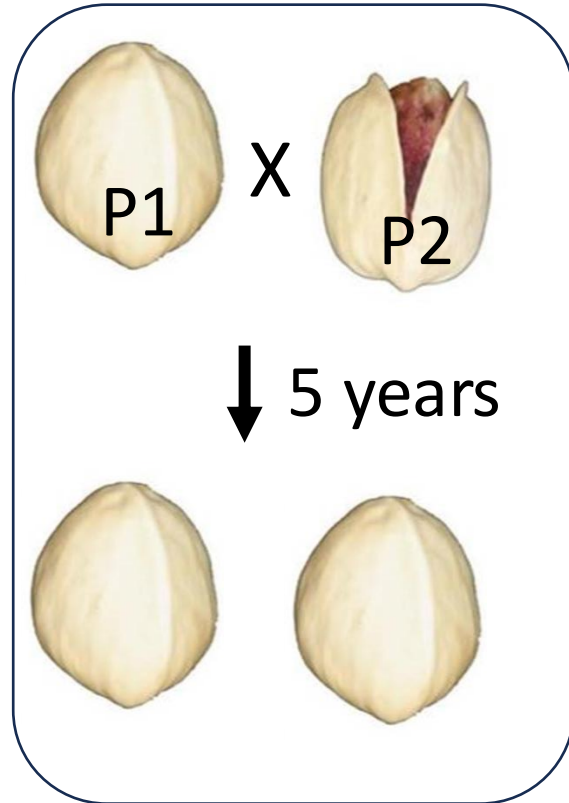


↓ 5 years

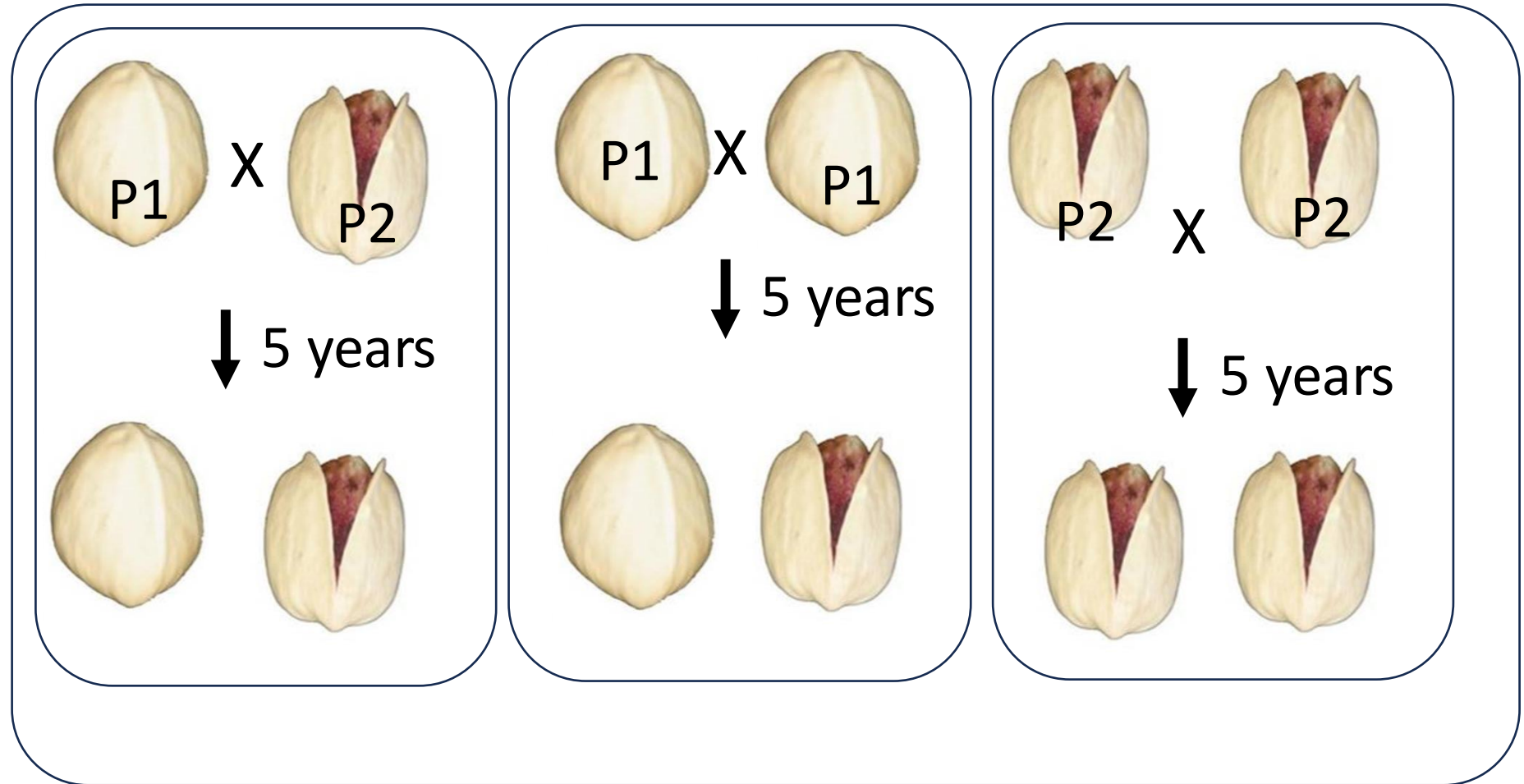


Unknown ☹

Scenario 1



Scenario 2



Hypothesis:
Open is dominant
Rejected

What are the genotypes of the parents in these two scenarios?

10 min Break :)



What the deal with
dominance?

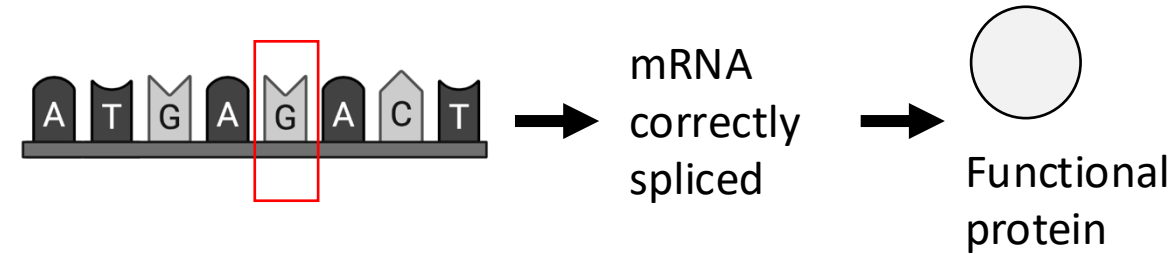
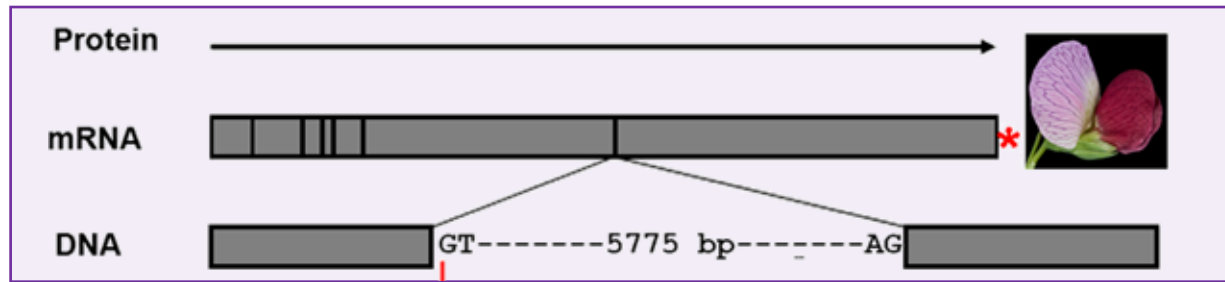
What the deal with dominance?

Identification of Mendel's White Flower Character

Roger P. Hellens¹, Carol Moreau², Kui Lin-Wang¹, Kathy E. Schwinn³, Susan J. Thomson⁴, Mark W. E. J. Fiers⁴, Tonya J. Frew⁴, Sarah R. Murray⁴, Julie M. I. Hofer², Jeanne M. E. Jacobs⁴, Kevin M. Davies³, Andrew C. Allan¹, Abdelhafid Bendahmane⁵, Clarice J. Coyne⁶, Gail M. Timmerman-Vaughan⁴, T. H. Noel Ellis^{2*}

¹ The New Zealand Institute for Plant and Food Research Ltd, Auckland, New Zealand, ² Department of Crop Genetics, John Innes Centre, Norwich, United Kingdom, ³ The New Zealand Institute for Plant and Food Research Ltd, Palmerston North, New Zealand, ⁴ The New Zealand Institute for Plant and Food Research Ltd, Christchurch, New Zealand, ⁵ URGV, Evry, France, ⁶ United States Department of Agriculture-Agricultural Research Service Western Regional Plant Introduction Station, Pullman, Washington, United States of America

P

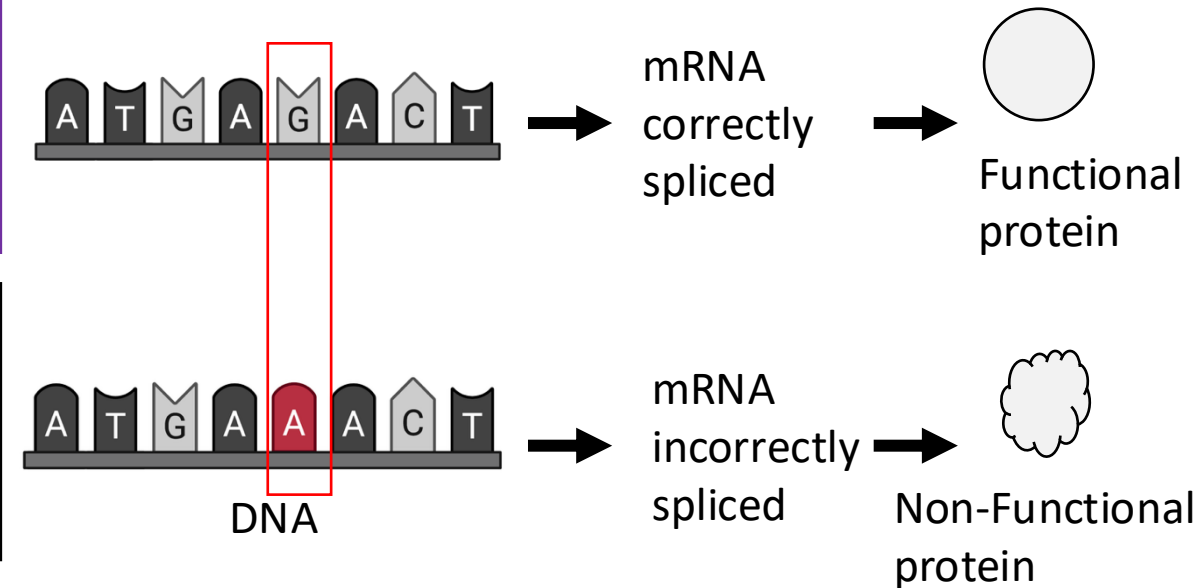
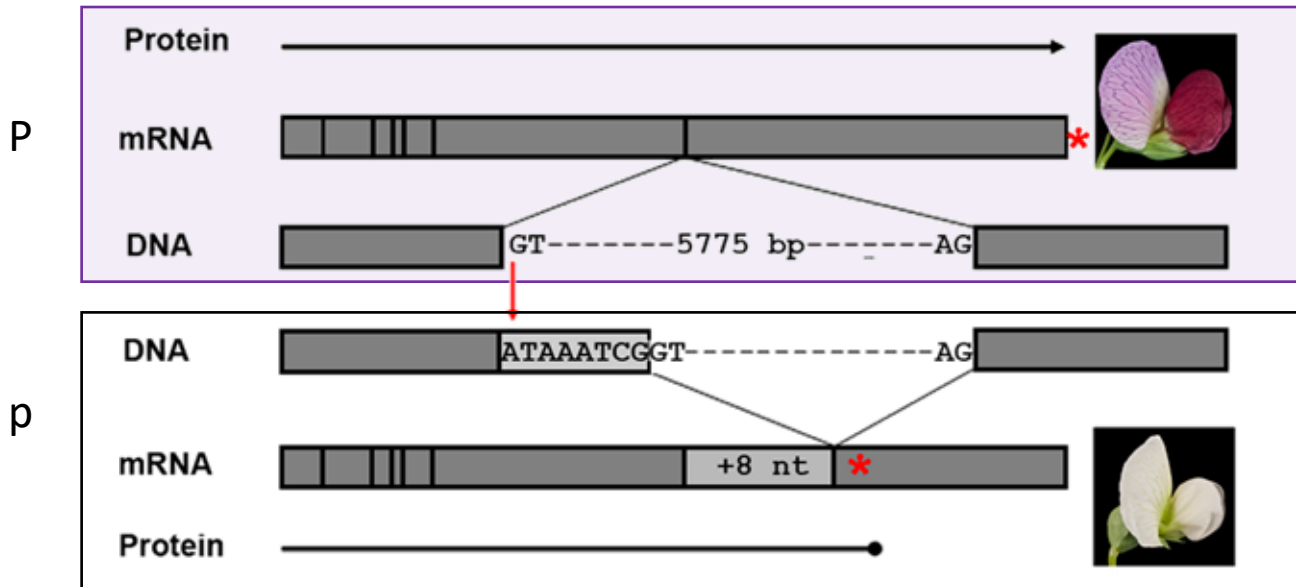


What the deal with dominance?

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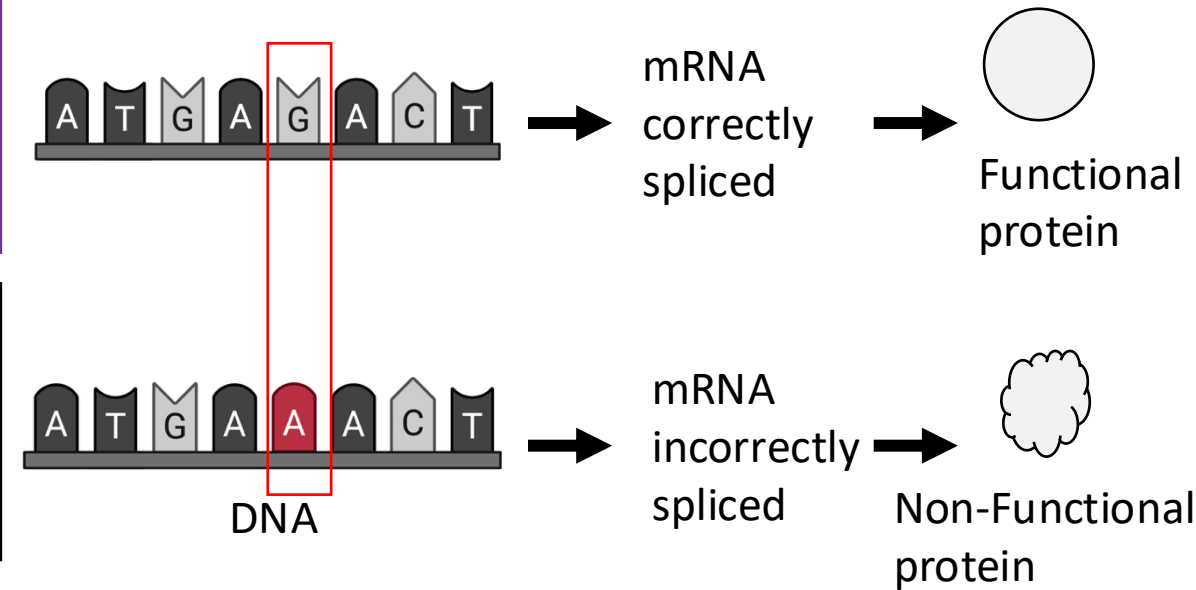
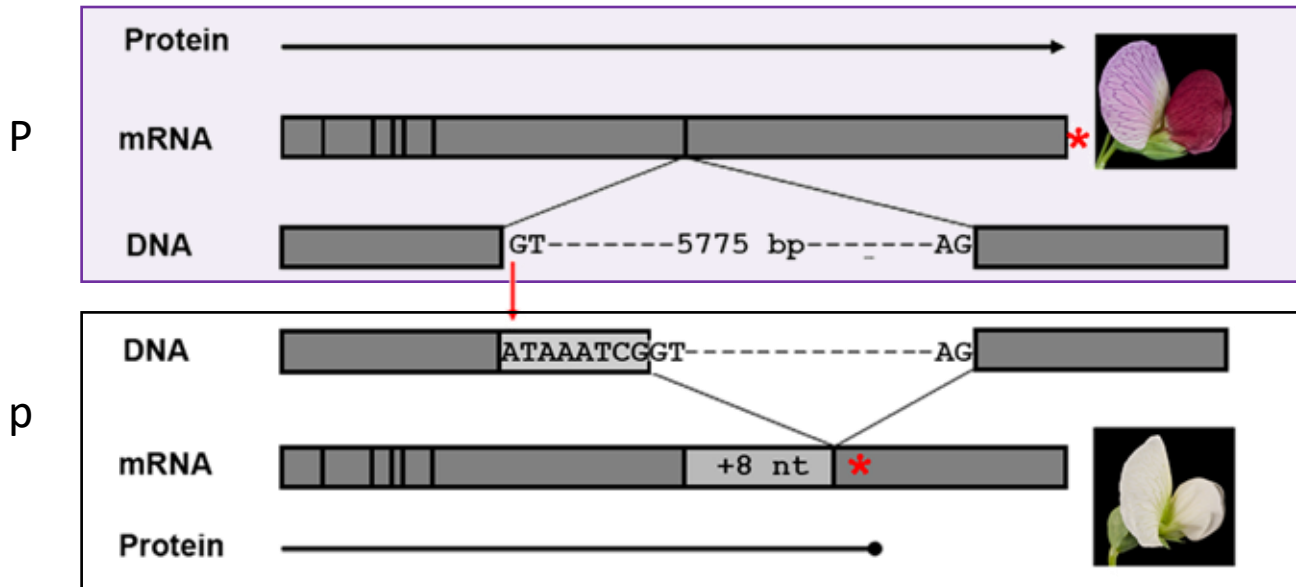
Example of molecular basis of dominance: Only 1 functional allele is needed to make Purple phenotype

What the deal with dominance?

Identification of Mendel's White Flower Character

Roger P. Hellens¹, Carol Moreau², Kui Lin-Wang¹, Kathy E. Schwinn³, Susan J. Thomson⁴, Mark W. E. J. Fiers⁴, Tonya J. Frew⁴, Sarah R. Murray⁴, Julie M. I. Hofer², Jeanne M. E. Jacobs⁴, Kevin M. Davies³, Andrew C. Allan¹, Abdelhafid Bendahmane⁵, Clarice J. Coyne⁶, Gail M. Timmerman-Vaughan⁴, T. H. Noel Ellis^{2*}

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Extract
DNA



Sequence
"P" locus

A/A

What do you
predict is this
plant's
phenotype?

G/G

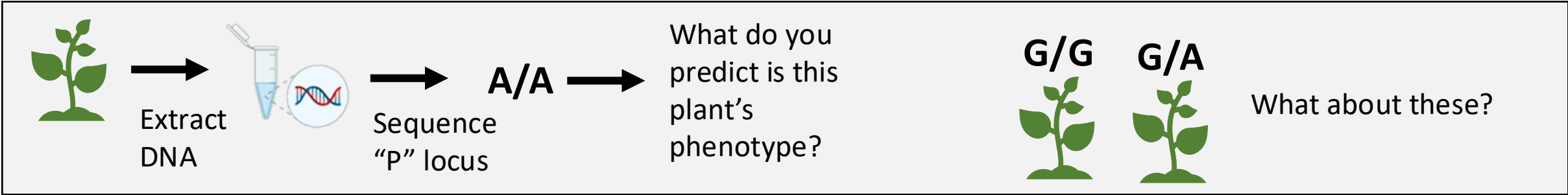
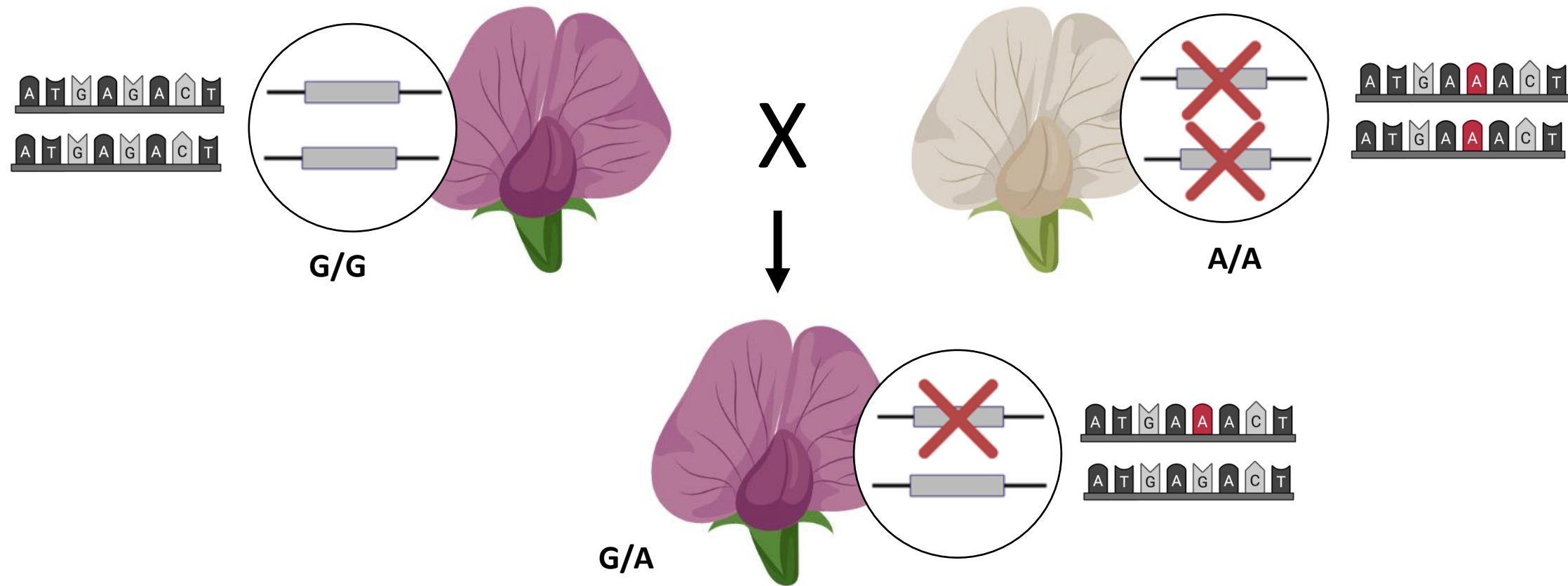


G/A



What about these?

Synthesizing Mendel and modern genetics with DNA sequencing technology

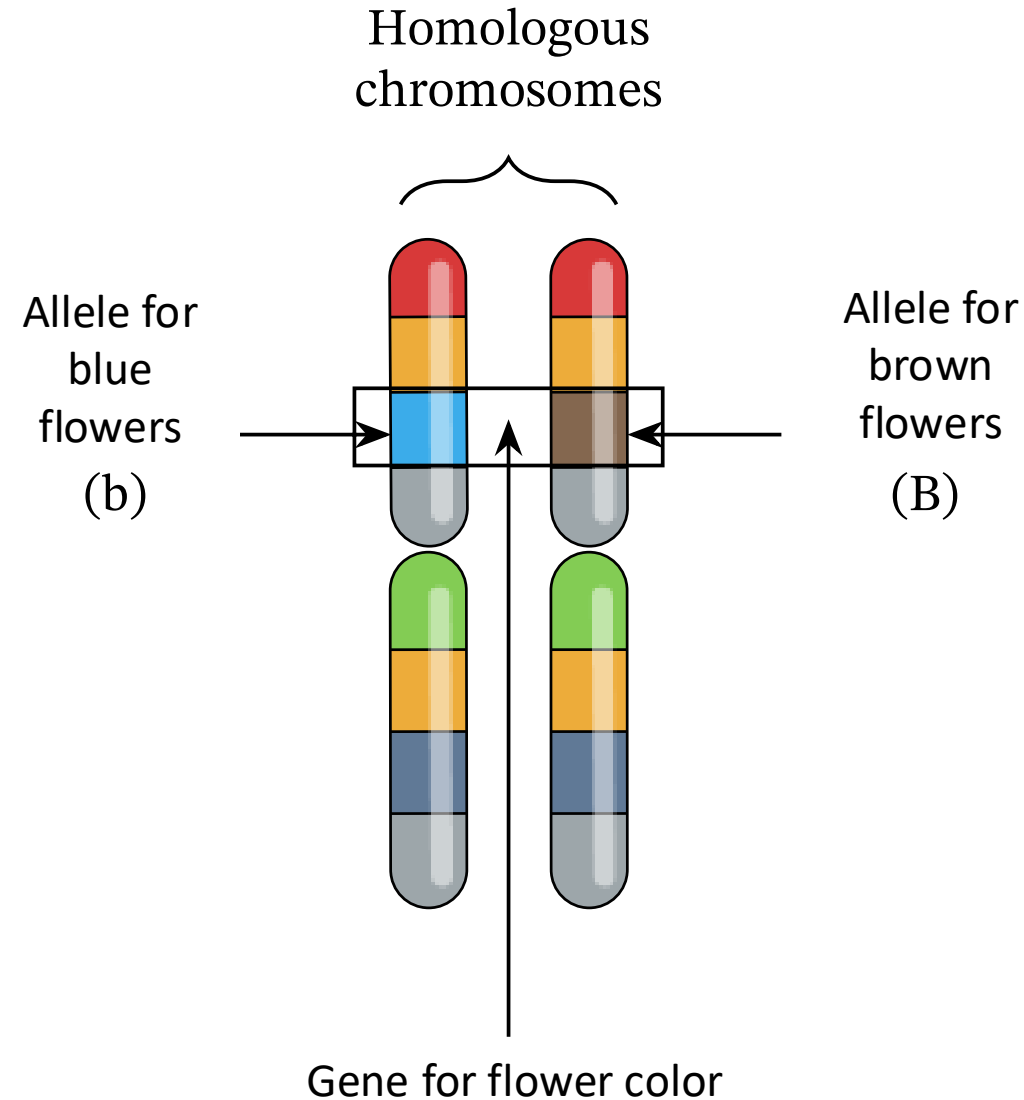


Gene vs allele

Genes are genome loci that control a trait (e.g. flower color)

Alleles are different version of a gene.
could be described by:

- phenotype (e.g. blue flower, brown flower, etc)
- Base-pair ("T", "C")
- Mendelian dominance notation ("B", "b")





P/P, G/G



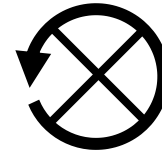
p/p, g/g

Parent 1



Parent 2

X



F1

Self pollination



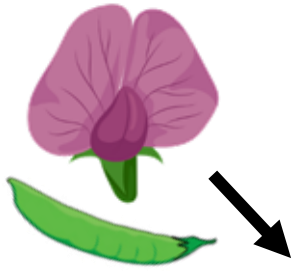
What is the expected ratio of phenotypes in the offspring?

F2

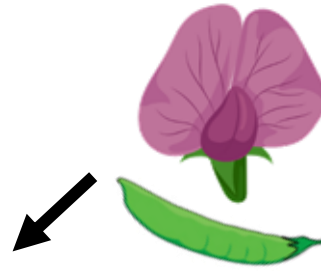


Hypothesis: Green pea allele and purple flower allele are inherited independently.

P/p, G/g



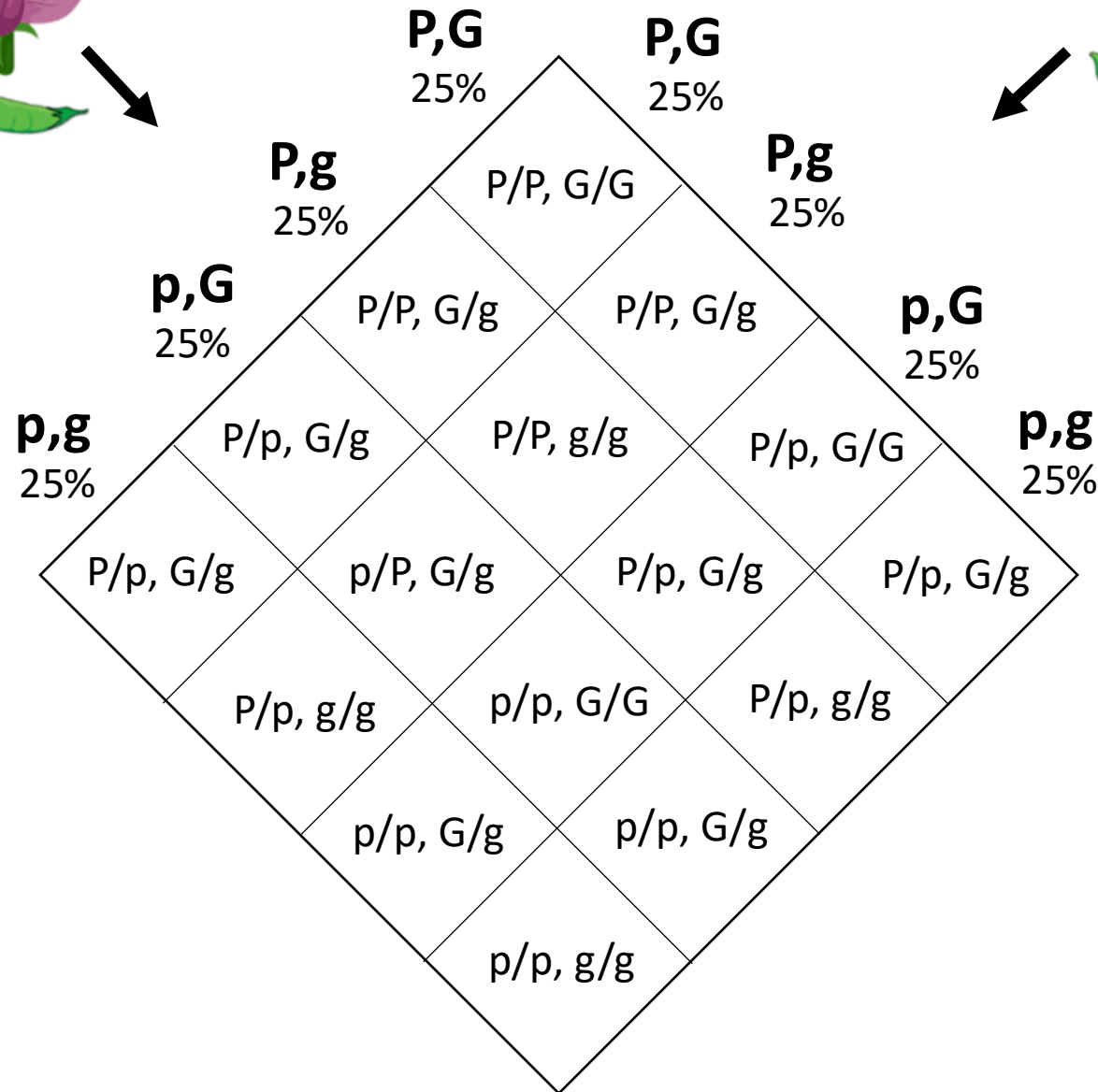
P/p, G/g



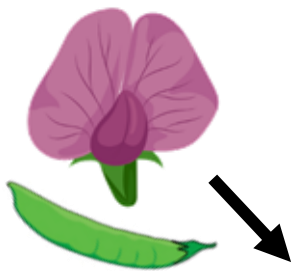
Di-hybrid Cross

Phenotypic ratios

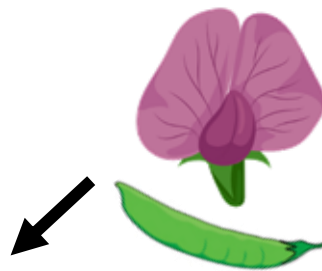
What is the expected ratio of phenotypes in the offspring?



P/p, G/g



P/p, G/g

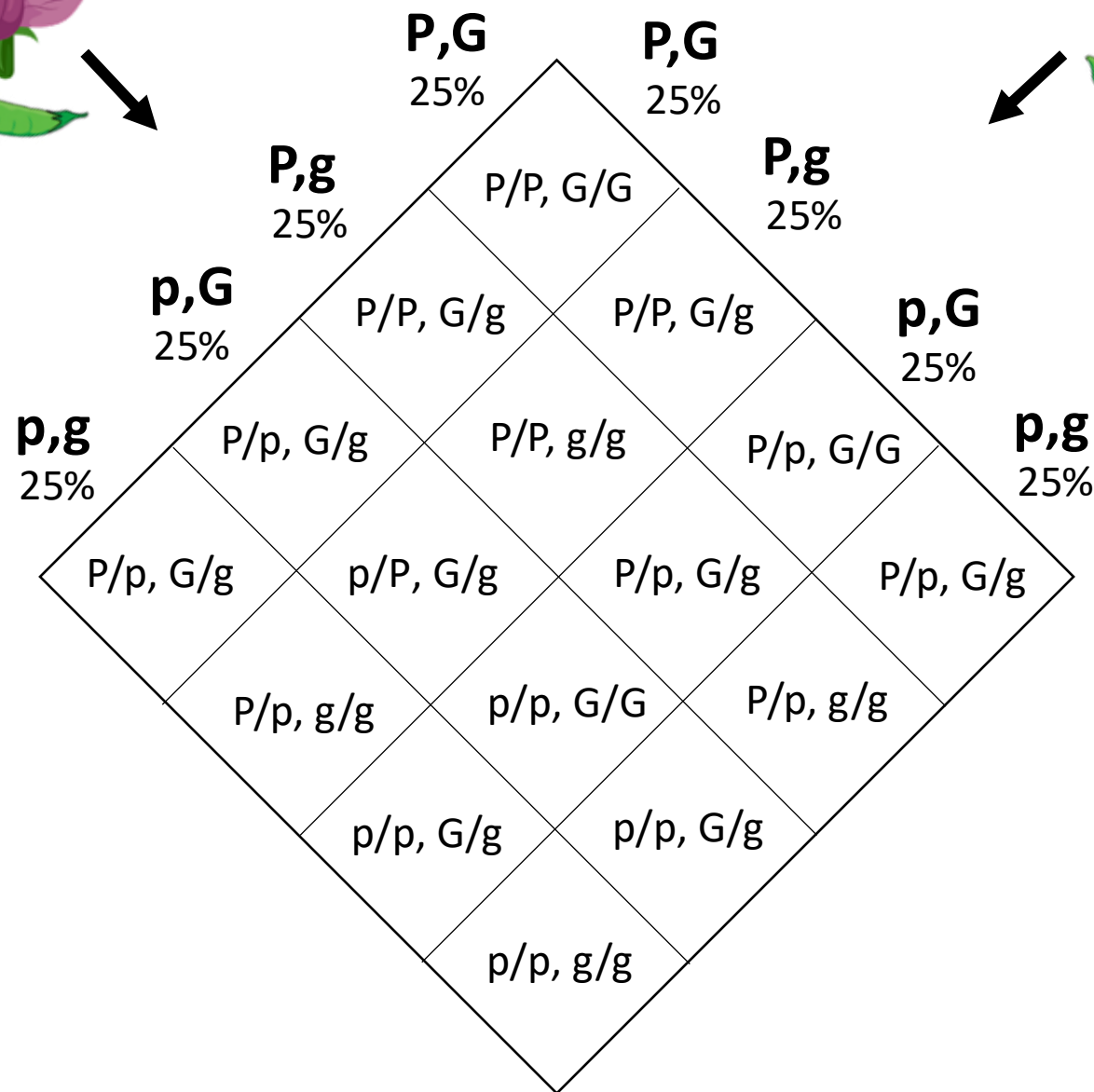


Di-hybrid Cross

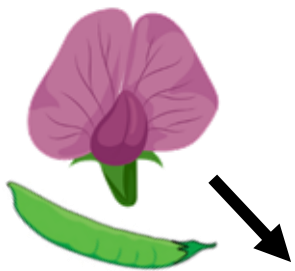
Phenotypic ratios



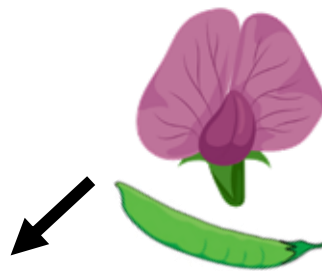
(9:3:3:1)



P/p, G/g



P/p, G/g

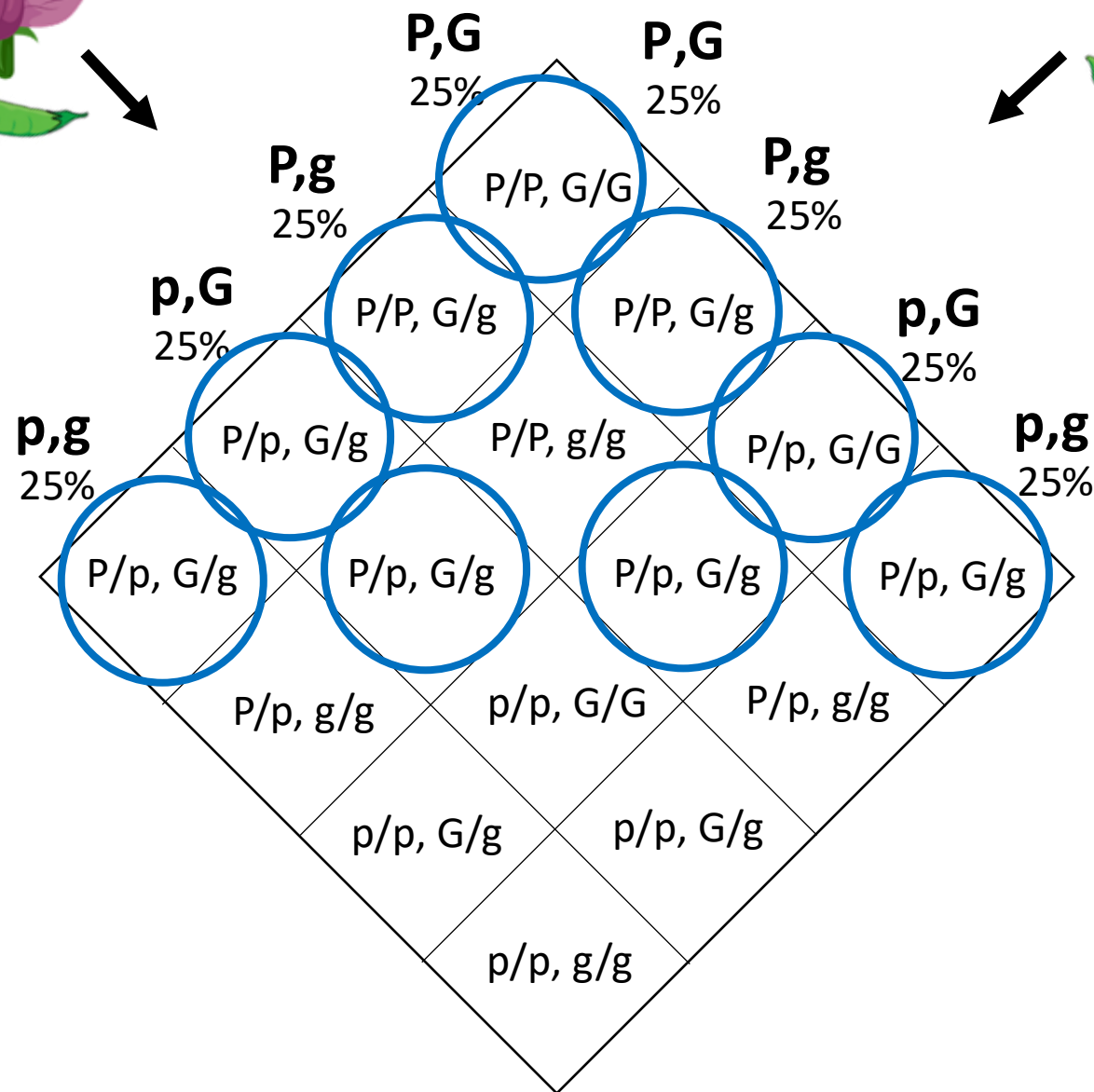


Di-hybrid Cross

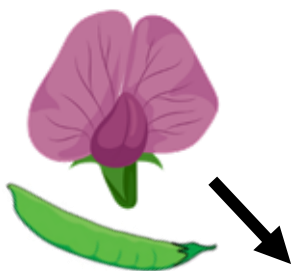
Phenotypic ratios



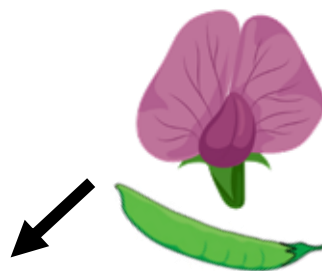
(9:3:3:1)



P/p, G/g

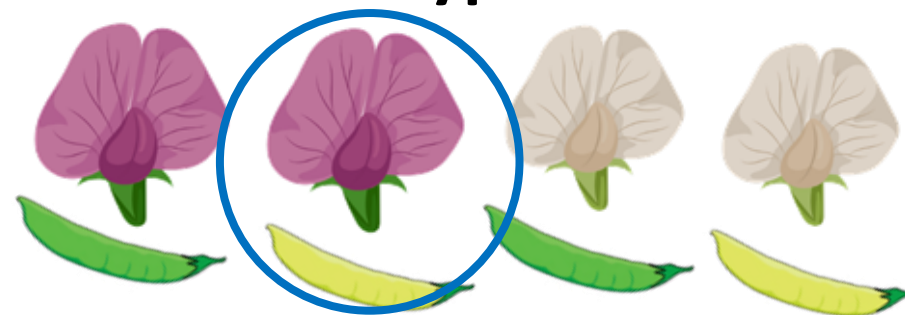


P/p, G/g

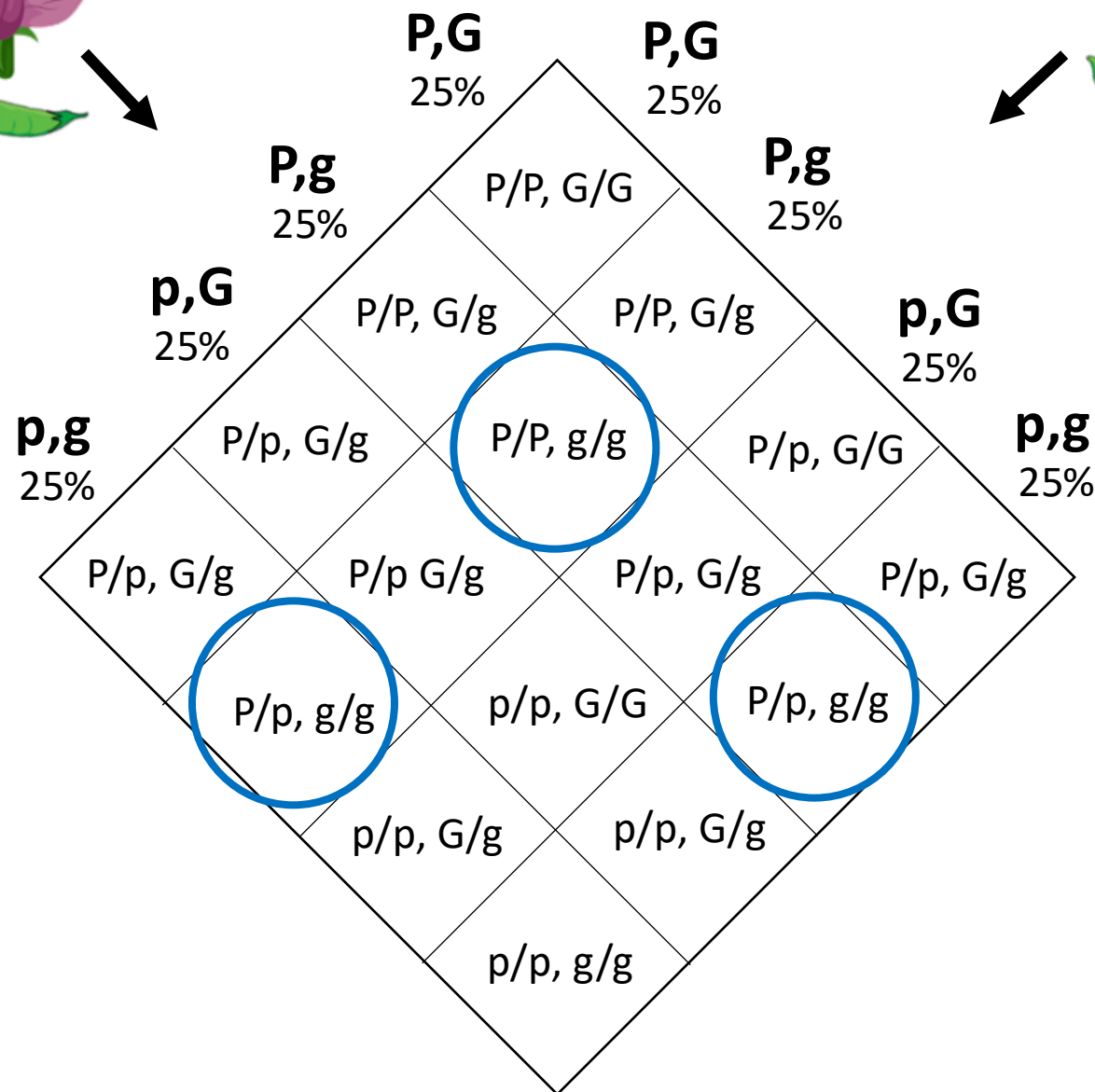


Di-hybrid Cross

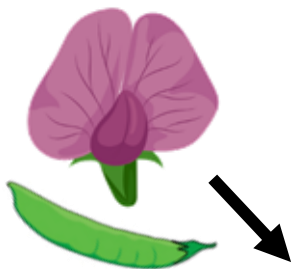
Phenotypic ratios



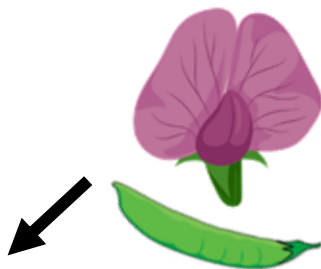
(9:3:3:1)



P/p, G/g

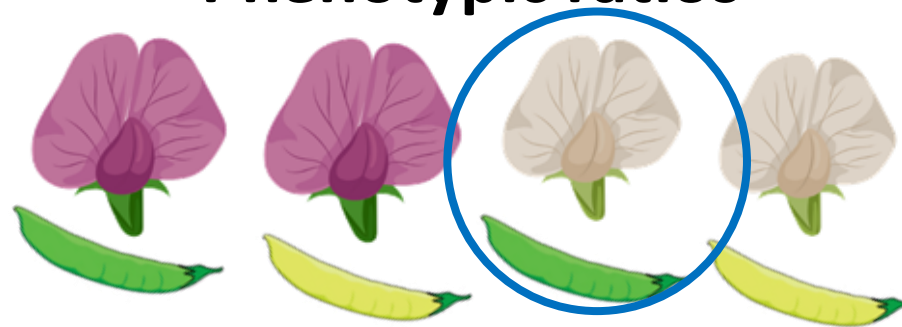


P/p, G/g

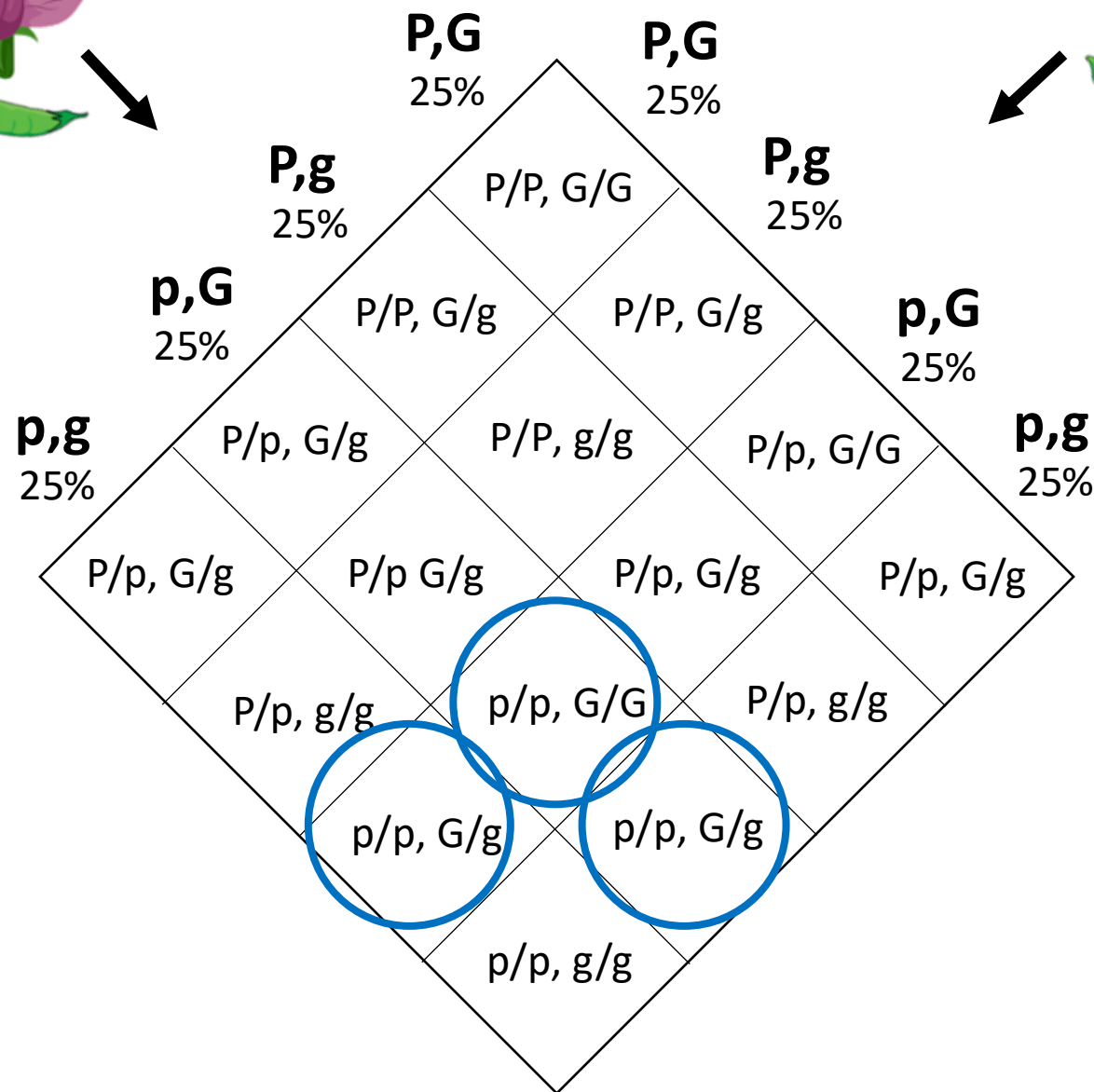


Di-hybrid Cross

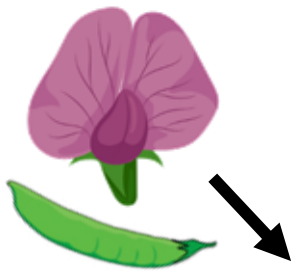
Phenotypic ratios



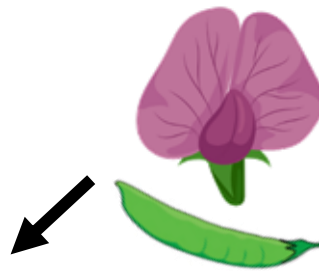
(9:3:3:1)



P/p, G/g



P/p, G/g



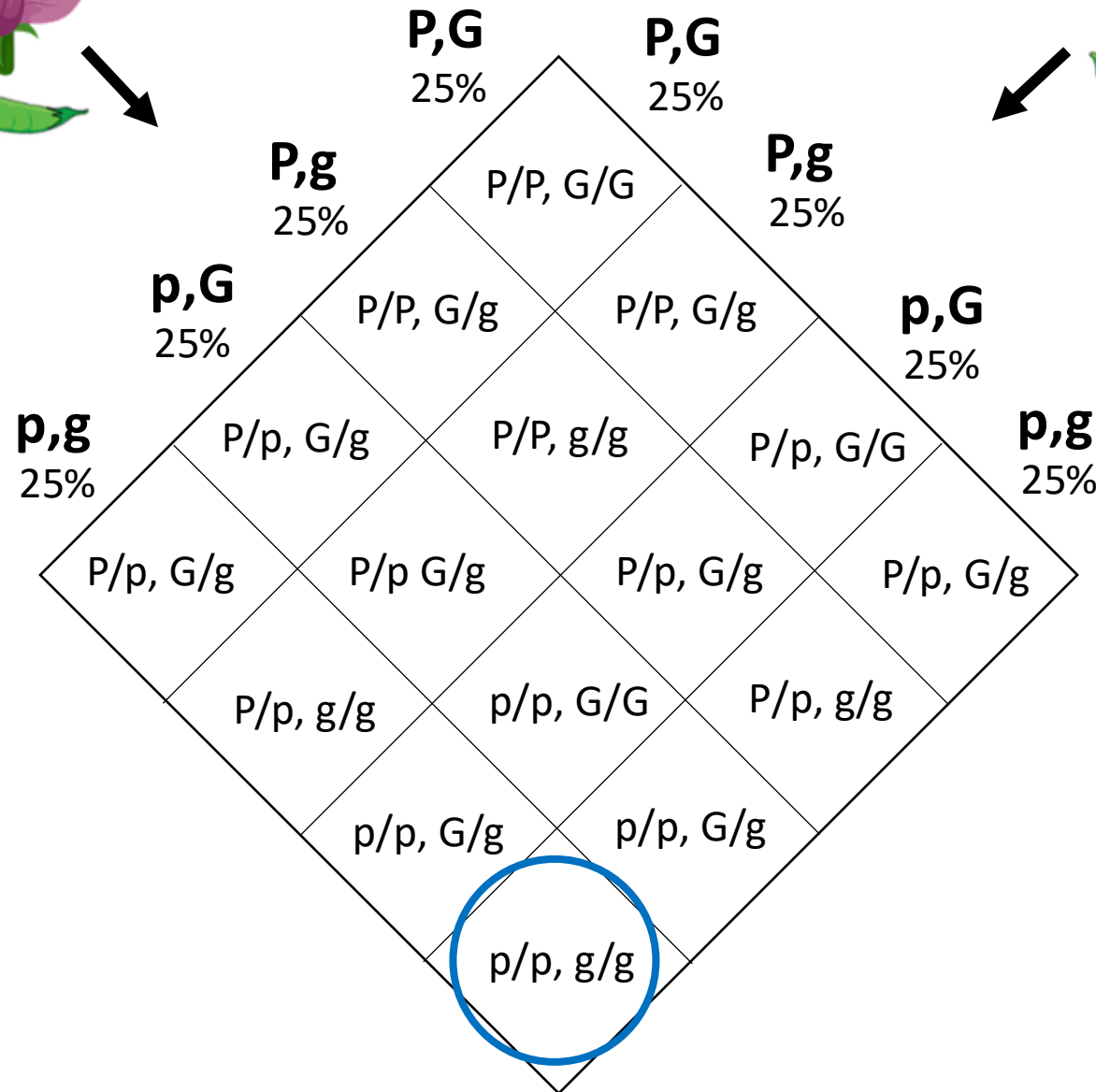
Di-hybrid Cross

Phenotypic ratios



(9:3:3:1)

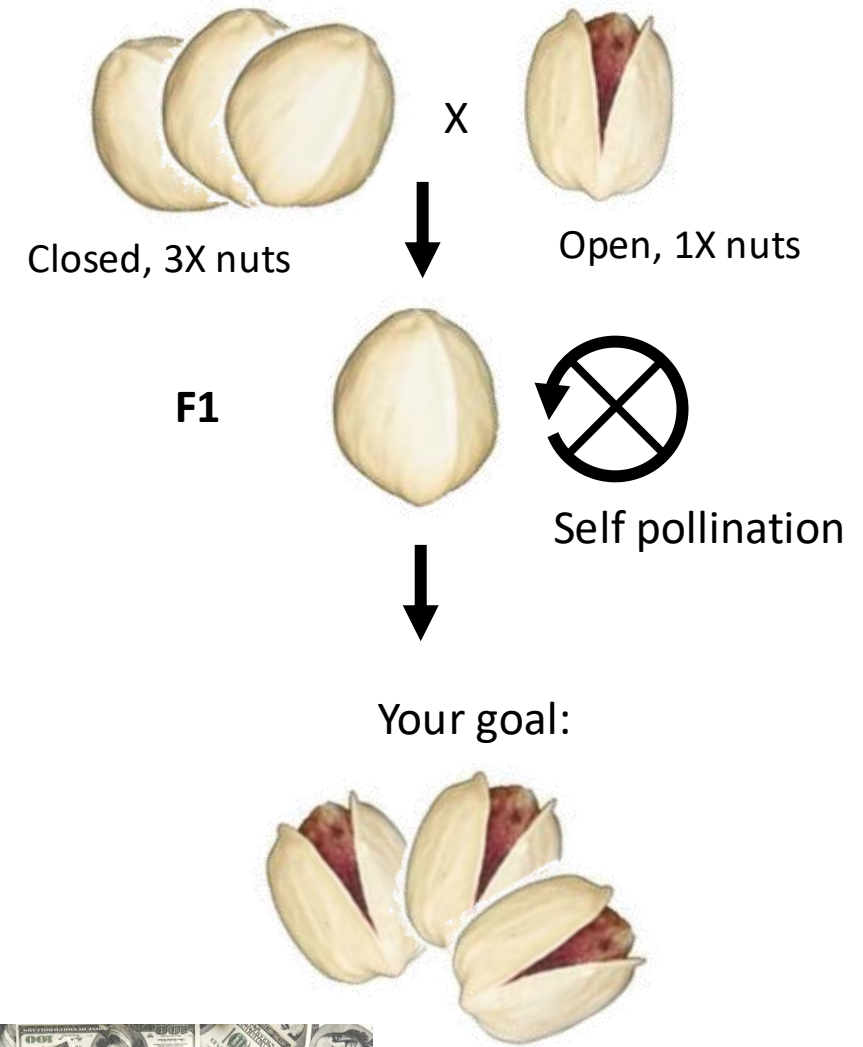
$$\text{Probability} = 1/4 * 1/4 = \mathbf{1/16}$$



A photograph of a long, straight row of mature trees, likely olive trees, in a field. The trees have dense green foliage and thick, gnarled trunks. They are planted in neat rows, with a dirt path or road running alongside them on the right. The ground is dry and sandy. The sky is blue with scattered white clouds. The overall scene is bright and sunny.

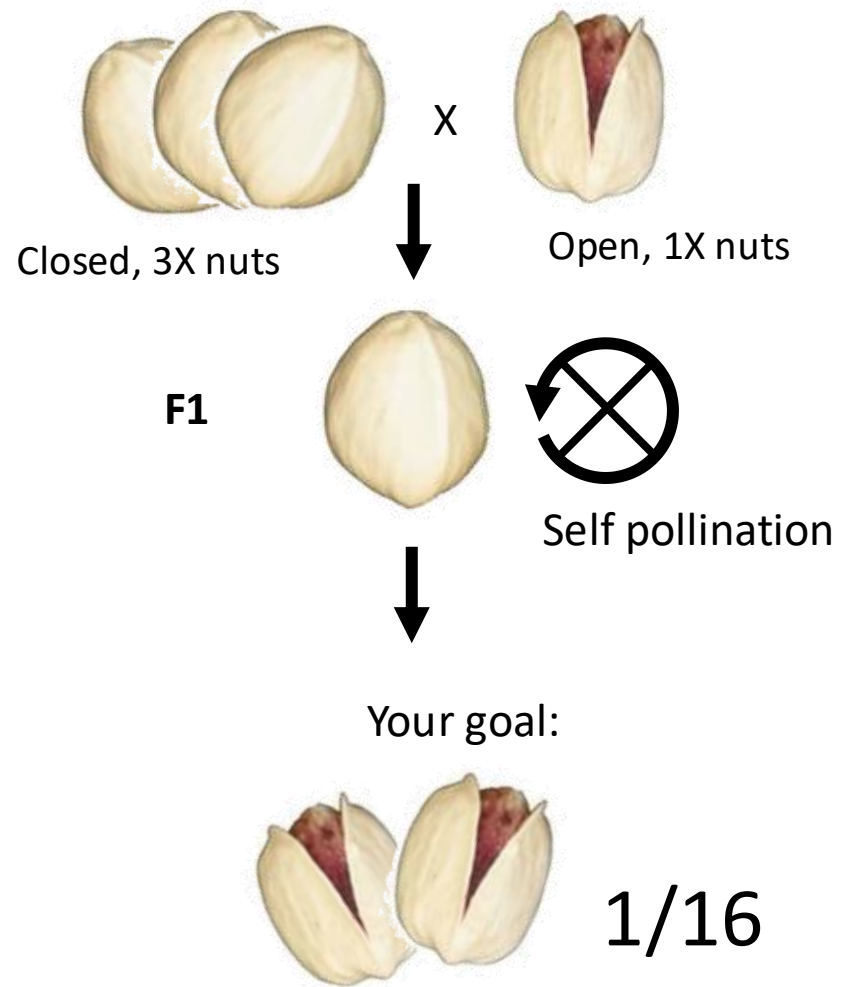
Applying Mendel's Theory
(with modern genetic tools)

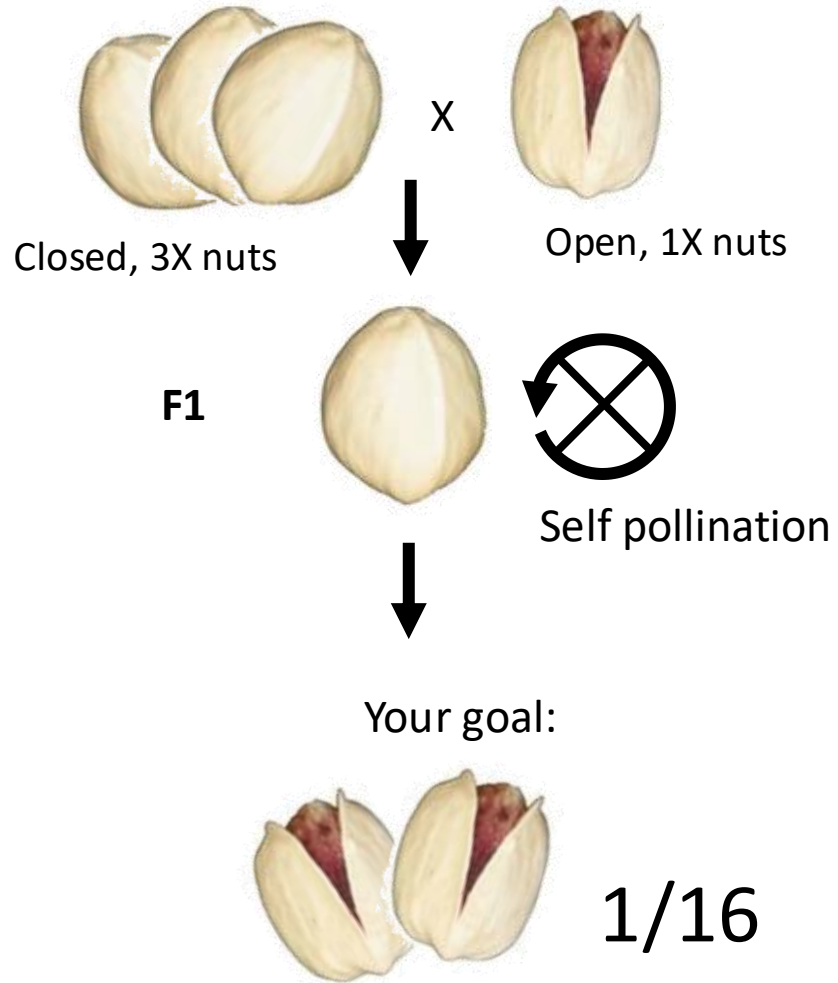
- You have an F1 from a cross between a tree with closed nuts that produces 3x yield and a tree with easy-open nuts but less yield.
- The easy-open phenotype is recessive and you know the gene: It is caused by a C -> T mutation in gene 1 (T causes easy-open phenotype).
- The 3X phenotype is also recessive and you know the gene: It is caused by a G -> C mutation in gene 2 (C causes 3X yield)
- Let's assume pistachio has same mating system as peas (i.e., can self plants).
- You propose to the California Pistachio Research Board to develop a pistachio variety with open nuts and 3X yield.
- **It costs \$10 dollars to sprout a pistachio nut, extract DNA from the seedling, and sequence the 2 genes controlling these traits.**
- **It costs \$500 dollars to grow a tree for 5 years from seedling until it is old enough to produce nuts.**



What is your proposed budget to produce expected 10 trees with easy-open nuts and 3X yield?

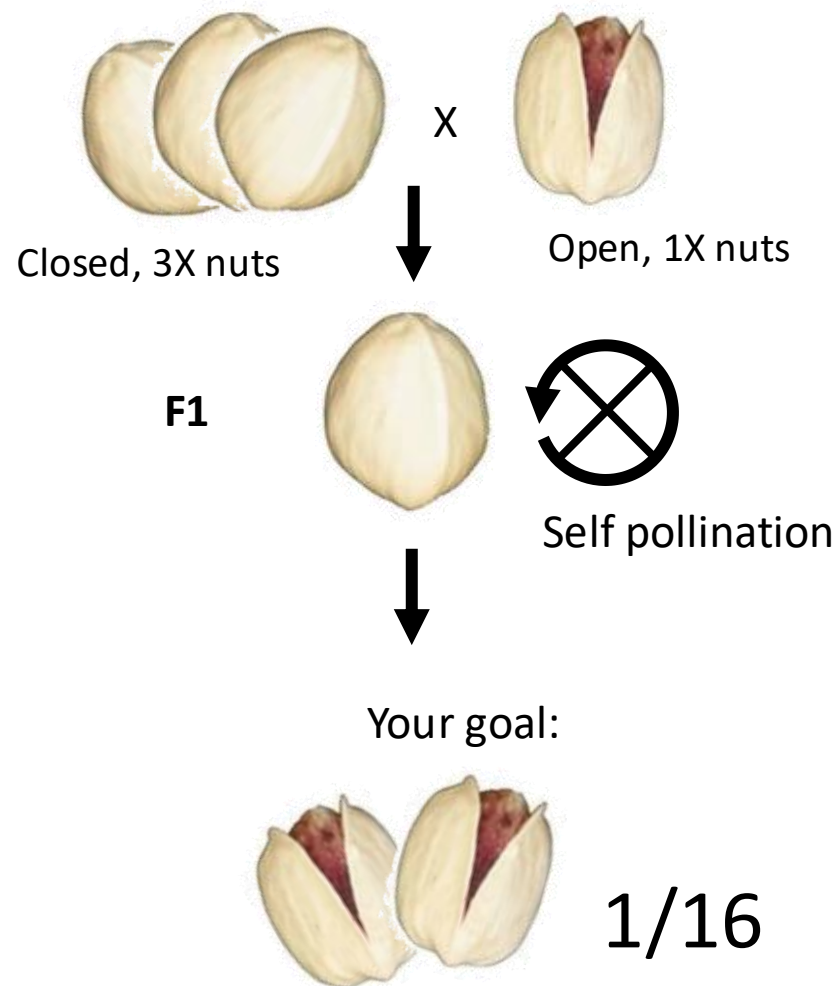






Without molecular markers:

To get (expected) 10 trees you need
160 F2 plants X **\$500** per tree grown
= \$80,000



with molecular markers:

Germinate 160 F2 plants and sequence for the genotype (x \$10): **\$1600**

+

Throw out ~150 and grow the 10 with predicted phenotype and sequence for the genotype (x \$500): **\$5000**

=\$6,600

(vs \$80,000 without Marker)



Crosses with more genes...

Monohybrid

$A/a \times A/a$

3 possible genotypes

A/A

a/a

A/a

Crosses with more genes...

Monohybrid

$A/a \times A/a$

3 possible genotypes

A/A

a/a

A/a

Dihybrid

$A/a, B/b \times A/a, B/b$

9 possible genotypes

$A/A, B/B$ $A/a, B/B$ $a/a, B/B$

$A/A, B/b$ $A/a, B/b$ $a/a, B/b$

$A/A, b/b$ $A/a, b/b$ $a/a, b/b$

Crosses with more genes...

Monohybrid

$A/a \times A/a$

3 possible genotypes

A/A
 a/a
 A/a

Dihybrid

$A/a,B/b \times A/a,B/b$

9 possible genotypes

$A/A, B/B$ $A/a, B/B$ $a/a, B/B$
 $A/A, B/b$ $A/a, B/b$ $a/a, B/b$
 $A/A, b/b$ $A/a, b/b$ $a/a, b/b$

Trihybrid

$A/a,B/b,C/c \times A/a,B/b,C/c$

27 possible genotypes

...

How many
genotypes from a
decahybrid (10
genes) cross?

Crosses with more genes...

One gene

Monohybrid

$A/a \times A/a$

3 possible genotypes

A/A
 a/a
 A/a

Two genes

Dihybrid

$A/a,B/b \times A/a,B/b$

9 possible genotypes

$A/A, B/B$ $A/a, B/B$ $a/a, B/B$
 $A/A, B/b$ $A/a, B/b$ $a/a, B/b$
 $A/A, b/b$ $A/a, b/b$ $a/a, b/b$

Three genes

Trihybrid

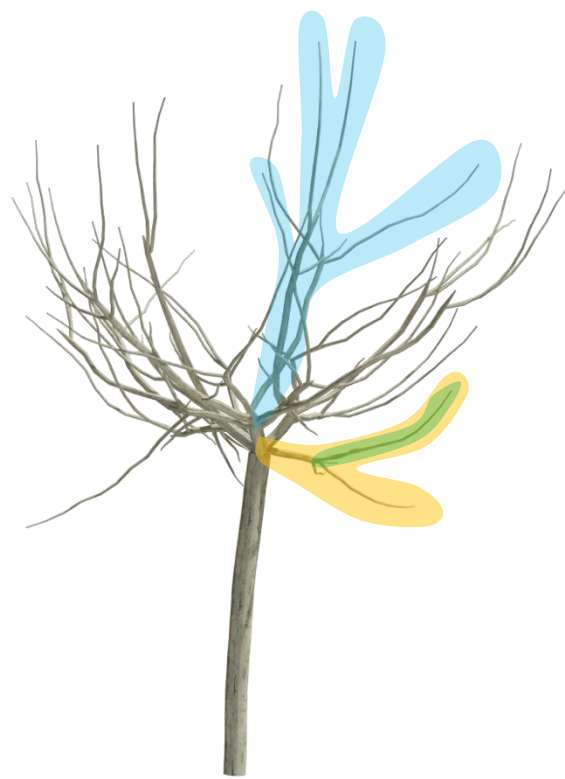
$A/a,B/b,C/c \times A/a,B/b,C/c$

27 possible genotypes

...

How many
genotypes from a
decahybrid (10
genes) cross?

genotypes = 3^n
 n = number of genes



Transmission Genetics 2: Breaking Mendel's Rules

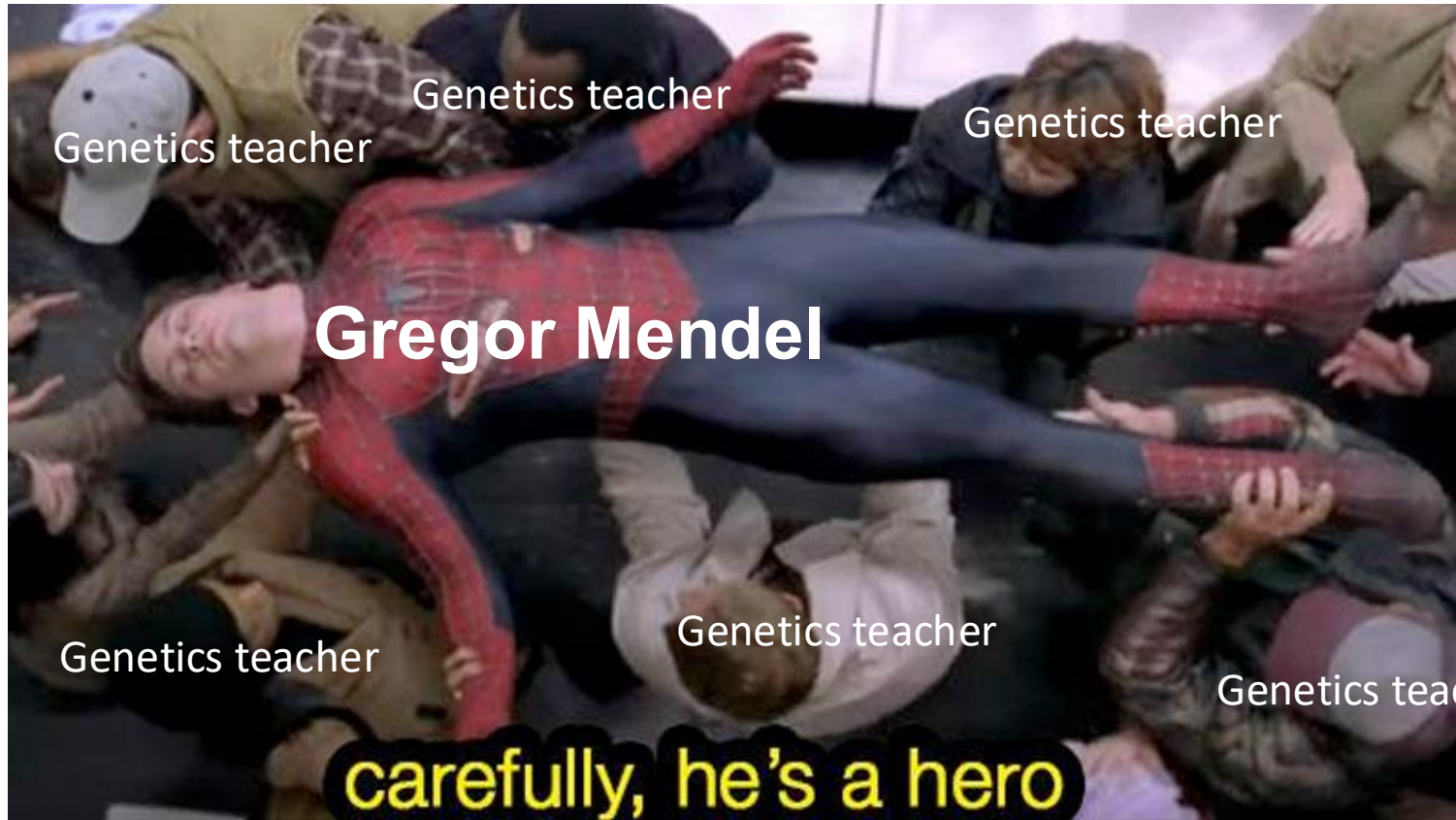
Dominance

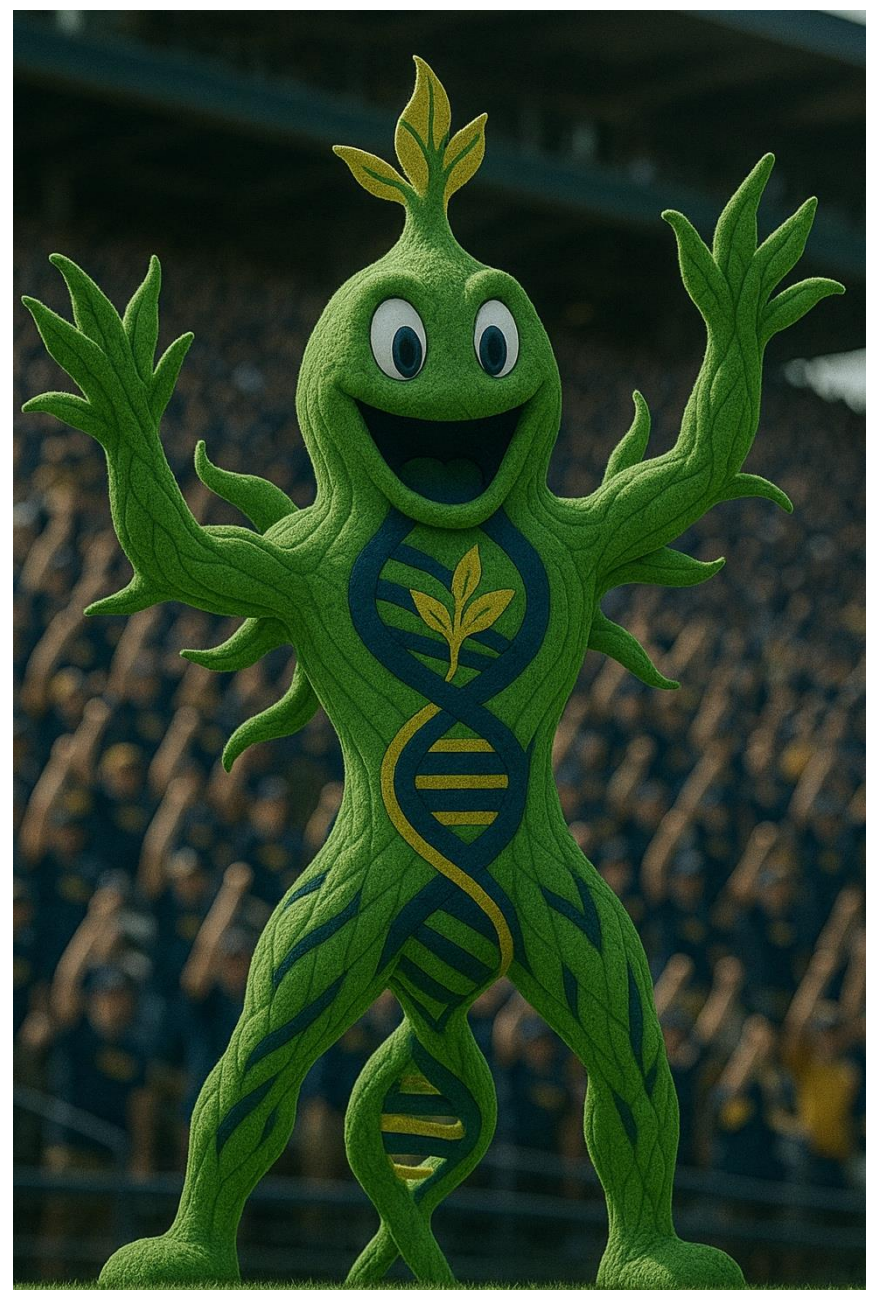


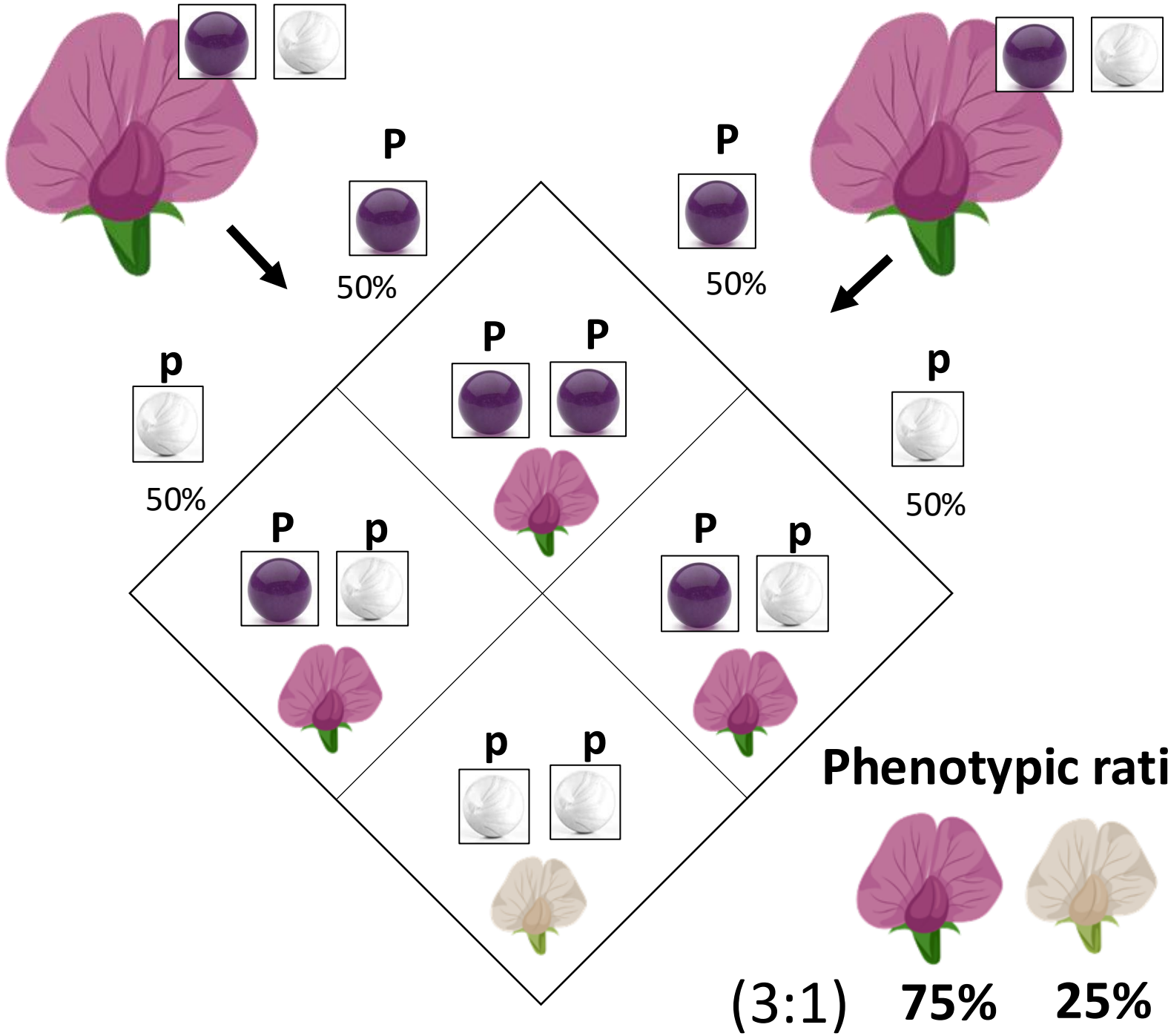
**Independent
assortment of genes**



**Random
segregation of
alleles**







Monohybrid Cross

P/p x P/p

	p	P
p	P/p	P/p
P	P/p	p/p

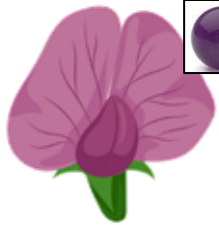
Genotypic ratios

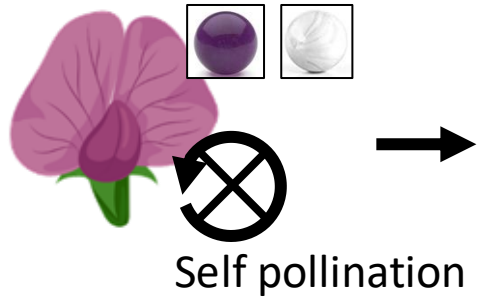
PP: 25%

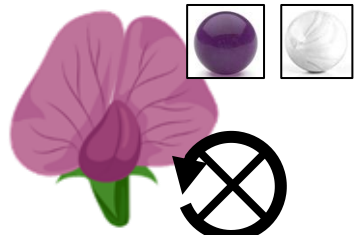
Pp: 50%

pp=25%

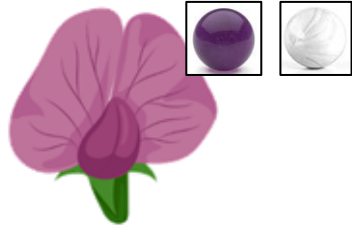
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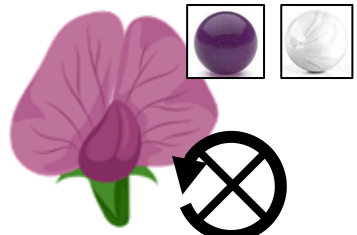




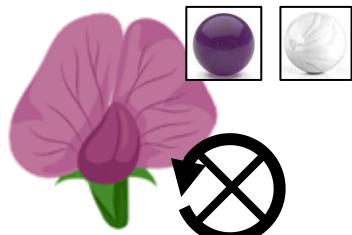
Self pollination



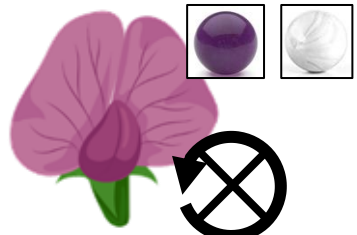
What are the chances of this?



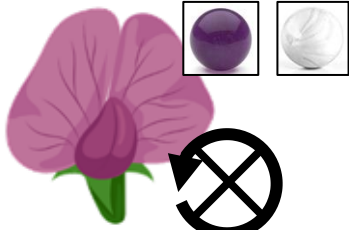
Self pollination



Self pollination



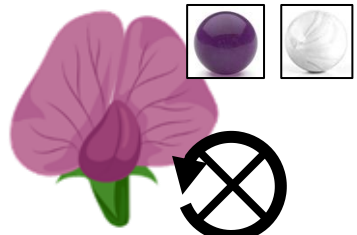
Self pollination



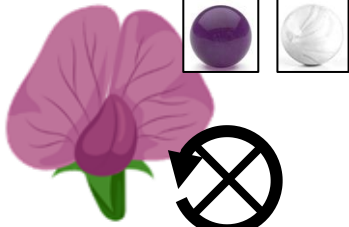
Self pollination



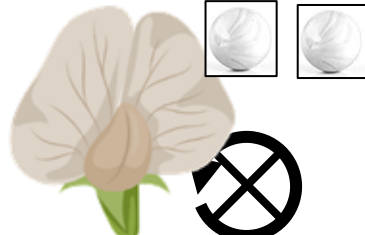
What are the chances of this?



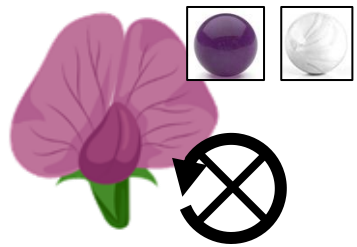
Self pollination



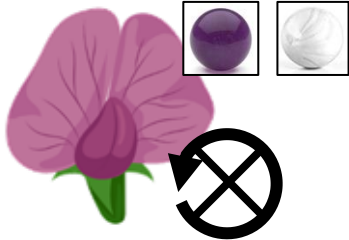
Self pollination



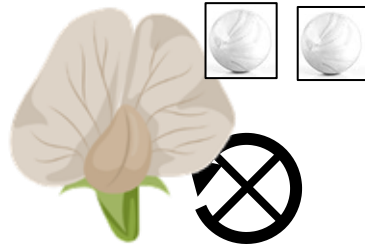
Self pollination



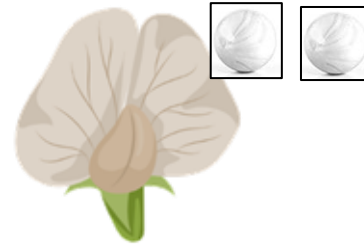
Self pollination



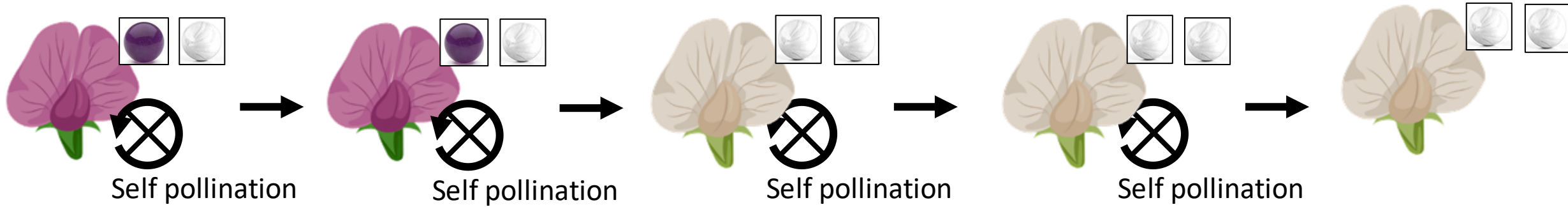
Self pollination

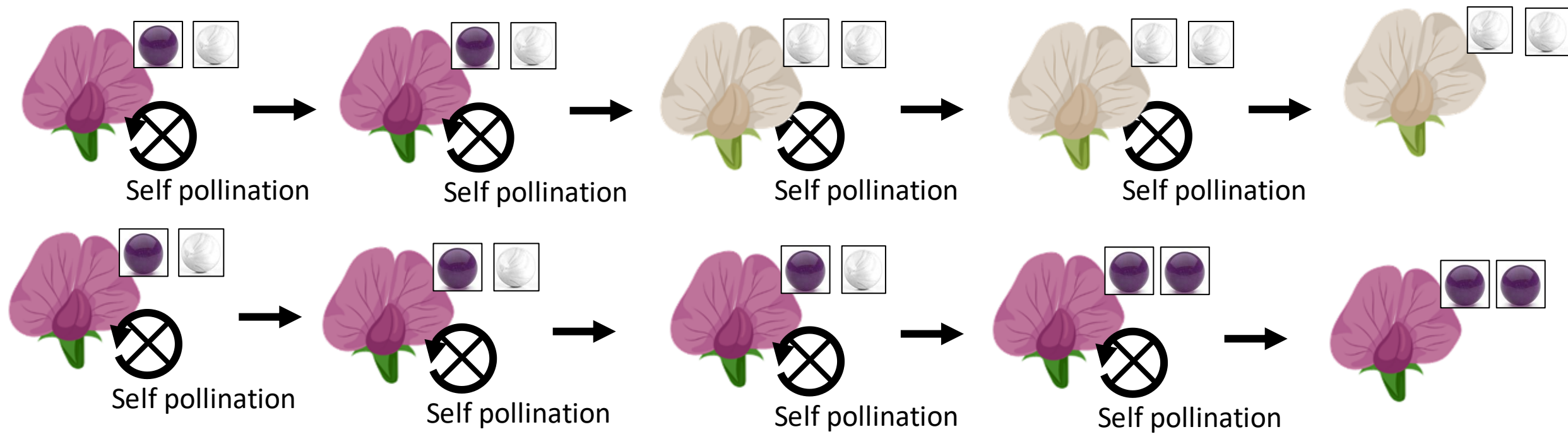


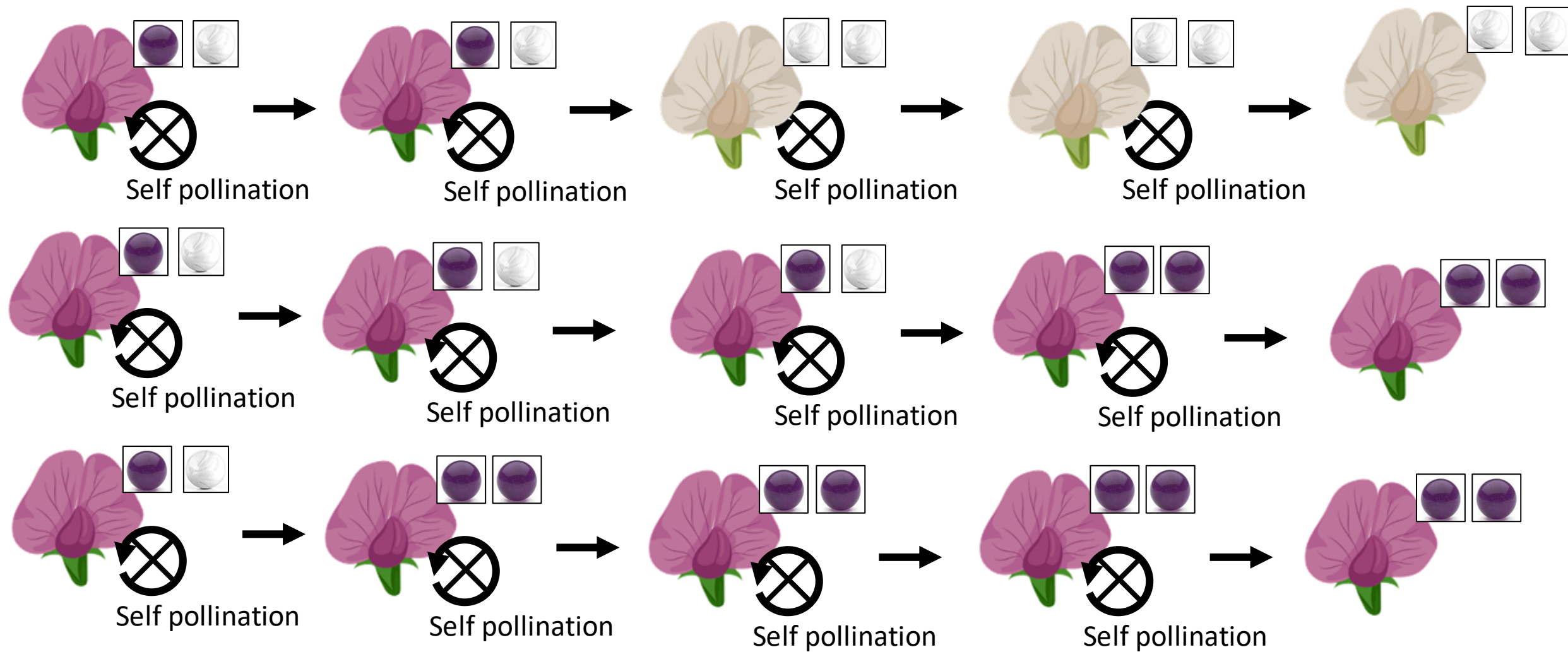
Self pollination

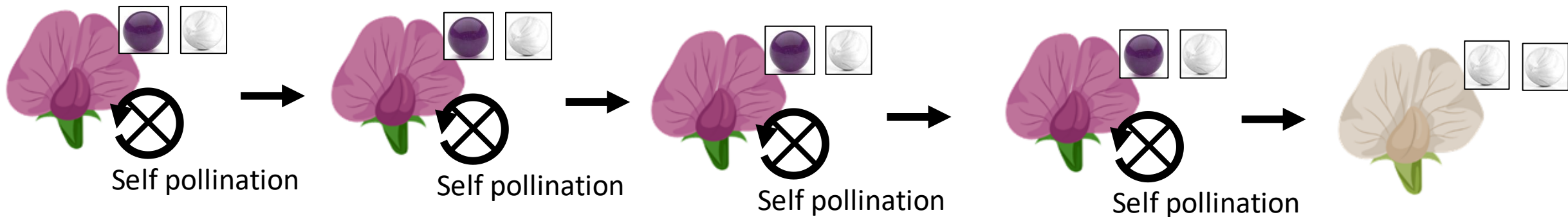
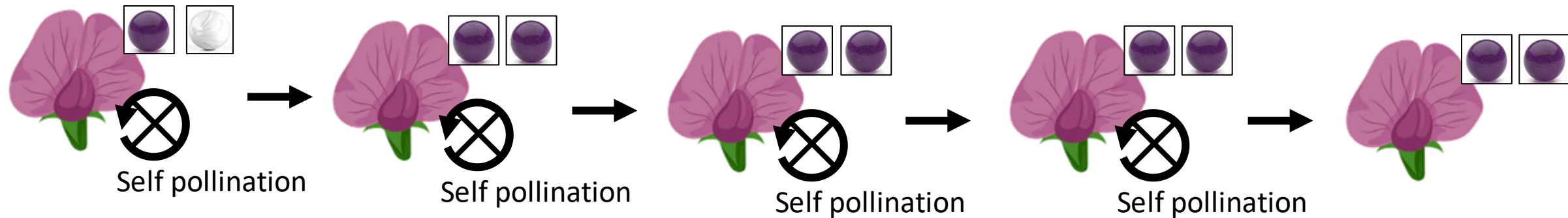
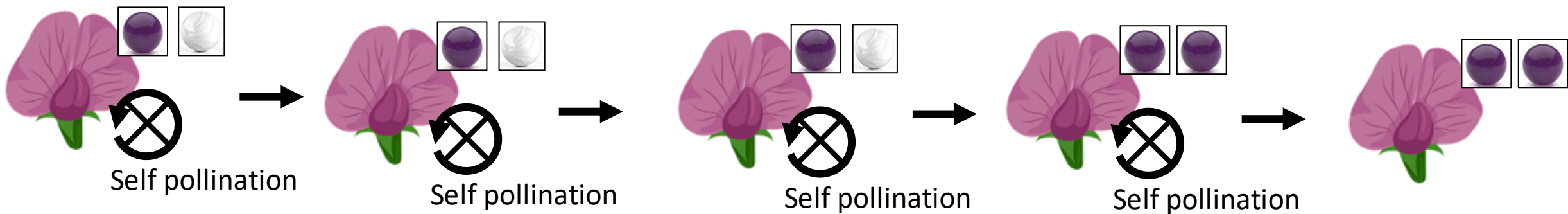
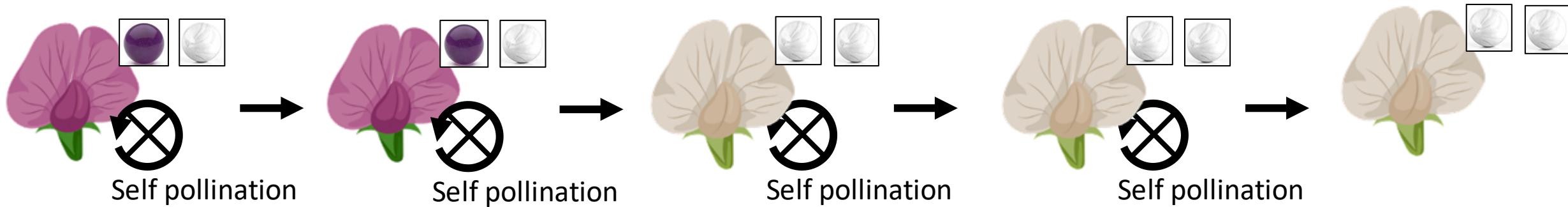


What are the chances of this?









Inbreeding leads to homozygosity

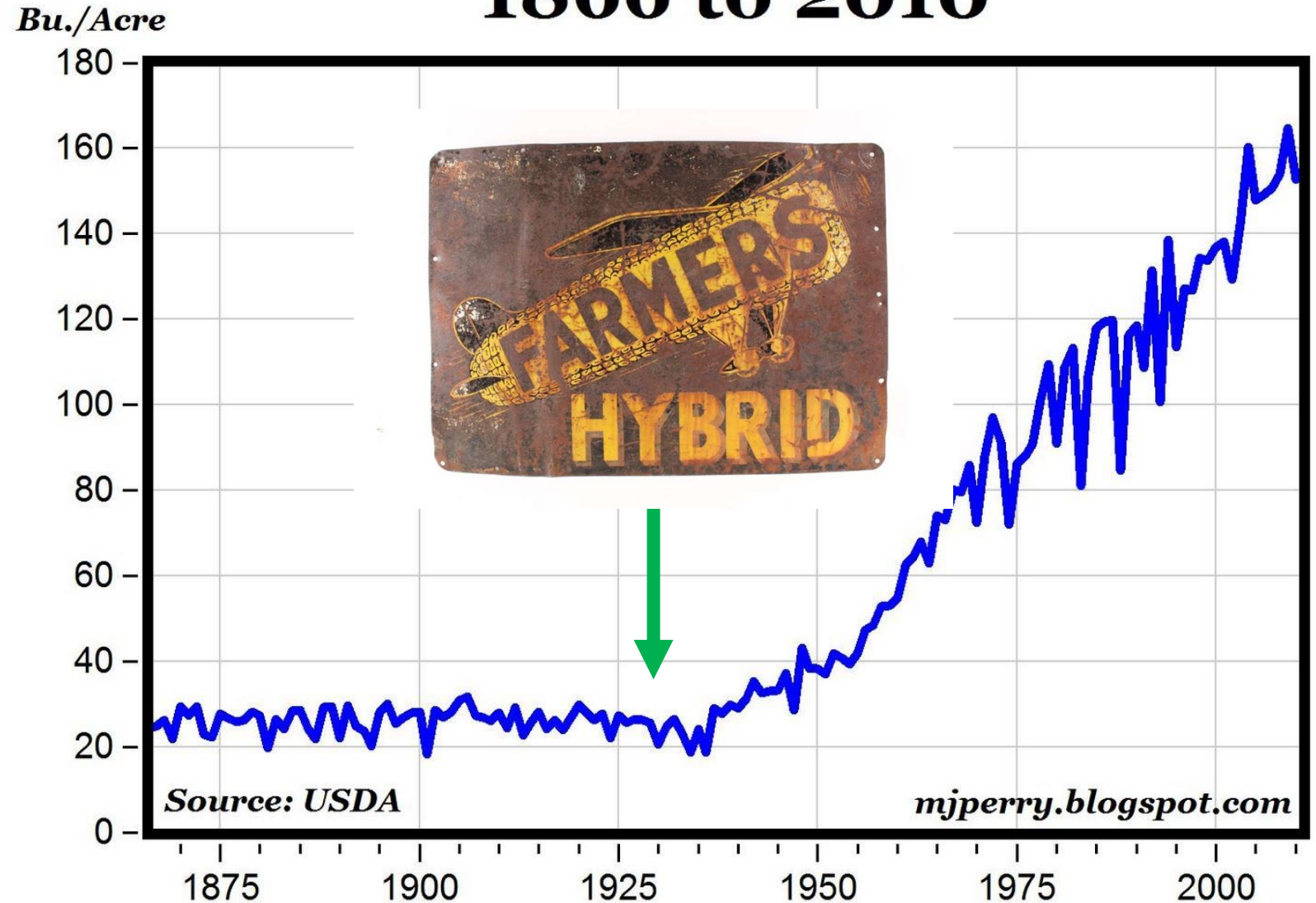
$$\% \text{ chance heterozygous} = (0.5^G) \times 100\%$$

G = number of generations inbreeding

Example: **After 4 generations of inbreeding (selfing): 6.25%**
chance of being heterozygous $(0.5 \times 0.5 \times 0.5 \times 0.5) \times 100\%$
(chance of being **homozygous** $100 - 6.25 = 93.75\%$)

**The \$Billion
value of
Mendelian
inheritance**

U.S. Corn: Yield per Acre 1866 to 2010



Inbred line

Disease resistance

A/A

b/b



X



Inbred line

a/a

B/B Ear size

Inbred line

Disease resistance A/A
 b/b

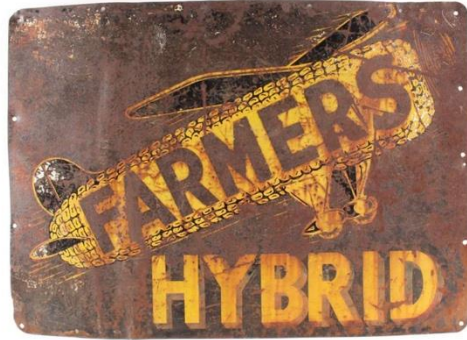


X



Inbred line

a/a
 B/B Ear size



F1

Hybrid – seed sold to farmers



A/a
 b/B



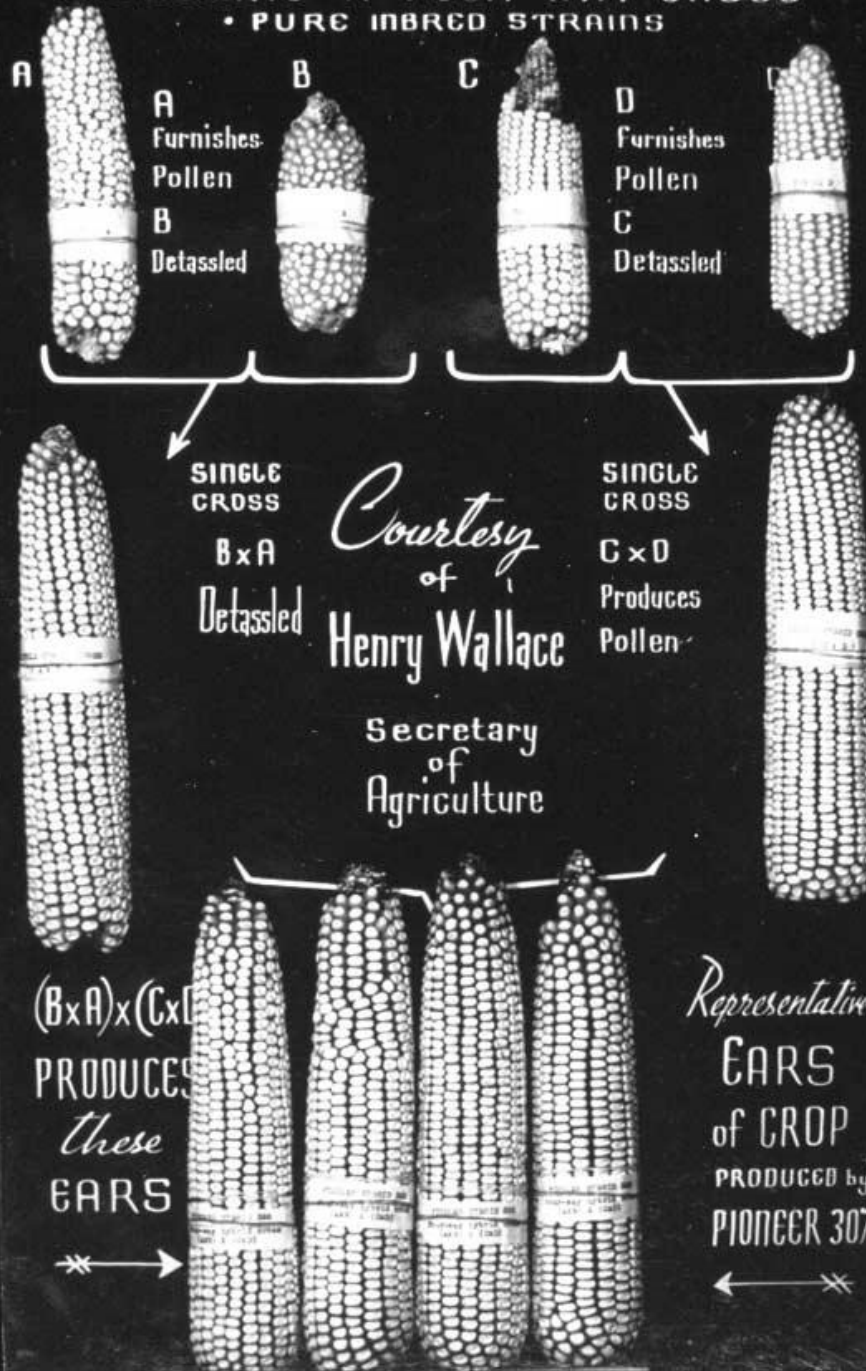
What would happen if the farmer saves some seed and plants it to grow the next year?

Take home messages

- Transmission Genetics is important for breeding because it gives rules of inheritance needed to predict phenotype.
- Alleles are passed randomly from one generation to the next (due to meiosis)
- Segregation of alleles creates an enormous amount of diversity
- Dominance describes when a heterozygote shows the phenotype of one of the alleles.
- Inbreeding leads to homozygosity
- Crossing inbred lines is how breeders produce hybrid seed

PRODUCING A FOUR-WAY CROSS

• PURE INBRED STRAINS



Before true inbreds there were “double-cross hybrids”...

“While the plants produced from double-cross hybrid seed are not as uniform and high-yielding as those for a single-cross, they exhibited greater vigor and performance than the open-pollinated corn varieties that preceded them.”

Pioneer’s Website